5G RAN

Positioning Feature Parameter Description

Issue Draft B

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Contents

1 Change History

1.1 5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised descriptions of the RF module requirements for SRS field strength-based positioning in TDD scenarios. For details, see **5.3.3 Hardware**.

Revised descriptions of the benefits of precise positioning (LampSite NR) in TDD scenarios. For details, see **6.2.1 Benefits**.

Revised descriptions of data preparation and MML-based configurations for the inter-base-station-transmission-based beacon-free function in TDD scenarios. For details, see **6.4.1.1 Data Preparation** and **6.4.1.2 Using MML Commands**.

Revised other descriptions in this document in TDD scenarios, which do not involve changes in meanings and therefore are not detailed.

1.2 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 06 (2025-12-14).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added mutually exclusive relationships among "SRS and PUSCH/PUCCH concurrency compatibility", "proactive avoidance of remote interference", and "positioning-dedicated SRS transmission in uplink slots". For details, see **5.3.2.2 Mutually Exclusive Functions** and **7.1.3.2.2 Mutually Exclusive Functions**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added a mutually exclusive relationship between PUSCH coordinated power control and "positioning-dedicated SRS transmission in uplink slots". For details, see **7.1.3.2.2 Mutually Exclusive Functions**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between SRS field-strength-based positioning and time-domain power saving BWP. For details, see **5.3.2.3 Function Impacts**.	None	Low-frequency TDD	DBS3900 LampSite and DBS5900 LampSite
Added a reserved parameter controlling the support for NR T_{ADV} measurement and reporting in uplink E-CID-based positioning to the disuse list. For details, see **4.1.1.2 E-CID-based Positioning**.	Added the **RSVDSWPARAM2_BIT19** option of the reserved parameter **NRCellRsvd**.***RsvdSwParam2*** to the disuse list.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised descriptions of networking requirements for LPHAP in TDD scenarios. For details, see **7.2.3.5 Networking**.

Revised descriptions of RF module requirements for precise positioning (LampSite NR) in TDD scenarios. For details, see **6.3.3 Hardware**.

Revised other descriptions in this document in TDD scenarios, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

Trial Features

Trial features are features that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these features can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial features shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial features. Trial features are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial features may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial features but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial features are free, Huawei is not liable for any trial feature malfunctions or any losses incurred by using the trial features. Huawei does not promise that problems with trial features will be resolved in the current version. Huawei reserves the rights to convert trial features into commercial features in later R/C versions. If trial features are converted into commercial features in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial features. If a customer fails to purchase such a license, the trial feature(s) will be invalidated automatically when the product is upgraded.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-050102	Basic Positioning	**4 Basic Cellular-based Positioning** **5 SRS Field Strength-based Positioning**
FOFD-050207	Precise Positioning (LampSite NR) (trial)	**6 Precise Positioning (LampSite NR) (Trial)**
FOFD-070210	Precise Positioning (NR) (trial)	**7 Precise Positioning (NR) (Trial)**

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Basic cellular-based positioning	In SA networking, all UEs report enhanced cell ID (E-CID)-based positioning information. In SA and NSA hybrid networking, only SA UEs report E-CID-based positioning information. In NSA networking, UEs cannot report E-CID-based positioning information.	**4 Basic Cellular-based Positioning**
SRS field strength-based positioning	Supported only in SA networking	**5 SRS Field Strength-based Positioning**
Precise positioning (LampSite NR) (trial)	Supported only in SA networking	**6 Precise Positioning (LampSite NR) (Trial)**
Precise positioning (NR) (trial)	Supported only in SA networking	**7 Precise Positioning (NR) (Trial)**

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Basic cellular-based positioning	Supported only in low frequency bands	**4 Basic Cellular-based Positioning**
SRS field strength-based positioning	Supported only in low frequency bands	**5 SRS Field Strength-based Positioning**
Precise positioning (LampSite NR) (trial)	Supported only in low frequency bands	**6 Precise Positioning (LampSite NR) (Trial)**
Precise positioning (NR) (trial)	Supported only in low frequency bands	**7 Precise Positioning (NR) (Trial)**

3 Overview

Positioning is a technique used to determine a UE's geographic location by measuring radio signals on a mobile communications network. This document describes basic cellular-based positioning, SRS field strength-based positioning, and precise positioning.

Applicable Scenarios

Basic cellular-based positioning applies to 5G to customer (5GtoC) scenarios:

- Public safety services, such as the emergency response to crimes or vehicle accidents/breakdowns. In such emergencies, callers may be unable to accurately report their locations. This function helps emergency responders determine exactly where they reach the callers in need for a rapid response.
- Individual safety services, such as elder and child care.
- Cellular mobile network design optimization, during which UE's location data facilitates value-added service provisioning on the 5G core network (5GC). With such data, network resources are dynamically and intelligently monitored and allocated, enhancing network performance and service quality.

SRS field strength-based positioning applies to indoor 5G to business (5GtoB) campus scenarios. For example:

- Asset tracking and inventory checking, such as material stocktaking, equipment tracking, and production resource scheduling for smart manufacturing
- Personnel positioning for assistance and out-of-bounds alarms based on user locations to ensure personnel safety

Precise positioning (LampSite NR) applies to indoor 5GtoB campus/factory scenarios. For example:

- Asset tracking and inventory checking, such as material stocktaking, equipment tracking, and production resource scheduling for smart manufacturing
- Personnel permission management, such as electronic fencing and unauthorized entry alerts
- Navigation services, such as real-time location display and accurate location tracking of personnel

Precise positioning (NR) applies to 5GtoB campus/factory scenarios, including manufacturing (high-rooftop indoor area), campus road, and warehousing and logistics scenarios. For example:

- Personnel positioning for assistance and out-of-bounds alarms based on user locations to ensure personnel safety
- Production resource positioning, such as production resource or material positioning in manufacturing scenarios to facilitate unified management and scheduling

Network Architecture

Figure 3-1 shows the architecture of positioning in NR.

When the access and mobility management function (AMF) receives a request for a positioning service associated with a particular target UE from another entity (for example, a gateway mobile location center (GMLC) or UE) or the AMF itself decides to initiate such a positioning request, the AMF sends the request to a location management function (LMF) over the NL1 interface. After receiving the positioning request, the LMF transfers positioning-related assistance data to the target UE over the NL1, N2, and NR-Uu interfaces and requests positioning-related measurement quantities from the UE or gNodeB to calculate the UE's location. The LMF then returns the calculated location to the AMF over the NL1 interface. If a positioning service is requested by an entity (for example, a GMLC or UE) other than the AMF, the AMF sends the positioning result to this entity. For details, see **4 Basic Cellular-based Positioning**.

In precise positioning (LampSite NR), after receiving an LCS client-initiated positioning request, the GMLC sends the request to the AMF over the NL2 interface. After receiving the positioning request, the AMF selects the LMF that serves the UE performing positioning services, queries and obtains related measurement quantities from the gNodeB over the N2 interface, and then transparently transmits the measurement quantities to the LMF. The LMF calculates the UE's location and then returns it to the AMF over the NL1 interface. The AMF transparently transmits the calculated location to the GMLC, and then the GMLC sends it to the positioning request-initiating LCS client. For details, see **6 Precise Positioning (LampSite NR) (Trial)**.

In precise positioning (NR), after receiving an LCS client-initiated positioning request, the GMLC sends the request to the AMF over the NL2 interface. After receiving the positioning request, the AMF selects the LMF that serves the UE performing positioning services, queries and obtains related measurement quantities from the gNodeB over the N2 interface, and then transparently transmits the measurement quantities to the LMF. The LMF calculates the UE's location and then returns it to the AMF over the NL1 interface. The AMF transparently transmits the calculated location to the GMLC, and then the GMLC sends it to the positioning request-initiating LCS client. For details, see **7 Precise Positioning (NR) (Trial)**.

Figure 3-1 NR positioning network architecture

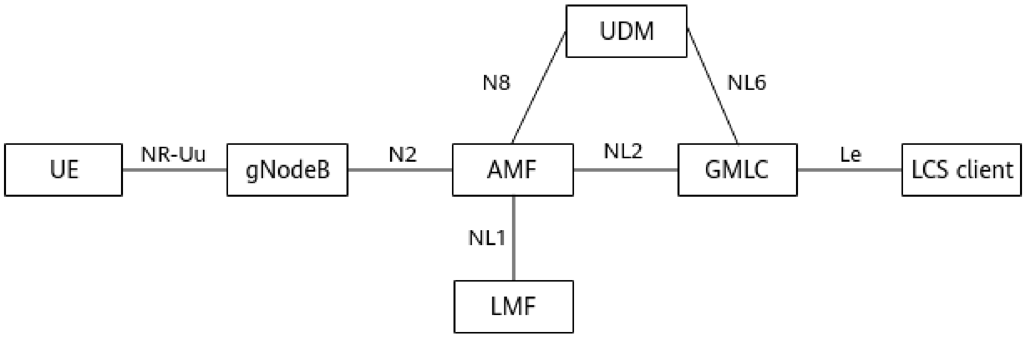


Table 3-1 Functions of positioning-involved NEs

NE	Function
gNodeB	When requested by an LMF, the gNodeB performs the following operations: ● Terminates the NR Positioning Protocol A (NRPPa) and receives or sends NRPPa-related positioning messages over the N2 interface. ● Performs positioning-related measurements and reports measurement quantities to the LMF. For example, in uplink E-CID-based positioning, the gNodeB reports required measurement quantities that include the synchronization signal based reference signal received power (SS-RSRP) and synchronization signal based reference signal received quality (SS-RSRQ) of the serving cell and its neighboring cells.
User equipment (UE)	● Terminates the LTE Positioning Protocol (LPP) and exchanges positioning messages with other entities. ● Reports its positioning capabilities to the LMF. ● Reports UE-assisted/LMF-based positioning-related measurement results or UE-based calculation results to the LMF.
Access and mobility management function (AMF)	● Receives and manages positioning requests from a UE, GMLC, or itself. ● Selects a suitable LMF. ● Interworks with gNodeBs for positioning based on NRPPa. ● Transparently transmits positioning messages to the LMF, gNodeB, UE, and other entities.

NE	Function
Location management function (LMF)	• Receives and processes positioning requests or positioning-related data requests from the AMF. • Reports positioning results or related positioning data to the AMF. • Selects a single positioning method or a combination of multiple positioning methods. • Determines positioning-related measurements suited to the selected positioning method. • Calculates positioning-related assistance data and delivers the data to the target UE. In E-CID-based positioning, no assistance data is required. • Calculates the location and determines the positioning accuracy.
Gateway mobile location center (GMLC)	• Obtains UE latitude and longitude and reports this information to the LCS client after receiving UE positioning requests from the LCS client. • Serves as the operation platform of the positioning system to manage subscriber data, service data, service subscription information, and service provider (SP) data and provide accounting and authentication for value-added services.
Unified data management (UDM)	Stores positioning subscription information and route information of subscribers.
LoCation service (LCS) client	A logical functional entity that may reside inside the PLMN (for example, an O&M tool) or outside the PLMN (for example, a third-party positioning server not deployed by an operator). The LCS client requests the location information of one or more target UEs by signaling a specified set of parameters, such as QoS.

NOTE

- A gNodeB and an LMF exchange positioning information in compliance with the NRPPa protocol, which is terminated at the gNodeB and LMF.
- An LMF and a UE exchange positioning information in compliance with the LPP protocol, which is terminated at the LMF and UE. The gNodeB transparently transmits NAS messages between the LMF and the UE.

Positioning Methods

Huawei NR positioning supports the following methods:

Basic cellular-based positioning:

- Cell ID-based positioning: The AMF obtains the UE's location directly through the gNodeB. For details, see **4.1.1.1 Cell ID-based Positioning**.

- E-CID-based positioning: The LMF determines to use this positioning method based on UE capabilities and requests required measurement results from the UE or gNodeB to calculate the UE's location. For details, see **4.1.1.2 E-CID-based Positioning**.

SRS field strength-based positioning:

- Field strength triangulation-based positioning: The gNodeB measures the RSRP values of uplink SRSs from a UE to at least three pRRUs and sends the results to the LMF. Then, the LMF calculates the UE's location based on the RSRP differences between every two pRRUs and their known locations. For details, see **5.1.1.1 Field Strength Triangulation-based Positioning**.
- Field strength fingerprint-based positioning: The gNodeB measures the RSRP values of uplink SRSs from a UE to multiple pRRUs and sends the results to the LMF. Then, the LMF determines the UE's location by matching the results against the signal characteristics in the SRS fingerprint database. For details, see **5.1.1.2 Field Strength Fingerprint-based Positioning**.

Precise positioning (LampSite NR):

- U-TDOA-based positioning: U-TDOA-based high-precision positioning and one-dimensional U-TDOA-based positioning are included. For details, see **6.1.1.1 U-TDOA-based Positioning**.
- TOA fingerprint-based positioning: The gNodeB measures the TOA values of uplink SRSs from a UE to multiple pRRUs and sends the results to the LMF. Then, the LMF determines the UE's location by matching the results against the TOA characteristics in the SRS fingerprint database. For details, see **6.1.1.2 TOA Fingerprint-based Positioning**.
- LPHAP: This function takes effect for RedCap UEs. For details, see **6.1.1.3 LPHAP**.
- Positioning-dedicated beacon-free function: The positioning accuracy objective can be achieved without deploying beacon UEs. For details, see **6.1.1.4 Positioning-dedicated Beacon-free Function**.

Precise positioning (NR):

- U-TDOA- and AoA-based high-precision positioning: The gNodeB measures the TOA and AoA values of uplink SRSs from a UE to at least two or three TRPs in different networking scenarios and sends the results to the LMF. The LMF then calculates the UE's location by using an appropriate algorithm based on the received TOA and AoA values of multiple TRPs. For details, see **7.1 U-TDOA- and AoA-based High-Precision Positioning**.
- LPHAP: This function takes effect for RedCap UEs. For details, see **7.2 LPHAP**.

4 Basic Cellular-based Positioning

4.1 Principles

4.1.1 Positioning Methods

Basic cellular-based positioning supports:

- Cell ID-based positioning
- E-CID-based positioning

E-CID-based positioning is an enhancement to cell ID-based positioning.

4.1.1.1 Cell ID-based Positioning

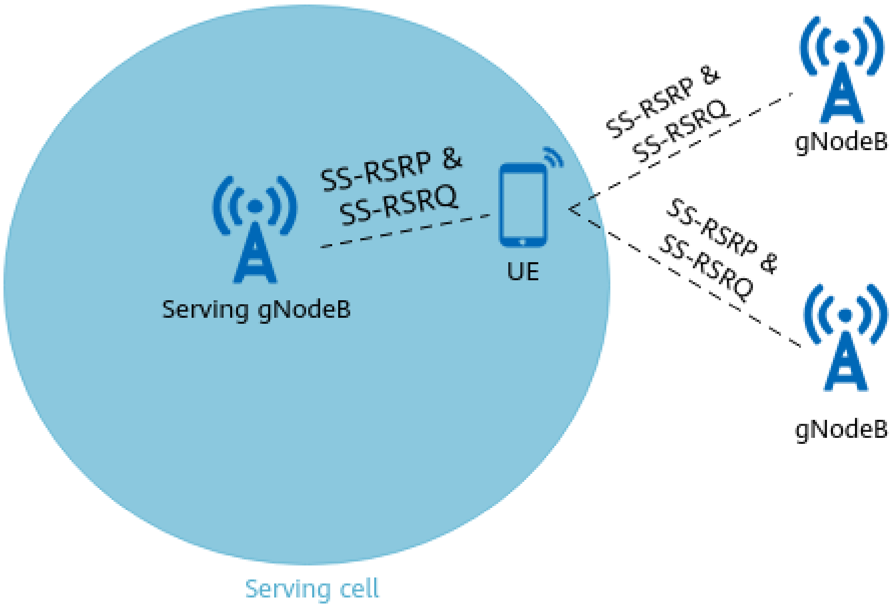
Cell ID-based positioning is a method introduced to locate a UE based on the cell ID. When initiating a registration, location update, or service setup, the UE transmits an RRC connection setup request message that includes the cell ID. The NR system uses this ID to estimate the location of a UE.

Cell ID-based positioning is not under switch control. This positioning method requires coordination of only the gNodeB and AMF, and does not require any positioning-related measurements by UEs or any other NEs. As such, a response can be quickly received once a positioning request is initiated. However, the positioning accuracy varies with the cell radius. A smaller cell radius delivers higher positioning accuracy, and a larger cell radius delivers the opposite. The deviation reaches several hundreds of meters to several kilometers. As a result, this method cannot satisfy the QoS requirements of most positioning services. For details about the positioning procedure, see **4.1.3.1 Cell ID-based Positioning Procedure**.

4.1.1.2 E-CID-based Positioning

Figure 4-1 shows how E-CID-based positioning works. Based on the SS-RSRP and SS-RSRQ values of the serving cell and its neighboring cells, the LMF calculates the distance from a UE to the antenna of each cell. In this way, the estimated UE's location is a point in the serving cell.

Figure 4-1 Principles of E-CID-based positioning



E-CID-based positioning is controlled by the **ECID_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter. Based on whether the gNodeB or UE sends measurement quantities to the LMF, E-CID-based positioning is classified into uplink E-CID-based positioning and downlink E-CID-based positioning.

- Uplink E-CID-based positioning

  In uplink E-CID-based positioning, the gNodeB sends measurement quantities to the LMF for location calculation. Based on the measurement quantities, the LMF estimates the distance between the UE and the antenna of each gNodeB to determine the geographic location of the UE. The measurement quantities include UE-measured SS-RSRP and SS-RSRQ values of the serving cell and its neighboring cells, NR CGI, NR TAC, base-station-measured NR T_{ADV} (standing for the timing advance), and other information.

  For details about the positioning procedure, see **4.1.3.2 Uplink E-CID-based Positioning Procedure**.

- Downlink E-CID-based positioning

  In this positioning method, a UE sends the measurement quantities to the LMF to estimate the location of the UE through the LPP message. The measurement quantities include the UE-measured SS-RSRP and SS-RSRQ values of the serving cell and its neighboring cells. Based on the results, the LMF estimates the distance between the UE and the antenna of each gNodeB and then determines the estimated UE location. Downlink E-CID-based positioning has the similar principle as uplink E-CID-based positioning, except that the UE, not the gNodeB, sends measurement quantities. For details about the positioning procedure, see **4.1.3.3 Downlink E-CID-based Positioning Procedure**.

4.1.2 Evaluation Standards

The accuracy of cell ID-based positioning and E-CID-based positioning is measured by the horizontal positioning accuracy, which is closely related to the cell radius. A smaller cell radius delivers higher positioning accuracy, and a larger cell radius delivers the opposite. If a cell has a radius of more than 2000 m, the positioning accuracy deviation may exceed 1000 m.

4.1.3 Positioning Procedure

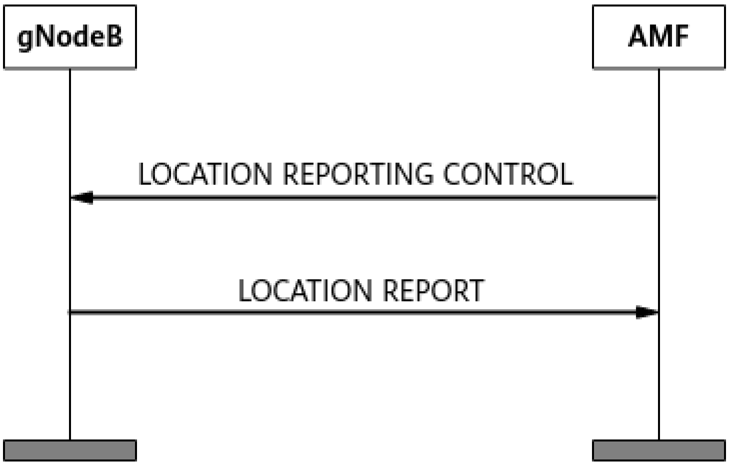
4.1.3.1 Cell ID-based Positioning Procedure

Cell ID-based positioning can be triggered through a location reporting or User Location Information procedure.

Triggered Through a Location Reporting Procedure

Figure 4-2 shows the location reporting procedure for cell ID-based positioning.

Figure 4-2 Triggered through a location reporting procedure



1. The AMF sends a LOCATION REPORTING CONTROL message to the gNodeB, requesting a UE location report.
2. The gNodeB returns a LOCATION REPORT message, including information about the UE's serving cell, such as the tracking area identity (TAI) and NR CGI, based on the AMF-signaled request type (direct reporting or condition-triggered reporting).

NOTE

When the **LOCATION_REPORTING_COMPAT_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter is selected, the base station performs the following operations based on the value of *Event Type* in *Location Reporting Request Type* of the Location Reporting Control message:

- If *Event Type* is **change of serving cell and UE presence in the Area of Interest**, the base station supports location information reporting.
- If *Event Type* is **Additional Cancelled Location Reporting Reference ID List**, the base station supports stopping location information reporting in batches.

Otherwise, the base station ignores the preceding IEs and does not perform corresponding processing.

Triggered Through a User Location Information Procedure

During PDU session management, UE context management, UE mobility management, and NAS transport message management, the gNodeB sends a message containing the *User Location Information* IE to the core network over the NG interface. This way, the core network can obtain information about the UE's serving cell to estimate UE location.

As defined in 3GPP TS 38.413, the *User Location Information* IE is contained in the following messages:

PDU session management messages

- PDU SESSION RESOURCE RELEASE RESPONSE
- PDU SESSION RESOURCE MODIFY RESPONSE
- PDU SESSION RESOURCE NOTIFY
- PDU SESSION RESOURCE MODIFY INDICATION
- PDU SESSION RESOURCE SETUP RESPONSE

UE context management messages

- UE CONTEXT RELEASE COMPLETE
- UE CONTEXT MODIFICATION RESPONSE
- RRC INACTIVE TRANSITION REPORT

UE mobility management messages

- HANDOVER NOTIFY
- PATH SWITCH REQUEST

NAS transport messages

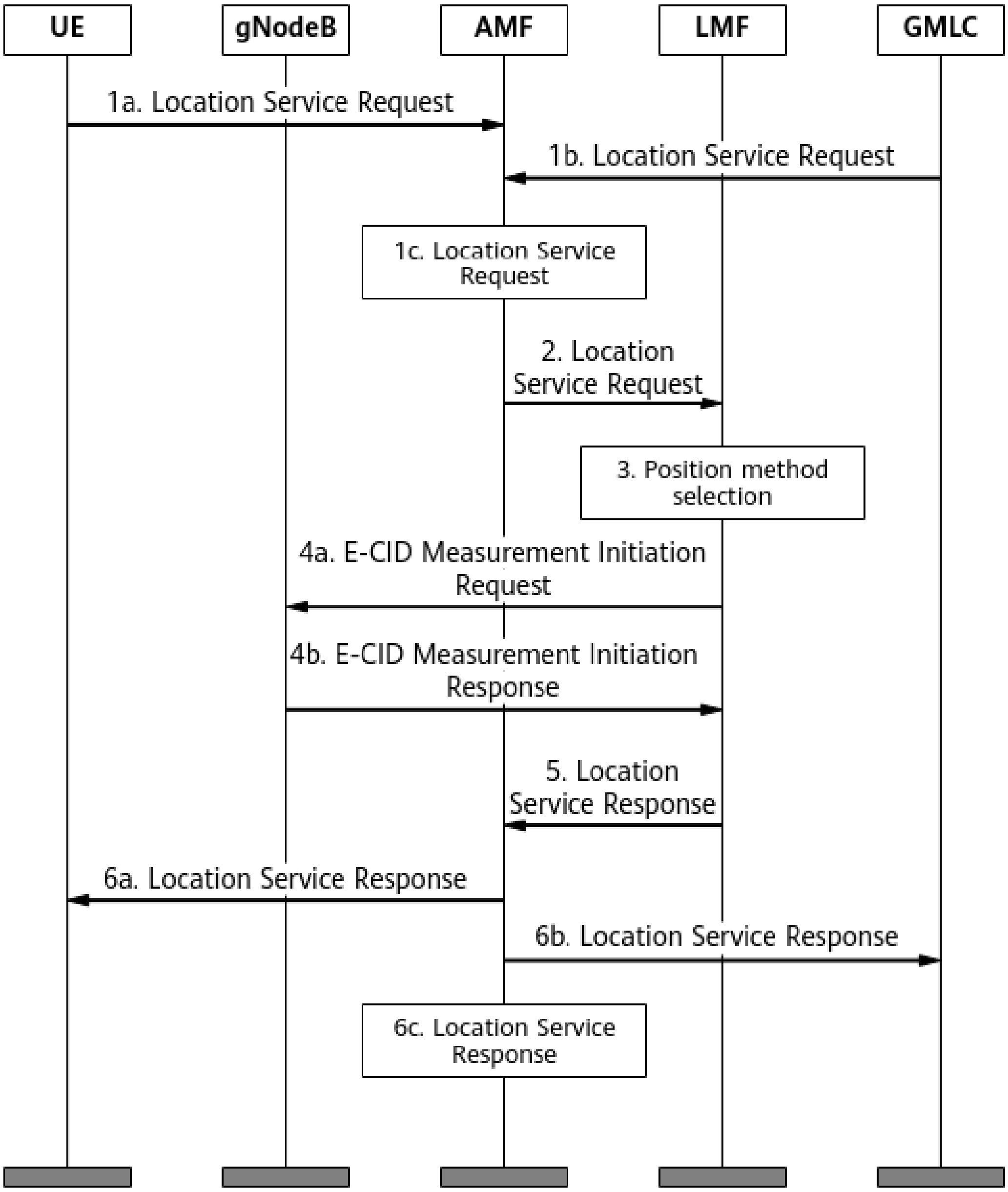
- INITIAL UE MESSAGE
- UPLINK NAS TRANSPORT

NOTE

If the PDU SESSION RESOURCE SETUP RESPONSE message needs to carry the *User Location Information* IE to the AMF, the **ULI_IN_PDU_SESS_SETUP_RSP_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter needs to be selected.

4.1.3.2 Uplink E-CID-based Positioning Procedure

Figure 4-3 Uplink E-CID-based positioning procedure



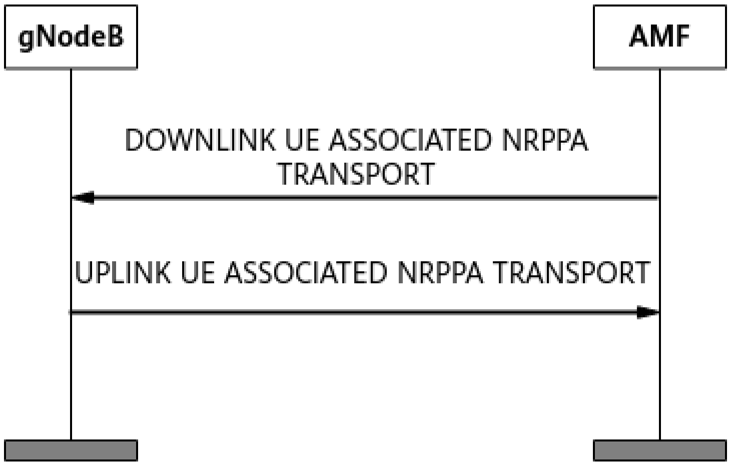
The uplink E-CID-based positioning procedure is described as follows:

1. A UE, GMLC, or AMF initiates a positioning request.
   - 1a. If a mobile originated location request (MO-LR) is initiated, a UE sends a NAS message to the serving AMF to request positioning services, such as UE location or positioning-related assistance data provisioning.
   - 1b. If a mobile terminated location request (MT-LR) is initiated, a GMLC sends a positioning request to the serving AMF to request positioning services for a specific UE, such as UE location provision.
   - 1c. If a network induced location request (NI-LR) is initiated, the PLMN serving a UE sends a positioning request to this UE. For example, the AMF initiates an emergency call for a UE.

2. The AMF sends the positioning request to the LMF.
3. The LMF selects the best-suited positioning method based on the following factors:
   - QoS requirements of the positioning application, including requirements for the horizontal positioning accuracy and latency
   - Positioning methods configured on the LMF
   - Whether the positioning function is activated in the serving cell of the UE
   - UE positioning capabilities
4. The LMF obtains positioning-related measurement quantities or assistance data.
   - 4a. The LMF sends an NRPPa message E-CID MEASUREMENT INITIATION REQUEST to the gNodeB, requesting for SS-RSRP and SS-RSRQ measurements.
   - 4b. The gNodeB sends an E-CID MEASUREMENT INITIATION RESPONSE message containing the SS-RSRP and SS-RSRQ values measured by the UE, NR T_{ADV} measured by the base station, NR CGI, and NR TAC to the LMF.

   The NRPPa messages involved in step 4 are exchanged between the gNodeB and AMF over the NG interface, as shown in **Figure 4-4**.
   - The AMF sends the gNodeB a DOWNLINK UE ASSOCIATED NRPPA TRANSPORT message containing the E-CID-based positioning request initiated by the LMF.
   - The gNodeB sends the measurement results to the AMF through an UPLINK UE ASSOCIATED NRPPA TRANSPORT message, and then the AMF forwards the results to the LMF.

Figure 4-4 NRPPa message exchanges over the NG interface

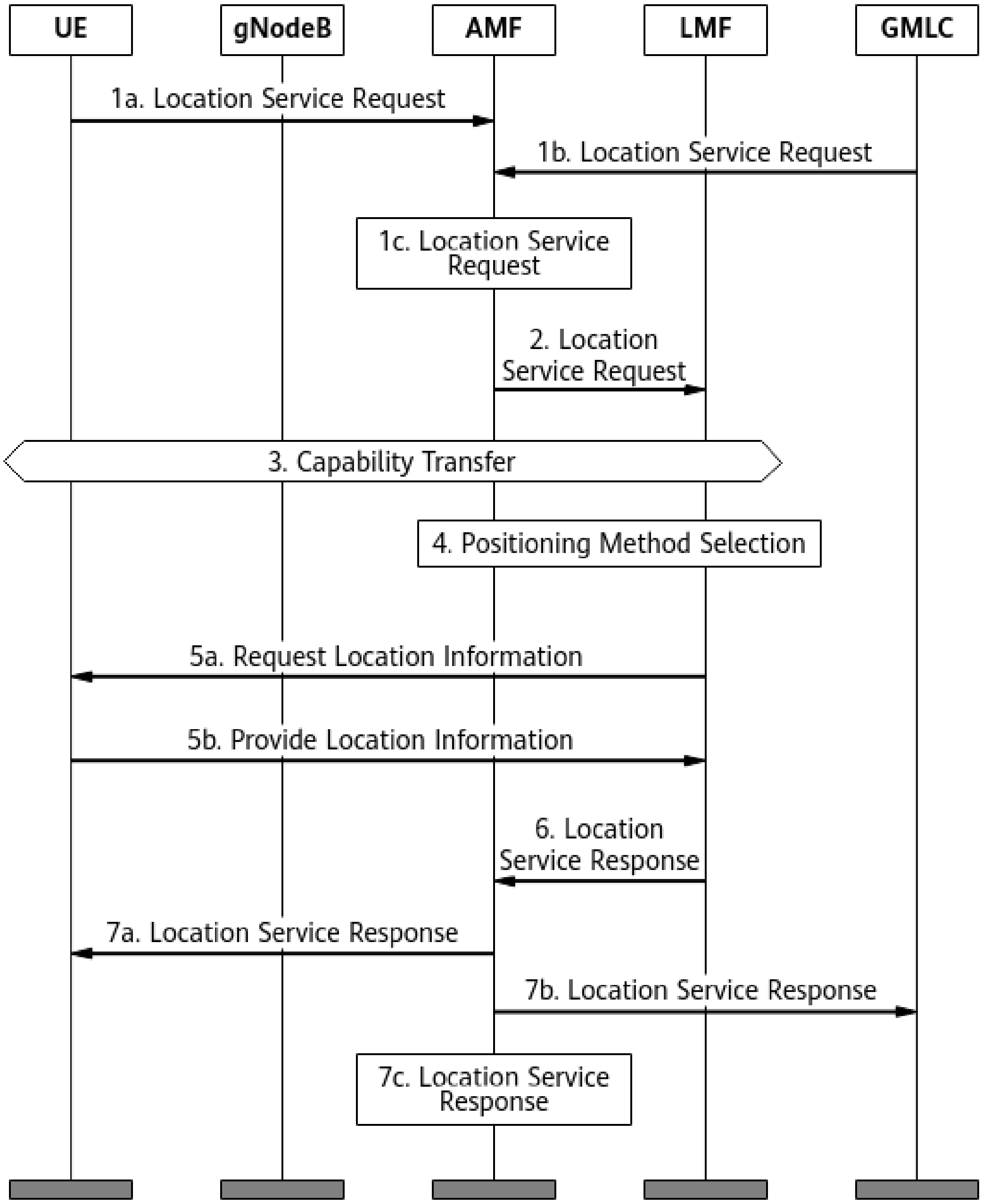


5. The LMF provides a positioning service response to the AMF, which includes details such as success or failure indication and deviation information.
6. The AMF reports the positioning results.
   - 6a. If step 1a is performed, the AMF transparently transmits the positioning results to the UE.
   - 6b. If step 1b is performed, the AMF transparently transmits the positioning results to the GMLC.

– 6c. If step 1c is performed, the AMF transparently transmits the positioning results to assist the service that triggers the positioning procedure.

4.1.3.3 Downlink E-CID-based Positioning Procedure

Figure 4-5 Downlink E-CID-based positioning procedure



Downlink E-CID-based positioning involves a procedure similar to that of uplink E-CID-based positioning, where the only differences occur in steps 3 and 5.

In step 3, the UE reports its positioning capabilities to the LMF.

In step 5a, the LMF sends a Request Location Information message (a NAS message for the gNodeB) to the UE to request location information. In step 5b, the UE completes required measurements and sends the results to the LMF through a Provide Location Information message (a NAS message for the gNodeB).

4.2 Network Analysis

4.2.1 Benefits

Basic cellular-based positioning is recommended in 5GtoC scenarios if UEs' locations need to be determined by measuring radio signals or operators need to provide services based on UEs' locations.

Note that this function involves the geographic locations of UEs during service provisioning or maintenance. As such, you must fully protect the personal data of users by complying with any and all applicable laws and user privacy policies in the concerned country.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

Network impacts: none

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	At present, UEs moving at a high speed cannot be positioned. Therefore, the positioning accuracy deteriorates in High-speed Railway Superior Experience scenarios.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

No requirements

4.3.4 Others

The AMF must comply with 3GPP TS 38.413 V15.0.0 or later and connects to the LMF to transmit NRPPa signaling between the LMF and the gNodeB. The LMF must comply with 3GPP TS 38.455 V16.1.0 or later.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-1 describes the parameters used for activation of E-CID-based positioning. No parameters are involved in function optimization. Cell ID-based positioning takes effect by default and no parameters are used for activation.

Table 4-1 Parameters used for activation (E-CID-based positioning)

Parameter Name	Parameter ID	Option	Setting Notes
LCS Switch	**NRCellAlgoSwitch.*LcsSwitch***	ECID_SW	It is recommended that this option be selected.
Compatibility Algorithm Switch	**gNodeBParam.*CompatibilityAlgoSwitch***	LOCATION_REPORTING_COMPAT_SW	Set this option based on the network plan.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling E-CID-based positioning
MOD NRCELLALGOSWITCH: NrCellId=7, LcsSwitch=ECID_SW-1;
//(Optional) Turning on the Location Reporting Compatibility Switch
MOD GNODEBPARAM: CompatibilityAlgoSwitch=LOCATION_REPORTING_COMPAT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the Location Reporting Compatibility Switch
MOD GNODEBPARAM: CompatibilityAlgoSwitch=LOCATION_REPORTING_COMPAT_SW-0;
//Turning off the E-CID-based positioning switch
MOD NRCELLALGOSWITCH: NrCellId=7, LcsSwitch=ECID_SW-0;
```


4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Cell ID-based Positioning

Verify whether cell ID-based positioning has been activated by following one of the following procedures:

Based on traced signaling messages in the location reporting procedure:

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left, choose **NG Interface Trace**.

Step 3 Enable the AMF to send a LOCATION REPORTING CONTROL message to the gNodeB, requesting a UE location report.

Step 4 On the **NG Interface Trace** page, check for the following messages:

- The LOCATION REPORTING CONTROL message sent from the AMF to the gNodeB
- The LOCATION REPORT message sent from the gNodeB to the AMF

If the above messages can be observed, cell ID-based positioning has been activated through the location reporting procedure.

----End

Based on traced signaling messages in the User Location Information procedure:

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left, choose **NG Interface Trace**.

Step 3 Trigger a PDU session or context release for a given UE.

Step 4 On the **NG Interface Trace** page, check for the following messages:

- The PDU SESSION RESOURCE RELEASE RESPONSE message sent from the gNodeB to the AMF, containing the *User Location Information* IE
- The PDU SESSION RESOURCE NOTIFY message sent from the gNodeB to the AMF, containing the *User Location Information* IE
- The UE CONTEXT RELEASE COMPLETE message sent from the gNodeB to the AMF, containing the *User Location Information* IE

If the above messages can be observed, cell ID-based positioning has been activated through the User Location Information procedure.

----End

Uplink E-CID-based Positioning

Verify whether uplink E-CID-based positioning has been activated based on signaling tracing or performance counters.

Based on signaling tracing:

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left, choose **NG Interface Trace**.

Step 3 Enable a UE, GMLC, or AMF to initiate a positioning request. After receiving the positioning request, the AMF sends the request to the LMF to trigger the uplink E-CID-based positioning procedure.

Step 4 On the **NG Interface Trace** page, check for the following messages:

- The DOWNLINK UE ASSOCIATED NRPPA TRANSPORT message sent from the AMF to the gNodeB, containing the *E-CID MEASUREMENT INITIATION REQUEST* IE
- The UPLINK UE ASSOCIATED NRPPA TRANSPORT sent from the gNodeB to the AMF, containing the *E-CID MEASUREMENT INITIATION RESPONSE* IE

If the above messages can be observed, uplink E-CID-based positioning has been activated.

----End

Based on related performance counters:

1. On the MAE-Access, start a task related to cell algorithm measurement to obtain the values of the **N.Ecid.MeasInit.Att** and **N.Ecid.MeasInit.Succ** counters.
2. Enable a UE to initiate a positioning request with specific QoS requirements during the measurement period. After receiving the positioning request, the AMF sends the request to the LMF to trigger the uplink E-CID-based positioning procedure.
3. Check the values of the **N.Ecid.MeasInit.Att** and **N.Ecid.MeasInit.Succ** counters. If all of them return a non-zero value, uplink E-CID-based positioning has been activated.

Downlink E-CID-based Positioning

Verify whether downlink E-CID-based positioning has been activated on the LMF.

4.4.3 Network Monitoring

After E-CID-based positioning is enabled, the feature performance can be monitored by using the following counters.

Counter ID	Counter Name	Counter Description
1911823059	**N.Ecid.MeasInit.Att**	Number of E-CID measurement initialization attempts

Counter ID	Counter Name	Counter Description
1911823058	**N.Ecid.MeasInit.Succ**	Number of successful E-CID measurement initializations

5 SRS Field Strength-based Positioning

5.1 Principles

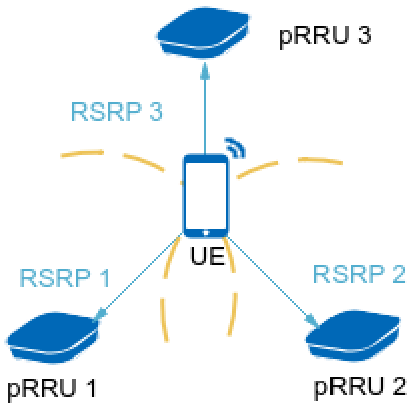
5.1.1 Positioning Methods

SRS field strength-based positioning: applies to line-of-sight (LOS) and non-line-of-sight (NLOS) scenarios not requiring a high positioning accuracy.

5.1.1.1 Field Strength Triangulation-based Positioning

In field strength triangulation-based positioning, the gNodeB measures the RSRP values of uplink SRSs from a UE to at least three pRRUs and sends the results to the LMF. The LMF then calculates the UE's location based on the signal strength differences between every two pRRUs and their known locations. For details, see **Figure 5-1**.

Figure 5-1 Field strength triangulation-based positioning



In positioning, the gNodeB's support for SRS RSRP measurement reporting is controlled by the **SRS_RSRP_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter. For details about the positioning procedure, see **5.1.3.1 Measurement Procedure for SRS Field Strength-based Positioning**.

To reduce SRS resource consumption of positioning service UEs, the **gNBRfspConfig**.***UserServiceType*** parameter can be set to **LCS** or the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter can be selected for UEs performing only positioning services. In this case, only 1-port CB SRS resources are allocated, but it is not mandatory that the full bandwidth be used.

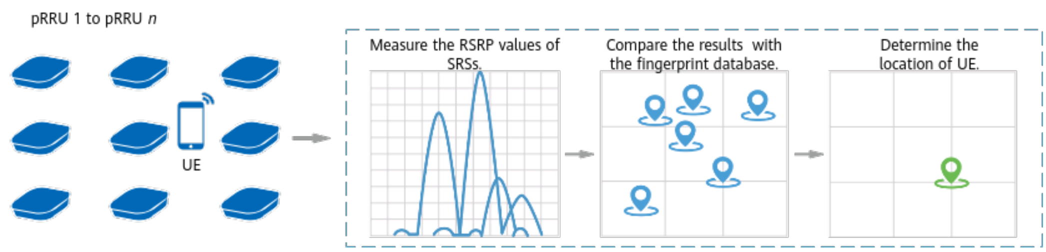
If three-pRRU measurement is not supported, the UE's location cannot be obtained based on field strength triangulation-based positioning. If there are two pRRUs that can be used for measurement, the relative relationship of distances on the line between a UE and the pRRUs can be calculated based on the field strength obtained through pRRU measurement. In this way, the UE's location can be obtained, providing basic positioning capabilities. If there is one pRRU that can be used for measurement, the pRRU coordinates are directly output to provide basic positioning capabilities.

In addition, positioning-dedicated SRS transmission in uplink slots needs to be enabled (by selecting the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter). After this function takes effect, the gNodeB allocates independent full-bandwidth SRS resources for positioning in fixed symbols of uplink slots.

5.1.1.2 Field Strength Fingerprint-based Positioning

In field strength fingerprint-based positioning, multiple pRRUs under the gNodeB measure the RSRP values of uplink SRSs sent from a UE and the results are sent to the LMF. The LMF then determines the location of the UE by matching the results against the SRS fingerprint database. **Figure 5-2** shows how this positioning method works. The SRS fingerprint database refers to the table of mapping between location coordinates and location-specific air interface characteristics. The location coordinates are measured manually. Then, the RSRP values of uplink SRSs from the UE at a specific location to the surrounding pRRUs are collected, and associated with the location coordinates to obtain the fingerprint at this location.

Figure 5-2 Field strength fingerprint-based positioning principles



In positioning, the gNodeB's support for SRS RSRP measurement reporting is controlled by the **SRS_RSRP_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter. For details about the positioning procedure, see **5.1.3.1 Measurement Procedure for SRS Field Strength-based Positioning**.

To reduce SRS resource consumption of positioning UEs, the **gNBRfspConfig**.***UserServiceType*** parameter can be set to **LCS** for UEs performing only positioning services. In this case, only 1-port CB SRS resources are allocated, but it is not mandatory to use the full bandwidth.

When fingerprint-based positioning is used, a fingerprint database needs to be generated first through collection. Traditionally, fingerprints need to be manually collected, which is time-consuming and labor-intensive for positioning of a site with a large footprint. Therefore, RSRP-based RadioSLAM automatic fingerprint collection is introduced to reduce the workload of fingerprint collection.

With RSRP-based RadioSLAM automatic fingerprint collection, a handheld UE traverses all pRRUs in the positioning area. Then, a fingerprint database is automatically generated based on the RSRP measurement quantity reported by the gNodeB to the LMF. The effect is the same as that of manual fingerprint collection at locations of fixed coordinates. On the gNodeB side, this function is controlled by the same option (the **SRS_RSRP_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter) as field strength triangulation-based positioning. On the LMF side, a fingerprint database is generated by using the RSRP-based RadioSLAM algorithm.

5.1.2 Evaluation Standards

The accuracy of SRS field strength-based positioning is measured by the horizontal positioning accuracy and is related to the pRRU deployment distance and physical environment. A shorter deployment distance and fewer obstacles lead to higher accuracy.

5.1.3 Positioning Procedure

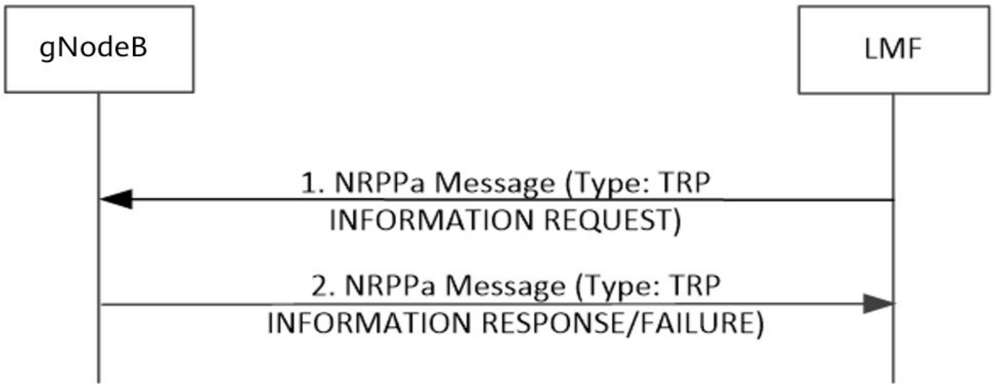
5.1.3.1 Measurement Procedure for SRS Field Strength-based Positioning

Positioning TRP Coordinate Configuration Function

The LMF sends a TRP INFORMATION REQUEST message to trigger the gNodeB to report TRP information through a TRP INFORMATION RESPONSE message. The reported information is used by the LMF for UE location calculation, as shown in **Figure 5-3**.

On the base station side, a pRRU's coordinate is configured by adding an LCS engineering parameter (**ADD LCSENGINEERINGPARA**) or LCS engineering parameter binding relationship (**ADD LCSENGPARABIND**). If the former method is used for configuring the coordinate of a pRRU's relative position, **LCSENGINEERINGPARA**.***MAPID*** or **LCSENGINEERINGPARA**.***RelativePosMapName*** can be selected, which is generally planned by a third-party platform in a unified manner. The map ID of relative position ranges from 0 to 511. If the number of maps managed and maintained by the third-party platform exceeds 512, it is recommended that the map name of relative position be used for configuring the coordinate of the pRRU relative to the origin. For details, contact Huawei technical support.

Figure 5-3 TRP information exchange procedure



NOTE

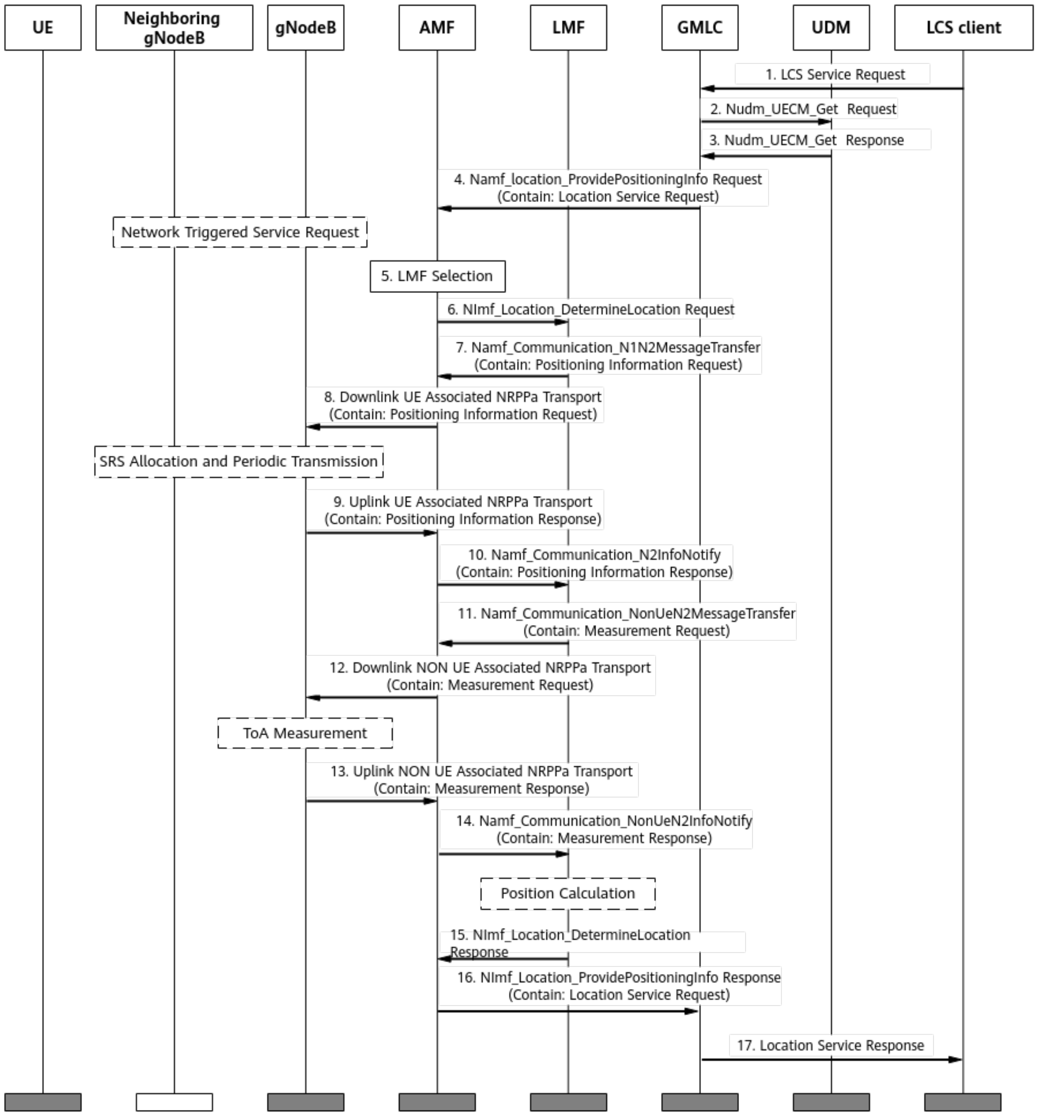
In the TRP INFORMATION REQUEST and TRP INFORMATION RESPONSE messages defined in section 9.2.24 "TRP ID" of 3GPP TS 38.455 V17.5.0, the value range of the carried **TRP ID** is (1,65535), which can be automatically generated or configured on the base station. To ensure that the value is unique at the base station level, the positioning TRP ID function is recommended. Specifically, run the MML command to configure the positioning TRP ID and set **gNodeBAlgo**.***PosTrpIdSwitch*** to **ON**. For details, see **Activation Command Examples** in **5.4.1.2 Using MML Commands**.

At least one of the parameters—**LCSENGINEERINGPARA**.***LOCATIONNAME***, **LCSENGINEERINGPARA**.***MAPID***, or **LCSENGINEERINGPARA**.***RelativePosMapName***—must be set to a valid value.

On-Demand Mode

In on-demand mode, each time the LCS client sends a positioning request, it needs to initiate the on-demand positioning procedure and obtains only one location per procedure. **Figure 5-4** shows the procedure for on-demand positioning.

Figure 5-4 On-demand positioning procedure



The on-demand positioning procedure is described as follows:

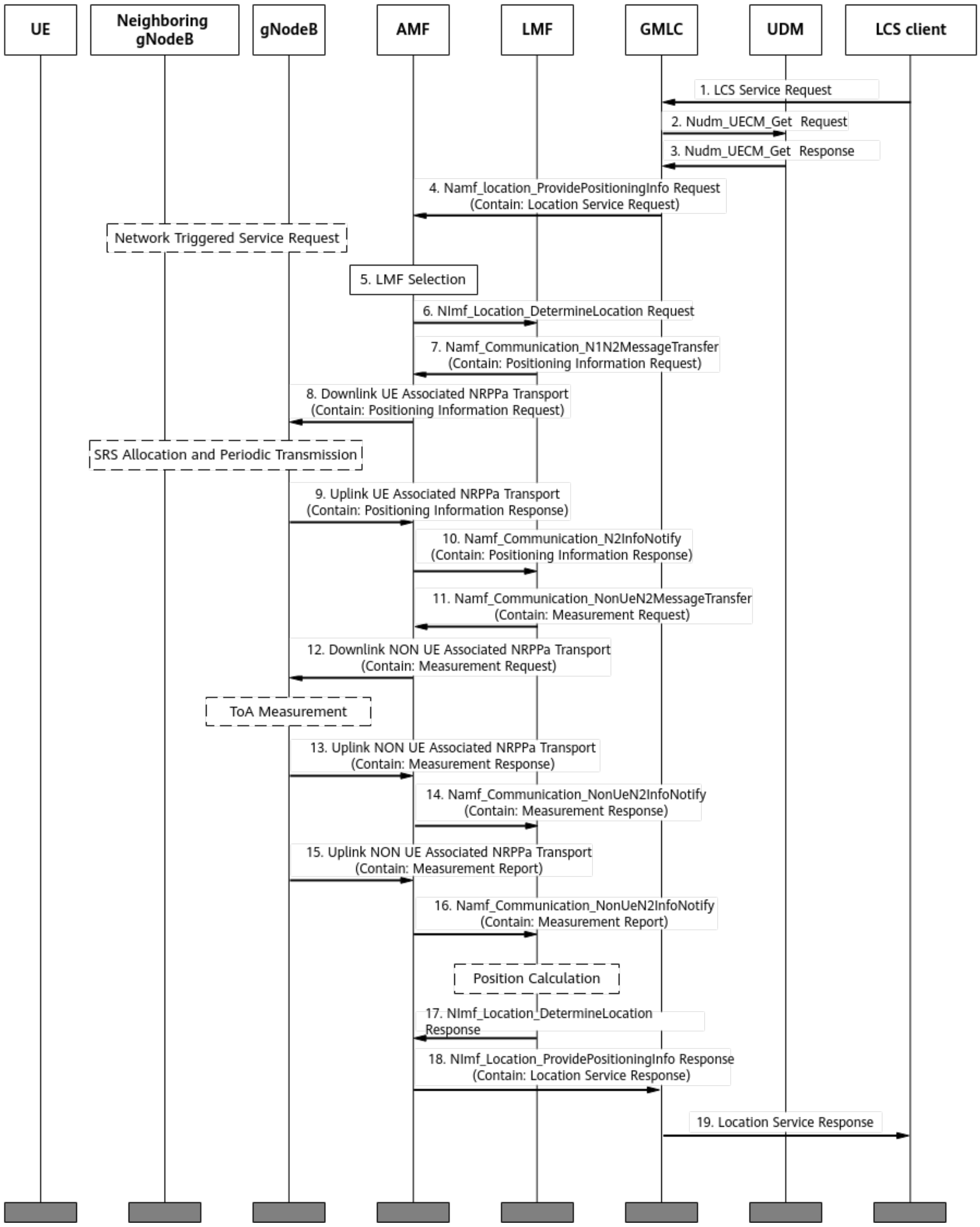
1. The LCS client sends the GMLC a positioning request where the target UE is identified by a generic public subscription identifier (GPSI) or subscription permanent identifier (SUPI).
2. The GMLC invokes a Nudm_UECM_Get service operation that carries the GPSI or SUPI information of the target UE to initiate a positioning request toward the home UDM of this UE.
3. The UDM returns the network addresses of the AMF that serves the UE.
4. The GMLC sends a positioning request to the AMF. If the UE is in idle state, the AMF initiates a network-triggered service request procedure to establish a signaling connection with the UE before proceeding to step 5. If the UE is in connected state, step 5 is directly performed.
5. The AMF selects an LMF to serve the target UE.
6. The AMF sends a positioning request to the LMF.

7. The LMF sends a Positioning Information Request message to the AMF, requesting SRS resource information for the UE performing positioning services.
8. The AMF transparently transmits the Positioning Information Request message to the gNodeB.
9. The gNodeB sends the AMF a Positioning Information Response message that contains information about the SRS resources for the positioning service UE.
10. The AMF transparently transmits the Positioning Information Response message to the LMF.
11. After obtaining information about the SRS resources for the UE, the LMF sends a Measurement Request message to the AMF. The message contains information such as the SRS configuration, measurement reporting mode, and positioning measurement quantity. In on-demand mode, the *Report Characteristics* IE is set to **OnDemand**.
12. The AMF transparently transmits the Measurement Request message to the gNodeB. The TRPs involved in measurements may be under one or more gNodeBs, and the AMF routes the message to all the corresponding gNodeBs.
13. After receiving the Measurement Request message, the gNodeB measures the SRS time-frequency position of the UE to obtain the RSRP measurement result, and reports the measurement result to the AMF through a Measurement Response message.
14. The AMF transparently transmits the Measurement Response message to the LMF.
15. The LMF parses the Measurement Response message to obtain the RSRP measurement results. The location is calculated and sent to the AMF.
16. The AMF transparently transmits the calculated location to the GMLC.
17. The GMLC sends it to the positioning request-initiating LCS client.

Periodic Mode

In periodic mode, after the LCS client initiates a positioning request and the positioning signaling procedure is established, the gNodeB periodically reports the measurement results, enabling the LCS client to periodically obtain the location of UE. **Figure 5-5** illustrates this procedure.

Figure 5-5 Periodic positioning procedure



The procedure for periodic positioning is described as follows:

- Steps 1 to 12 and steps 17 to 19 in **Figure 5-5** are similar to those in the procedure for on-demand positioning. The major difference lies in steps 11 and 12 where the *Report Characteristics* IE contained in the Measurement Request message is set to **Periodic** and the Measurement Request message carries the *Measurement Periodicity* IE.
- Unlike those in on-demand positioning, the Measurement Response messages in steps 13 and 14 of **Figure 5-5** do not contain measurement results. Instead, they serve only as the responses indicating successful processing of Measurement Request messages.

- Steps 15 and 16 of **Figure 5-5** are unique to the procedure for periodic positioning. After measuring the TOA and RSRP, the gNodeB periodically sends the AMF a Measurement Report message containing the measurement results. The AMF transparently transmits the message to the LMF. The LMF parses the Measurement Report message to obtain the TOA and RSRP measurement results, and then proceeds with the subsequent calculation procedures.

5.2 Network Analysis

5.2.1 Benefits

This function is recommended in 5GtoB scenarios if a UE's geographical location needs to be determined by measuring radio signals or an operator needs to provide UE-location-based services.

Note that this function involves the geographical locations of UEs during service provisioning or maintenance. As such, you must fully protect the personal data of users by complying with any and all applicable laws and user privacy policies in the concerned country or territory.

5.2.2 Impacts

The impacts between this function and other functions are described in **5.3.2.3 Function Impacts**.

When SRS field strength-based positioning and positioning-dedicated SRS transmission in uplink slots are enabled, SRSs for positioning UEs occupy symbol resources in uplink slots, which has the following impacts:

- The value of the **N.UL.Last.Symbol13.Pwr** counter may increase because the detection symbol is changed from symbol 13 to symbol 12.
- The UE access delay and handover delay may increase because the demodulation performance of the PUCCH and PRACH deteriorates.
- The PUCCH SINR distribution (**N.UL.SINR.PUCCH.Index0** to **N.UL.SINR.PUCCH.Index7**) may change.
- The number of UEs scheduled in the uplink per TTI and the total number of UEs scheduled in the uplink in a cell may decrease.
- The average uplink cell throughput (**Cell Uplink Average Throughput (DU)**) may decrease.
- The values of counters related to CCE usage may change.
- The total duration of uplink data transmission at the RLC layer in a cell (**N.RLC.ThpTime.UL.Cell**) and the duration of uplink data transmission at the RLC layer in a cell (**N.ThpTime.UL**) may increase.
- The average number of PRBs occupied by the PUSCH (**N.PRB.PUSCH.Used.Avg**), total number of PRBs that are paired for uplink MIMO (**N.ChMeas.MIMO.UL.Pair.PRB**), and total number of layers that are paired for uplink MIMO (**N.ChMeas.MIMO.UL.Pair.Layer**) may change.
- Reconfiguring the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter will cause the cell to restart.

5.3 Requirements

5.3.1 Licenses

There are no license requirements for basic functions.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
SRS period configuration	**NRDUCellSrs**.***SrsPeriod***	*Channel Management*	When SRS field strength-based positioning is enabled, the SRS period must be 40 ms or longer.
Positioning-dedicated SRS transmission in uplink slots	**SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter	*Positioning*	When the **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter is selected, SRS field-strength-based positioning requires positioning-dedicated SRS transmission in uplink slots to be enabled.
NR DU Cell Networking Mode	**NRDUCell**.***NrDuCellNetworkingMode***	None	When SRS field strength-based positioning is enabled, the **NRDUCell**.***NrDuCellNetworkingMode*** parameter must be set to **NORMAL_CELL**.

5.3.2.2 Mutually Exclusive Functions

Mutually exclusive functions of SRS field strength-based positioning (specified by the **SRS_RSRP_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter)

Function Name	Function Switch	Reference	Description
User Service Type	**gNBRfspConfig**.***UserServiceType***	None	• When the **gNBRfspConfig**.***RfspIndex*** parameter is set to **0**, the corresponding **gNBRfspConfig**.***UserServiceType*** parameter must be set to **DS**. • When positioning-dedicated SRS transmission in uplink slots is enabled for a cell served by a base station, the **gNBRfspConfig**.***UserServiceType*** parameter cannot be set to **DS_LCS** for the base station.
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	SRS field strength-based positioning cannot work in hyper cells.
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Cell Combination cannot work with SRS field strength-based positioning.
Joint Transmission Cell Combination	• **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE** • **NRDUCellMultiTrp**.***DataTransMode*** set to **JOINT_MODE**	*Cell Combination*	Joint Transmission Cell Combination cannot work with SRS field strength-based positioning.
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	High-speed cells do not support SRS field strength-based positioning.
SRS period adaptation	**SRS_PERIOD_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	*Channel Management*	SRS period adaptation cannot work with SRS field strength-based positioning.

Function Name	Function Switch	Reference	Description
Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	Intra-base-station UL CoMP cannot work with SRS field strength-based positioning.
Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	Intra-base-station DL CoMP cannot work with SRS field strength-based positioning.
Intra-base-station overlapping area measurement	**INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	Intra-base-station overlapping area measurement cannot work with SRS field strength-based positioning.
Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	Co-MM cannot work with SRS field strength-based positioning.
Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	Coordinated interference management cannot work with SRS field strength-based positioning.
Uplink assistance	**UL_ASSISTANCE_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	None	SRS field strength-based positioning cannot work with uplink assistance.
U-TDOA- and AoA-based high-precision positioning	**AOA_UTDOA_HYBRID_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	U-TDOA- and AoA-based high-precision positioning and SRS-based field strength-based positioning cannot be deployed on the same base station.
PUSCH scheduling in self-contained slots	**PUSCH_SCH_IN_S_SLOT_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	None	SRS field strength-based positioning cannot work with PUSCH scheduling in self-contained slots.

Function Name	Function Switch	Reference	Description
Intra-frequency beam statistics	**INTRA_FREQ_BEAM_STAT_SW** option of the **NRDUCellCollabServ**.***NeighborNrCellMgmtSw*** parameter	None	Intra-frequency beam statistics cannot work with SRS field strength-based positioning.
GP symbol extension	**GP_SYMBOL_EXTENSION_SW** option of the **NRDUCellAlgoSwitch**.***AlgoCompatibilitySwitchExt*** parameter	None	GP symbol extension cannot work with SRS field strength-based positioning.

Mutually exclusive functions of positioning-dedicated SRS transmission in uplink slots (specified by the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter)

Function Name	Function Switch	Reference	Description
User Service Type	**gNBRfspConfig**.***UserServiceType***	None	• When the **gNBRfspConfig**.***RfspIndex*** parameter is set to **0**, the corresponding **gNBRfspConfig**.***UserServiceType*** parameter must be set to **DS**. • When positioning-dedicated SRS transmission in uplink slots is enabled for a cell served by a base station, the **gNBRfspConfig**.***UserServiceType*** parameter cannot be set to **DS_LCS** for the base station.
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Positioning-dedicated SRS transmission in uplink slots cannot work in hyper cells.
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Cell Combination cannot work with positioning-dedicated SRS transmission in uplink slots.

Function Name	Function Switch	Reference	Description
LTE TDD and NR dynamic spectrum sharing	**LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	None	LTE TDD and NR dynamic spectrum sharing cannot work with positioning-dedicated SRS transmission in uplink slots.
LTE and NR Uplink Spectrum Sharing	**LTE_NR_UL_SPECTRUM_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	None	LTE and NR Uplink Spectrum Sharing cannot work with positioning-dedicated SRS transmission in uplink slots.
PUSCH scheduling in self-contained slots	**PUSCH_SCH_IN_S_SLOT_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	None	PUSCH scheduling in self-contained slots cannot work with positioning-dedicated SRS transmission in uplink slots.
SRS period shortening for downlink 1T4R UEs	**DL_1T4R_SRS_PERIOD_SHORTEN_SW** option of the **NRDUCellPdschPre-code**.***SuMimoSrsPeriodShortSw*** parameter	None	SRS period shortening for downlink 1T4R UEs cannot work with positioning-dedicated SRS transmission in uplink slots.
SRS period shortening for downlink 2T4R UEs	**DL_2T4R_SRS_PERIOD_SHORTEN_SW** option of the **NRDUCellPdschPre-code**.***SuMimoSrsPeriodShortSw*** parameter	None	SRS period shortening for downlink 2T4R UEs cannot work with positioning-dedicated SRS transmission in uplink slots.
Speed turbo UE guarantee	**ST_UE_GUARANTEE_SW** option of the **NRCellAlgoSwitch**.***SpeedTurboSw*** parameter	None	Speed turbo UE guarantee cannot work with positioning-dedicated SRS transmission in uplink slots.
SRS interference coordination based on self-contained and uplink slots	**SRS_INTRF_COORD_S_U_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Channel Management*	SRS interference coordination based on self-contained and uplink slots cannot work with positioning-dedicated SRS transmission in uplink slots.

Function Name	Function Switch	Reference	Description
IAB Deployment	**NRDUCELL**.***DeployPosition*** set to **DEPLOY_IAB**	None	If the local cell is deployed in IAB mode, positioning-dedicated SRS transmission in uplink slots cannot be enabled.
Uplink narrowband interference avoidance	**UL_NARROWBAND_INTRF_AVOID** option of the **NRDUCellIntrfIdent**.***IntrfOptimizationMethod*** parameter	None	Uplink narrowband interference avoidance cannot work with positioning-dedicated SRS transmission in uplink slots.
SRS capability optimization	**SRS_CAPB_OPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	None	SRS capability optimization cannot work with positioning-dedicated SRS transmission in uplink slots.
Light-load time sequence optimization	**LIGHT_LOAD_TIME_SEQ_OPT_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter	None	Light-load time sequence optimization cannot work with positioning-dedicated SRS transmission in uplink slots.
Differentiated eMBB QoS	**EMBB_QOS_DIFF_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter	None	Differentiated eMBB QoS cannot work with positioning-dedicated SRS transmission in uplink slots.
Location Name	**LCSENGINEERINGPARA**.***LOCATIONNAME***	None	Either **Location Name** or **Map ID** (controlled by **LCSENGINEERINGPARA**.***MAPID***) must be set to a valid value.
Bi-Carrier	**BI_CARRIER_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	None	When the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is selected, the **BI_CARRIER_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter cannot be selected.

Function Name	Function Switch	Reference	Description
SRS resource allocation for UEs performing downlink large-packet services	**DL_BIG_PKT_SRS_RES_ALLOC_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	None	When the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo.*LcsMeasureAlgoSw*** parameter is selected, SRS resource allocation for UEs performing downlink large-packet services cannot be enabled.
Uplink symbol power saving	**UL_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	Uplink symbol power saving cannot work with positioning-dedicated SRS transmission in uplink slots.
Intra-frequency beam statistics	**INTRA_FREQ_BEAM_STAT_SW** option of the **NRDUCellCollabServ**.***NeighborNrCellMgmtSw*** parameter	None	Intra-frequency beam statistics cannot work with positioning-dedicated SRS transmission in uplink slots.
Intra-frequency assistant cell	**NRDUCell**.***SupplementaryCellInd*** set to **INTRA_FREQ_ASSISTANT_CELL**	None	Positioning-dedicated SRS transmission in uplink slots cannot work in intra-frequency assistant cells.
TDD-only uplink intra-FR1 inter-band CA	**FR1_INTER_BAND_TT_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	None	Positioning-dedicated SRS transmission in uplink slots cannot work with TDD-only uplink intra-FR1 inter-band CA.
Wideband SRS measurement collection	**WIDEBAND_SRS_COLLECT_SW** option of the **NRCellFeatureSwitch**.***CellMeasCollectSwitch*** parameter	None	Positioning-dedicated SRS transmission in uplink slots cannot work with wideband SRS measurement collection.
Downlink avoidance enhancement mode 2	**DL_AVOID_ENH_MODE2_SW** option of the **NRDUCellTimeIntrf**.***TimeIntrfAvoidSw*** parameter	None	Positioning-dedicated SRS transmission in uplink slots cannot work with downlink avoidance enhancement mode 2.

Function Name	Function Switch	Reference	Description
SRS and PUSCH/ PUCCH concurrency compatibility	**SRS_PUSCH_CONCURRENCY_COMPT_SW** option of the **NRDUCellCarrMgmt**.***CaCompatibilitySwitch*** parameter	*Carrier Aggregation*	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, positioning-dedicated SRS transmission in uplink slots cannot be enabled together with proactive avoidance of remote interference.
Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	

5.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	At present, UEs moving at a high speed cannot be positioned. Therefore, the positioning accuracy deteriorates in High-speed Railway Superior Experience scenarios.
Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, downlink intra-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.

Function Name	Function Switch	Reference	Description
Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, downlink intra-FR inter-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.
Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*CA Experience Improvement*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, inter-gNodeB CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.
Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, uplink intra-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.

Function Name	Function Switch	Reference	Description
Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, uplink intra-FR inter-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.
DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	DRX does not take effect for UEs with SRS field strength-based positioning in effect.
Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	Power saving BWP does not take effect for UEs with SRS field strength-based positioning in effect.
Time-domain power saving BWP	**TDM_BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	Time-domain power saving BWP can take effect for UEs with SRS field strength-based positioning in effect if the UEs can report ***srs-PosResources-r15***.

Function Name	Function Switch	Reference	Description
Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	If the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is selected and the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected, the **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be selected. In this case, proactive avoidance of remote interference can take effect.
CA SRS carrier switching	**SRS_CARRIER_SWITCHING_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*CA Experience Improvement*	If both positioning-dedicated SRS transmission in uplink slots and CA SRS carrier switching are enabled: ● The setting of the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be the same between the PCC and SCCs. Otherwise, CA SRS carrier switching does not take effect. ● If PUCCH format 3 resources are insufficient, the value of **User Downlink Average Throughput (DU)** decreases slightly.

Function Name	Function Switch	Reference	Description
Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When both positioning-dedicated SRS transmission in uplink slots and intra-base-station UL CoMP are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, UL CoMP cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability and "positioning-dedicated SRS transmission in uplink slots" is enabled for the coordinating neighboring cell, UL CoMP can take effect for the UE and the performance gains are not affected. ● If a UE supports the positioning-dedicated SRS transmission capability and "positioning-dedicated SRS transmission in uplink slots" is not enabled for the coordinating neighboring cell, UL CoMP can take effect for the UE, but the performance gains decrease.

Function Name	Function Switch	Reference	Description
Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When both positioning-dedicated SRS transmission in uplink slots and intra-base-station DL CoMP are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, DL CoMP cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability, DL CoMP can take effect for the UE.

Function Name	Function Switch	Reference	Description
Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	When both positioning-dedicated SRS transmission in uplink slots and coordinated interference management are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, coordinated interference management cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability, coordinated interference management can take effect for the UE.

Function Name	Function Switch	Reference	Description
Super uplink	**SUPER_UL_TDM_SCH_SW** and **FLEX_SUPER_UPLINK_SW** options of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	When both positioning-dedicated SRS transmission in uplink slots and super uplink are enabled, the SUL cells must meet the SUL cell specifications of the corresponding baseband processing unit. Otherwise, positioning-dedicated SRS transmission in uplink slots cannot take effect. For details about the SUL cell specifications of different baseband processing units, see the table content where the **Frequency Band Type of a Cell** column includes **SUL** in **Product Specifications** > **BBU Technical Specifications** > **Technical Specifications of Baseband Processing Units** > **NR Technical Specifications of Baseband Processing Units** > **NR TDD Technical Specifications of Baseband Processing Units** in *3900 & 5900 Series Base Station Product Documentation*.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

SRS field strength-based positioning:

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR TDD-capable main control boards and UBBPg3 support this function. Other BBPs do not support this function. For details, see *LampSite BBU Hardware Description* in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 4T4R pRRUs support this function. For details, see the related pRRU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

The pRRU565*xAB* and pRRU575*xAB* RF modules must meet the following requirements:

- If a pRRU serves both an NR cell and an LTE cell working in the same frequency band, positioning-dedicated SRS transmission in uplink slots cannot be enabled.
- Up to four pRRUs can be deployed under a TRP.
- In intra-band NR multi-carrier scenarios, carriers under a TRP share the same sector configurations, and positioning-dedicated SRS transmission in uplink slots needs to be enabled for all cells.

NOTE

In LampSite scenarios, one TRP corresponds to one sector equipment group which may contain multiple pRRUs.

5.3.4 Cells

For SRS field strength-based positioning, the cell must meet the following conditions:

- **NRDUCell**.***UlBandwidth*** must be set to 80 MHz or 100 MHz.
- The cell frequency band must be set to n41, n77, n78, or n79.
- The cells are not in SUL mode (with **NRDUCell**.***DuplexMode*** set to a value other than **CELL_SUL**).
- LampSite cells are configured (with **NRDUCell**.***LampSiteCellFlag*** set to **YES**).

5.3.5 Others

The AMF must comply with 3GPP TS 38.413 V15.0.0 or later and connects to the LMF to transmit NRPPa signaling between the LMF and the gNodeB. The LMF must comply with 3GPP TS 38.455 V16.1.0 or later.

When positioning-dedicated SRS transmission in uplink slots is enabled and UEs capable of sending SRSs for positioning need to be positioned, the core network needs to support parsing of the *Positioning SRS* IE.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-2 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 5-1 Parameter used for activation (positioning TRP ID function)

Parameter Name	Parameter ID	Option	Setting Notes
Positioning TRP ID Switch	**gNodeBAlgo**.***PosTrpIdSwitch***	None	It is recommended that the positioning TRP ID function be enabled.

Table 5-2 Parameters used for activation (SRS field strength-based positioning)

Parameter Name	Parameter ID	Option	Setting Notes
User Service Type	**gNBRfspConfig**.***UserServiceType***	None	Set this parameter based on the planned UE service type.
LCS Switch	**NRCellAlgoSwitch**.***LcsSwitch***	SRS_RSRP_LOCATION_SW	Select this option when SRS field strength-based positioning is used.
SRS for Positioning in UL Slot Switch	**NRDUCellPosAlgo**.***LcsMeasureAlgoSw***	SRS_FOR_POSITION_ON_USLOT_SW	Select this option if UEs capable of sending SRSs for positioning need to be positioned.

Parameter Name	Parameter ID	Option	Setting Notes
LCS Measure Algorithm Switch	**NRDUCellPosAlgo**.***LcsMeasureAlgoSw***	COMMON_USER_LCS_SW	Select this option for a common UE (non-low-power-consumption positioning service UE) not configured with an RFSP index during registration.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- If RFSP indexes have been configured during UE registration, the **gNBRfspConfig**.***UserServiceType*** parameter can be configured based on RFSP indexes. The following gives an example of configuration:
  a. Planning by customers based on actual service requirements: Assume that UEs in campus A have two types of services. Two UE types need to be planned: Type-I UEs perform only data services and type-II UEs support positioning services and a small number of data services.
  b. SIM card registration with the operator: According to the preceding requirements, customers in campus A register SIM cards with the operator, and the operator assigns two **gNBRfspConfig**.***RfspIndex*** values (for example, **100** and **101**) which are configured in the users' registration information, to distinguish the two UE types.
  c. SIM card-service mapping by customers: After obtaining registered SIM cards from the operator, the customers in campus A determine and maintain the mapping between the two types of SIM cards and two types of services. For example, **100** is allocated to UEs performing only data services and **101** to UEs performing positioning services.
  d. MML-based configurations on the gNodeB: After obtaining the mapping between SIM cards and services from the campus customers, network maintenance engineers run MML commands and configure the **gNBRfspConfig**.***UserServiceType*** parameter based on the service types planned for the two types of SIM cards (with different **gNBRfspConfig**.***RfspIndex*** values). The default value of

gNBRfspConfig.***UserServiceType*** is **DS**. If SRS field strength-based positioning needs to be enabled, it is recommended that the **gNBRfspConfig**.***UserServiceType*** parameter be set to **LCS**.

```
//Adding gNodeB RFSP configurations with UserServiceType set to LCS for UEs performing only positioning services or UEs performing both positioning and MBB services and requiring positioning accuracy guarantee. This ensures that only 1-port CB SRS resources are allocated.
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=10, UserServiceType=LCS;
```

- If a common UE (non-low-power-consumption positioning service UE) is not configured with an RFSP index during registration, turn on the common UE positioning switch:

```
//Enabling positioning-dedicated SRS transmission in uplink slots
MOD NRDUCELLPOSALGO: NrDuCellId=1, LcsMeasureAlgoSw=SRS_FOR_POSITION_ON_USLOT_SW-1;
//Turning on the common UE positioning switch
MOD NRDUCELLPOSALGO: NrDuCellId=1, LcsMeasureAlgoSw=COMMON_USER_LCS_SW-1;
```

- Positioning TRP ID function
  a. Find the RRU or port associated with an engineering parameter ID based on the engineering parameter configuration of the positioning service.
  b. Find the **SectorEqmId** corresponding to the RRU or port based on **MO SECTOREQM**.
  c. Find the **NrDuCellTrpId** and **NrDuCellCoverageId** corresponding to **SectorEqmId** based on **MO NRDUCELLCOVERAGE**.
  d. Configure a positioning TRP ID (controlled by **NRDUCellLcsProperty**.***PosTrpId***) for every combination of **NrDuCellTrpId**, **NrDuCellCoverageId**, and engineering parameter ID. The positioning TRP ID ranges from 1 to 65535 and is unique within a gNodeB. This parameter takes effect only after the positioning TRP ID function is enabled. The following is an MML command example:

```
//Enabling the positioning TRP ID function
MOD GNODEBALGO: PosTrpIdSwitch=ON;
```

- SRS field strength-based positioning (field strength triangulation-based positioning/field strength fingerprint-based positioning)

```
//Enabling field strength triangulation-based/field strength fingerprint-based positioning
MOD NRCELLALGOSWITCH: NrCellId=1, LcsSwitch=SRS_RSRP_LOCATION_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Positioning TRP ID function

```
//Disabling the positioning TRP ID function
MOD GNODEBALGO: PosTrpIdSwitch=OFF;
```

- SRS field strength-based positioning (field strength triangulation-based positioning/field strength fingerprint-based positioning)

```
//Disabling field strength triangulation-based/field strength fingerprint-based positioning
MOD NRCELLALGOSWITCH: NrCellId=1, LcsSwitch=SRS_RSRP_LOCATION_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

SRS Field Strength-based Positioning

Verify whether SRS field strength-based positioning (field strength triangulation-based positioning/field strength fingerprint-based positioning) has been activated by using signaling tracing or performance counters.

Based on signaling tracing:

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left, choose **NG Interface Trace**.

Step 3 The LCS client initiates a positioning request. The AMF receives the positioning request. After receiving the response from the AMF, the LMF initiates the subsequent SRS field strength-based positioning measurement procedure.

Step 4 On the **NG Interface Trace** page, check for the following messages:

- A POSITIONING INFORMATION REQUEST or MEASUREMENT REQUEST message sent from the AMF to the gNodeB
- A POSITIONING INFORMATION RESPONSE, MEASUREMENT RESPONSE, or MEASUREMENT REPORT message sent from the gNodeB to the AMF

  If the preceding messages can be observed, SRS field strength-based positioning has been activated.

----End

Based on related performance counters:

Step 1 On the MAE-Access, start tasks for measuring counters related to **Measurement of gNodeB Performance** and **Measurement of Cell Performance**, including **N.Positioning.Info.Req.Att**, **N.Positioning.Info.Req.Succ**, **N.Positioning.Meas.Req.Att**, **N.Positioning.Meas.Req.Succ**, and **N.Positioning.Meas.Rpt**.

Step 2 Enable the LCS client to initiate a positioning request during the measurement period. After processing by the AMF, the LMF triggers the SRS field strength-based positioning measurement procedure.

Step 3 Check the counter values.

1. In on-demand positioning mode, SRS field strength-based positioning has been activated if none of the values of the following counters is 0: **N.Positioning.Info.Req.Att**, **N.Positioning.Info.Req.Succ**, **N.Positioning.Meas.Req.Att**, and **N.Positioning.Meas.Req.Succ**.
2. In periodic positioning mode, SRS field strength-based positioning has been activated if none of the values of the following counters is 0: **N.Positioning.Info.Req.Att**, **N.Positioning.Info.Req.Succ**, **N.Positioning.Meas.Req.Att**, **N.Positioning.Meas.Req.Succ**, and **N.Positioning.Meas.Rpt**.

----End

5.4.3 Network Monitoring

After SRS field strength-based positioning is enabled, the feature performance can be monitored by using the following counters.

Counter ID	Counter Name	Counter Description
1911829380	**N.Positioning.Info.Req.Att**	Number of positioning information requests
1911829379	**N.Positioning.Info.Req.Succ**	Number of positioning information responses
1911829377	**N.Positioning.Meas.Req.Att**	Number of positioning measurement requests
1911829376	**N.Positioning.Meas.Req.Succ**	Number of successful positioning measurements
1911829375	**N.Positioning.Meas.Rpt**	Number of positioning measurement reporting times
1911842612	**N.User.Positioning.Avg**	Average number of UEs performing positioning services

When a large number of positioning requests are initiated simultaneously, flow control may be triggered, causing some NRPPa messages to be discarded. The feature performance can be monitored by using the following counters.

Counter ID	Counter Name	Counter Description
1911829378	**N.NRPPa.Ng.Msg.Disc.FlowCtrl**	Number of times NRPPa messages over the NG interface are discarded due to flow control

6 Precise Positioning (LampSite NR) (Trial)

6.1 Principles

Precise Positioning (LampSite NR) is a trial feature.

6.1.1 Positioning Methods

Precise positioning methods include U-TDOA-based positioning, TOA fingerprint-based positioning, and LPHAP.

- U-TDOA-based positioning: applies to LOS scenarios requiring a high positioning accuracy of 1–3 meters.
- TOA fingerprint-based positioning: applies to LOS/NLOS scenarios requiring a positioning accuracy of 3–7 meters.
- LPHAP: applies to low-power-consumption scenarios.

For details about each scenario, see **6.3.5 Networking**.

NOTE

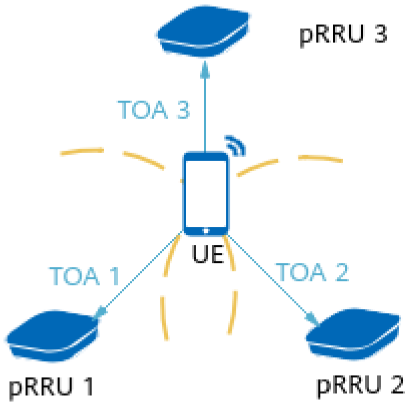
LOS means that radio signals are transmitted directly between the transmitter and the receiver (for example, the pRRU) without any obstacles.

6.1.1.1 U-TDOA-based Positioning

U-TDOA-based High-Precision Positioning

In U-TDOA-based high-precision positioning, the gNodeB measures the TOA values of the uplink SRSs from a UE to at least three pRRUs and sends the measurement results to the LMF. The LMF then calculates the UE's location based on the TDOA between every two pRRUs and their known locations, as shown in **Figure 6-1**.

Figure 6-1 U-TDOA-based high-precision positioning



U-TDOA-based high-precision positioning is controlled by the **UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter. For details about the procedure, see **6.1.3.1 U-TDOA-based Positioning Procedure**.

U-TDOA-based high-precision positioning depends on TOA high-precision measurements. The TOA measurement accuracy is closely related to the bandwidth for SRS transmission. To ensure the U-TDOA-based positioning accuracy while reducing SRS resource consumption of positioning UEs, the **gNBRfspConfig**.***UserServiceType*** parameter can be configured based on the types of services running on UEs. The allocated bandwidth resource varies with the value of this parameter. Further details are as follows:

- For UEs mainly performing positioning services:
  - If they are common UEs (non-low-power-consumption positioning service UEs), the user service type can be set to **LCS** or the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter can be selected. In this way, full-bandwidth SRS resources are configured for the UEs to ensure high positioning accuracy for them.
  - If they are low-power-consumption positioning service UEs, the user service type needs to be set to **LPHAP_LCS**. In this way, full-bandwidth SRS resources are configured for the UEs in inactive state to ensure high positioning accuracy for them.
- For UEs mainly performing data services, the user service type can be set to **DS**, indicating that SRS resources are allocated based on the traditional SRS resource allocation algorithm. For UEs that are not configured with a service type, the user service type is **DS** by default.

If a UE performs both data and positioning services and requires positioning accuracy guarantee, the following parameters can be set to ensure full-bandwidth SRS resource transmission. Currently, U-TDOA-based high-precision positioning can be configured in 100 MHz cell bandwidth as follows:

- **NRDUCellSrs**.***SrsWideBandIndexCsrs***: Indicates the maximum bandwidth index Csrs of periodic wideband SRS.
- **NRDUCellSrs**.***SrsWideBandIndexBsrs***: Indicates the maximum bandwidth index Bsrs of periodic wideband SRS.
- **NRDUCellSrs**.***SrsNarrowBandIndexCsrs***: Indicates the maximum bandwidth index Csrs of periodic narrowband SRS.

- **NRDUCellSrs**.***SrsNarrowBandIndexBsrs***: Indicates the maximum bandwidth index Bsrs of periodic narrowband SRS.

If there are RedCap UEs on the network, it is recommended that positioning-dedicated SRS transmission in uplink slots be enabled (by selecting the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter). After this function takes effect, the gNodeB allocates independent full-bandwidth SRS resources for positioning in fixed symbols of uplink slots.

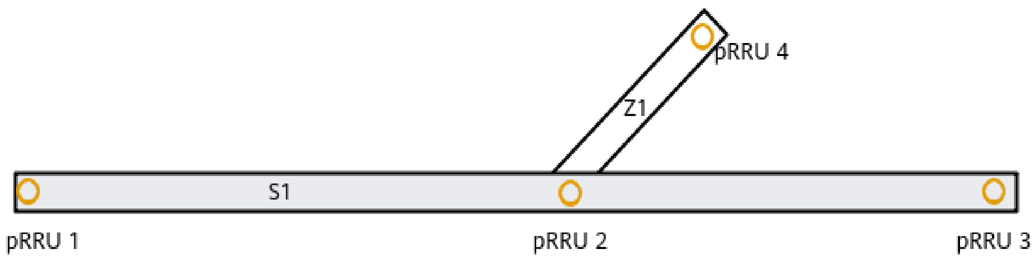
If more than four pRRUs are deployed in a cell, multi-channel TOA measurement is introduced to improve the accuracy of U-TDOA-based high-precision positioning. This function is controlled by the **MULTI_CHANNEL_TOA_MEAS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter. After this function takes effect, the number of channels used by the gNodeB to report TOA measurement results to the LMF increases. The LMF filters data based on the TOAs measured over multiple channels and the U-TDOA-based high-precision positioning algorithm is optimized to improve the positioning accuracy.

One-dimensional U-TDOA-based Positioning

In scenarios such as long narrow corridors (with a width of 3 m to 5 m), U-TDOA-based high-precision positioning deployment requirements are not met. Therefore, linear deployment mode is used. U-TDOA-based positioning is performed based on two or more linearly deployed pRRUs, and the UE's specific location range in the corridor can be calculated. This positioning policy is one-dimensional U-TDOA-based positioning.

For one-dimensional U-TDOA-based positioning, the target area range is determined through the pRRUs receiving the SRSs. Then, the target UE's location is calculated based on the locations of two pRRUs on both sides. Assume that the UE is located on the line between two pRRUs. The location of the UE is calculated based on the coordinates of the pRRUs that receive the signals and the difference between the distances from the UE to the two pRRUs. As shown in **Figure 6-2**, there is a long narrow corridor (with a width of 3 m to 5 m) consisting of main corridor S1 and branch corridor Z1. Four pRRUs (pRRU 1 to pRRU 4) are deployed. The target UE may be located at a specific position in the corridor. For one-dimensional positioning, the UE's location in S1 (distances to pRRU 1, pRRU 2, and pRRU 3) or Z1 (distances to pRRU 2 and pRRU 4) must be provided.

Figure 6-2 One-dimensional positioning principles



On the gNodeB, both this function and U-TDOA-based high-precision positioning are controlled by the **UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter. On the LMF, the UE's location is calculated based on the TOA values of the two pRRUs sent by the gNodeB.

If there are RedCap UEs on the network, it is recommended that positioning-dedicated SRS transmission in uplink slots be enabled (by selecting the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter). After this function takes effect, the gNodeB allocates independent full-bandwidth SRS resources for positioning in fixed symbols of uplink slots.

6.1.1.2 TOA Fingerprint-based Positioning

In TOA fingerprint-based positioning, the gNodeB measures the TOA values of uplink SRSs from a UE to multiple pRRUs and sends the results to the LMF. Then the LMF determines the UE's location by matching the results against the TOA characteristics in the SRS fingerprint database. The SRS fingerprint database refers to the table of mapping between map location coordinates and location-specific air interface TOA characteristics. The location coordinates are obtained manually. Then, the TOA values of uplink SRSs from a UE at a specific location to the surrounding pRRUs are collected, and associated with the location coordinates to obtain the fingerprint at this location.

In positioning, TOA measurement reporting on the gNodeB is controlled by the **UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter. For details about the positioning procedure, see **6.1.3.1 U-TDOA-based Positioning Procedure**.

To reduce SRS resource consumption of positioning UEs, the **gNBRfspConfig**.***UserServiceType*** parameter can be set to **LCS** for UEs performing only positioning services. Only 1-port CB SRS resources are allocated, but the use of full bandwidth is mandatory.

When fingerprint-based positioning is used, a fingerprint database needs to be formed first through collection. Traditionally, fingerprints need to be manually collected, which is time-consuming and labor-intensive for positioning of a site with a large footprint. Therefore, the TOA-based RadioSLAM automatic fingerprint collection function is introduced to reduce the workload of fingerprint collection.

With TOA-based RadioSLAM automatic fingerprint collection, a handheld UE traverses all pRRUs in the positioning area. Then, a fingerprint database is automatically generated based on the TOA/RSRP measurement quantities reported by the gNodeB to the LMF. The effect is the same as that of manual fingerprint collection at fixed coordinates. On the gNodeB side, this function is controlled by the **UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter, as U-TDOA-based positioning is. On the LMF side, the fingerprint database is generated based on the TOA-based RadioSLAM algorithm.

If there are RedCap UEs on the network, it is recommended that positioning-dedicated SRS transmission in uplink slots be enabled (by selecting the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter). After this function takes effect, the gNodeB allocates independent full-bandwidth SRS resources for positioning in fixed symbols of uplink slots.

6.1.1.3 LPHAP

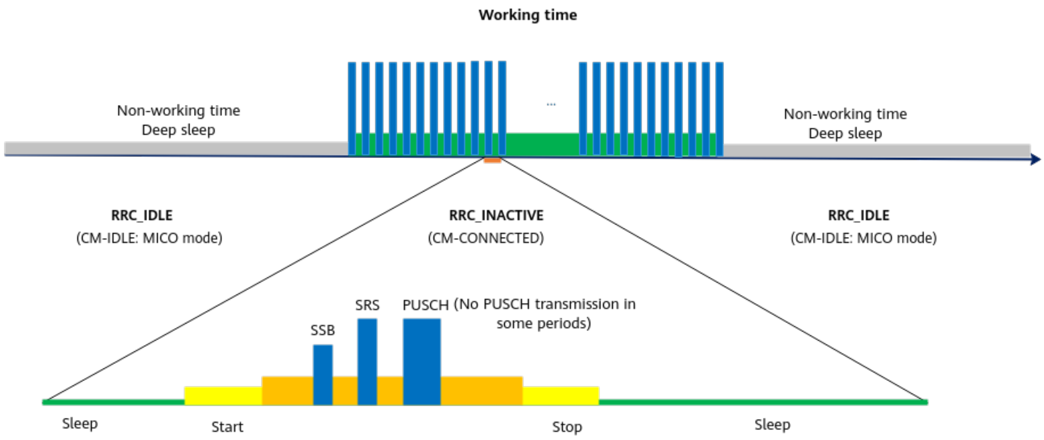
Positioning services in campuses/factories—such as valuable assets positioning—require a certain level of positioning accuracy and necessitate positioning tags powered by batteries that meet battery life requirements. Therefore, LPHAP is introduced.

LPHAP is controlled by the **LPHAP_REDCAP_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter. After this function takes effect, the following operations are performed for a RedCap UE:

- The LMF triggers periodic U-TDOA-based positioning on the UE.
- The UE needs to set up an RRC connection only once. After obtaining the PUSCH and SRS resources allocated by the gNodeB for LPHAP, the UE sends SRSs for positioning and performs PUSCH small data transmission (SDT) in the RRC_INACTIVE state, and then enters the sleep state.
- The gNodeB measures SRSs for positioning and reports the measurement results for location calculation by the LMF.
- After the next positioning period starts, the UE reads SSB information, completes time synchronization, sends positioning information, and then enters the sleep state again. This positioning procedure repeats periodically.

This function saves power by keeping the UE in the sleep state most of the time as shown in **Figure 6-3**.

Figure 6-3 LPHAP



During LPHAP, SRS bandwidth resources are selected for positioning services based on the user service type (**gNBRfspConfig**.***UserServiceType***). For UEs performing LPHAP services, this parameter needs to be set to **LPHAP_LCS** so that SRS resources for positioning are used to ensure positioning accuracy.

This function is recommended in 5GtoB scenarios where the geographical locations of low-power-consumption UEs need to be determined by measuring radio signals on the 5G network or where operators need to support the services that provide locations of low-power-consumption UEs and have certain low-power consumption requirements. This function saves UE power.

6.1.1.4 Positioning-dedicated Beacon-free Function

In U-TDOA-based positioning, pRRUs take different time to process received signals. As a result, an error is introduced to the measured TDOA (TOA *i* - TOA *j*) value between every two pRRUs, affecting the positioning accuracy. To solve this issue, the latency difference needs to be calibrated in advance.

Simple latency difference calibration can use the fixed-location beacon UE deployment solution. Specifically, a beacon UE at a known location is deployed in the positioning area to calibrate the latency difference in a real-time manner. However, in this solution, the beacon UE needs to be placed at a fixed location in the center of the positioning area, and sustained power supply is required. Therefore, it is difficult to promote the related positioning service on a large scale. As such, the positioning-dedicated beacon-free function is introduced. The positioning-dedicated beacon-free function is classified into the UE-based beacon-free function and inter-base-station-transmission-based beacon-free function.

UE-based Beacon-free Function

With the UE-based beacon-free function, data reported by positioning UEs are automatically collected. Signal subtypes with similar signal characteristics distribution rules in a positioning area are selected for latency difference calculation, and the latency difference calibration result is obtained after latency difference distribution rule aggregation. This solution can replace the fixed-location beacon UE deployment solution and solve the engineering issues of U-TDOA-based positioning deployment and beacon UE maintenance.

On the gNodeB, both the UE-based beacon-free function and the U-TDOA-based high-precision positioning function are controlled by the **UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter. On the LMF, this algorithm can be enabled through parameter configuration.

If there are RedCap UEs on the network, it is recommended that positioning-dedicated SRS transmission in uplink slots be enabled (by selecting the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter). After this function takes effect, the gNodeB allocates independent full-bandwidth SRS resources for positioning in fixed symbols of uplink slots.

NOTE

A beacon UE is a 5G SA UE with a known and fixed location. A beacon UE is required for every three or four adjacent pRRUs, and LOS transmission must be guaranteed between the beacon UE and each adjacent pRRU. The number of beacon UEs and their deployment positions are determined through simulation and planning based on deployment environment. After beacon UEs are deployed, the identifier and accurate deployment position of each beacon UE must be recorded and then configured on the LMF.

Inter-Base-Station-Transmission-based Beacon-free Function

The inter-base-station-transmission-based beacon-free function is controlled by the **NRDUCellPosAlgo**.***InterTrpTransmissionSwitch*** parameter. After this function is enabled, signals are exchanged between pRRUs for calculating the inter-pRRU differences in the time of processing received signals based on their known coordinates. During the TOA measurement of the to-be-positioned UE, the differences in the time of processing received signals are eliminated. With this function, the positioning accuracy requirement can be met without deploying beacon UEs.

The inter-base-station-transmission-based beacon-free function requires pRRU physical coordinates to be configured by setting LCS engineering parameters for pRRUs in the positioning area (**ADD LCSENGINEERINGPARA**) and LCS engineering parameter binding relationships (**ADD LCSENGPARABIND**) on the base station side.

6.1.2 Evaluation Standards

The accuracy of precise positioning is represented by the horizontal positioning accuracy. High-precision positioning of 1–3 meters in 90% of LOS transmissions is realized. The positioning accuracy is closely related to the proportion of LOS transmissions. Whether LOS transmissions are acquired is closely related to the physical environment and the distance between pRRUs. (The number of pRRUs and their installation positions need to be considered in the network planning and design.) A higher proportion of LOS transmissions leads to more accurate positioning.

6.1.3 Positioning Procedure

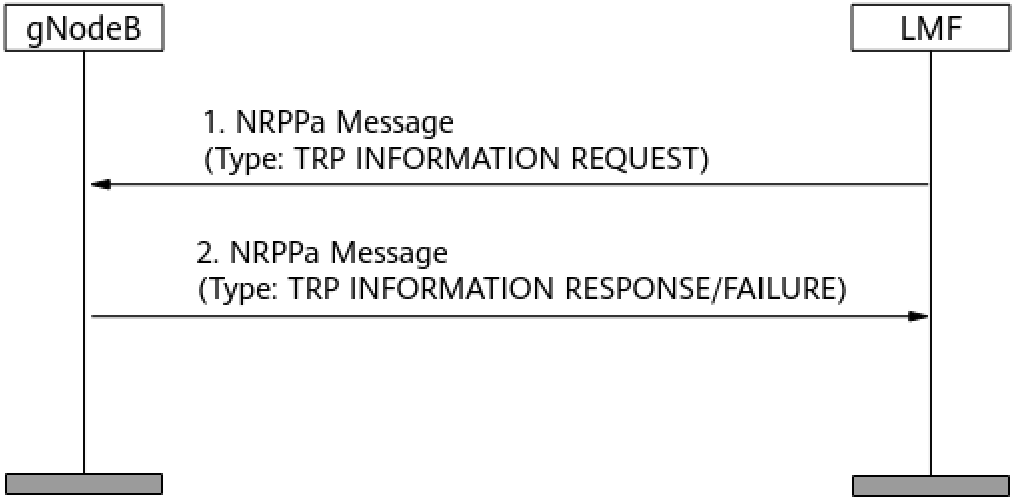
6.1.3.1 U-TDOA-based Positioning Procedure

The U-TDOA-based positioning procedure can be triggered on demand or periodically, as indicated in the request message sent from the core network.

Positioning TRP Coordinate Configuration

The LMF sends a TRP INFORMATION REQUEST message to trigger the gNodeB to report TRP information through a TRP INFORMATION RESPONSE message. The reported information is used by the LMF for UE location calculation, as shown in **Figure 6-4**. With this function, a pRRU's coordinate is configured through an LCS engineering parameter (**ADD LCSENGINEERINGPARA**) or LCS engineering parameter binding relationship (**ADD LCSENGPARABIND**) on the base station side. If an LCS engineering parameter is used for configuring the coordinate of a pRRU's relative position, **LCSENGINEERINGPARA**.***MAPID*** or **LCSENGINEERINGPARA**.***RelativePosMapName*** can be selected, which is generally planned by a third-party platform in a unified manner. The map ID of relative position ranges from 0 to 511. If the number of maps managed and maintained by the third-party platform exceeds 512, it is recommended that the map name of relative position be used for configuring the coordinate of the pRRU relative to the origin. For details, contact Huawei technical support.

Figure 6-4 TRP information exchange



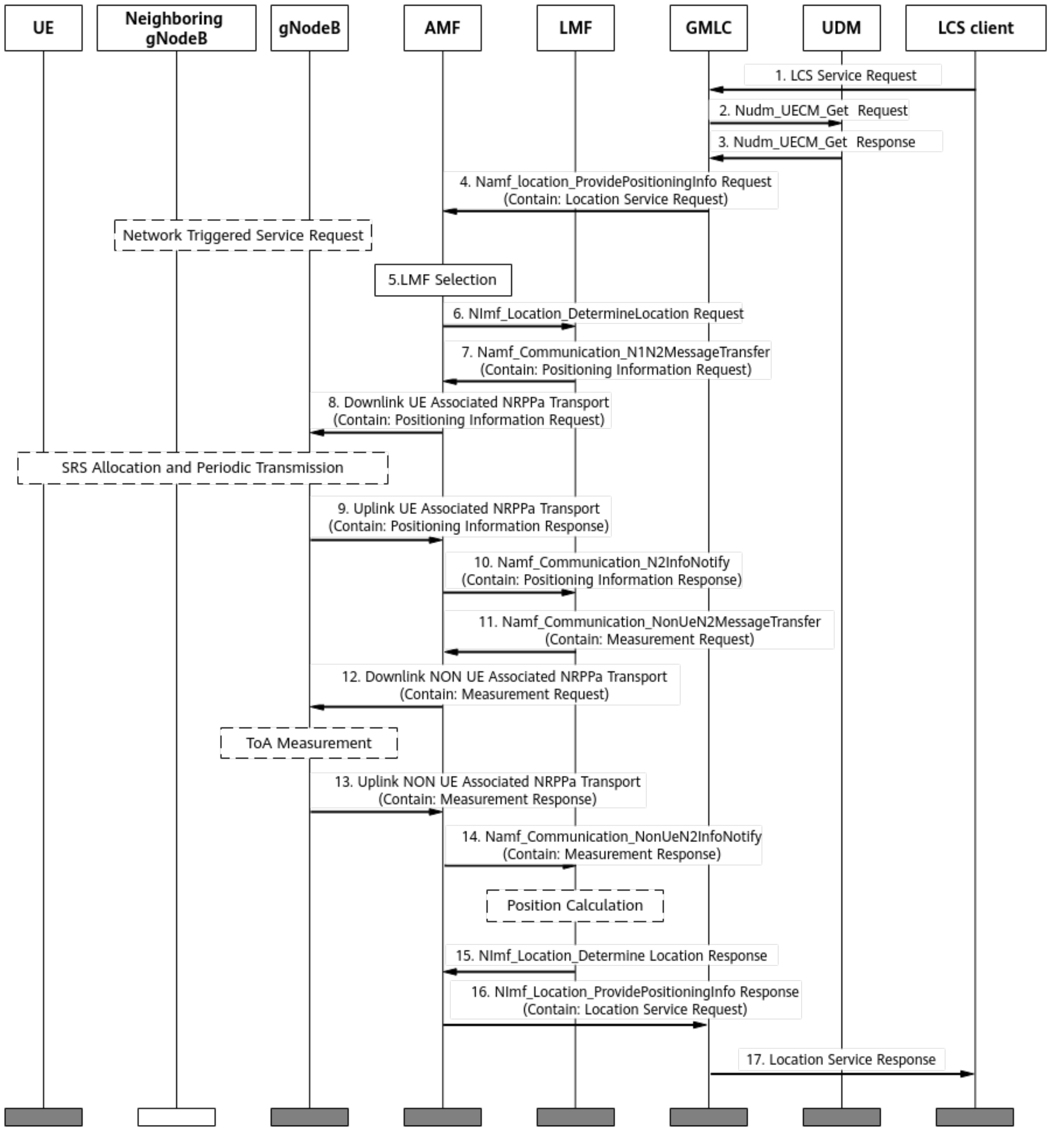
NOTE

- In the TRP INFORMATION REQUEST and TRP INFORMATION RESPONSE messages defined in section 9.2.24 "TRP ID" of 3GPP TS 38.455 V17.5.0, the value range of the carried **TRP ID** is (1,65535), which can be automatically generated or configured on the base station. To ensure that the value is unique at the base station level, the positioning TRP ID function is recommended. Specifically, run the MML command to configure the positioning TRP ID and set **gNodeBAlgo**.***PosTrpIdSwitch*** to **ON**. For details, see **Activation Command Examples** in **6.4.1.2 Using MML Commands**.
- At least one of the parameters—**LCSENGINEERINGPARA**.***LOCATIONNAME***, **LCSENGINEERINGPARA**.***MAPID***, or **LCSENGINEERINGPARA**.***RelativePosMapName***—must be set to a valid value.
- When configuring the coordinates of positioning TRPs, configure all TRPs in the same cell as positioning TRPs to ensure positioning accuracy.

On-Demand Mode

In on-demand mode, each time the LCS client sends a positioning request, it needs to initiate the procedure as shown in **Figure 6-5** and obtains only one UE location per procedure.

Figure 6-5 Procedure for on-demand positioning



The procedure for on-demand positioning is described as follows:

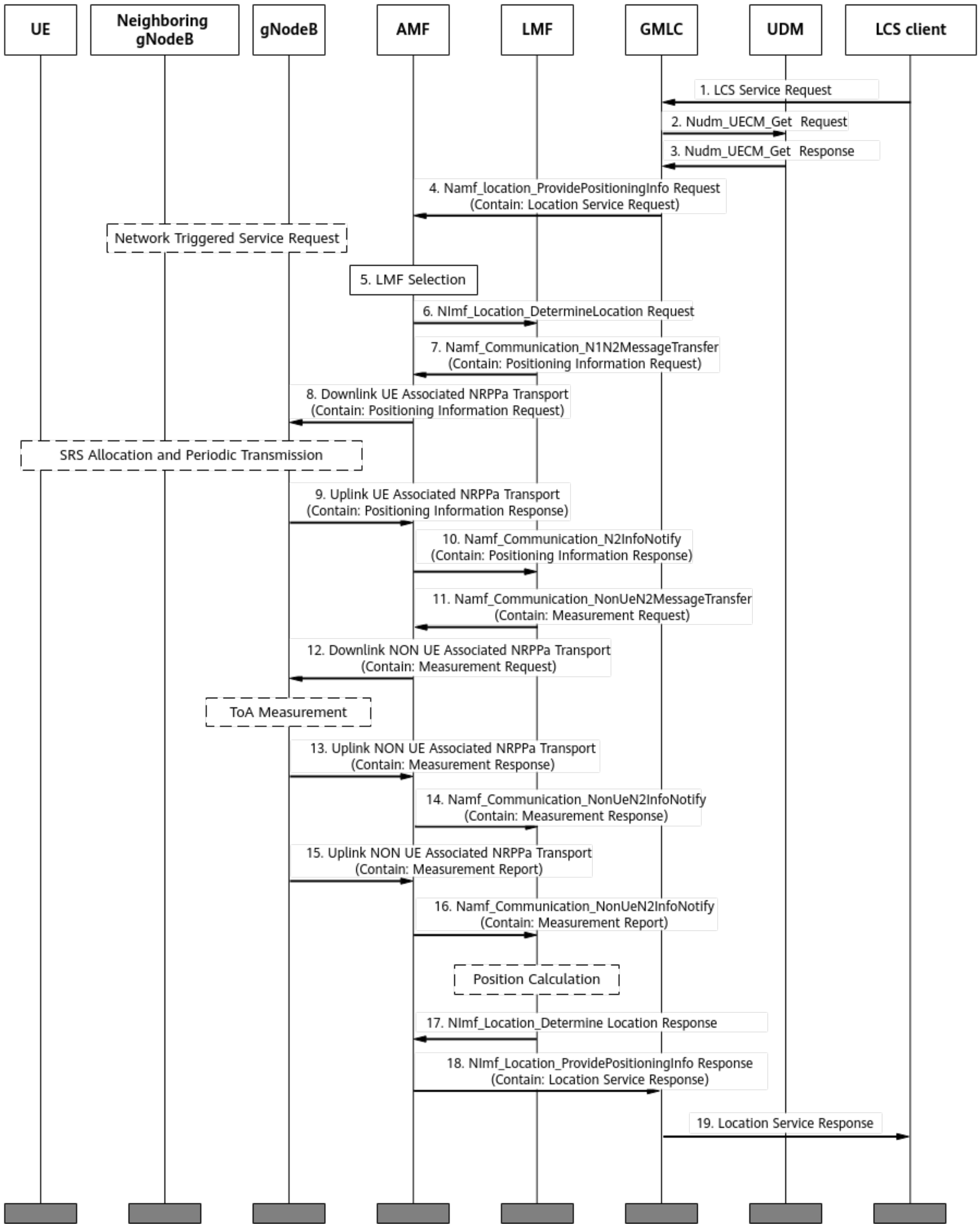
1. The LCS client sends the GMLC a positioning request where the target UE is identified by a GPSI or SUPI.
2. The GMLC invokes a Nudm_UECM_Get service operation that carries the GPSI or SUPI information of the target UE to initiate a positioning request toward the home UDM of this UE.
3. The UDM returns the network address of the AMF that serves the UE.
4. The GMLC sends a positioning request to the AMF. If the UE is in idle state, the AMF then initiates a network-triggered service request procedure to establish a signaling connection with the UE before proceeding to step 5. If the UE is in connected state, the AMF skips to step 5.
5. The AMF selects an LMF to serve the target UE.
6. The AMF sends a positioning request to the LMF.
7. The LMF sends a Positioning Information Request message to the AMF, requesting the SRS resource information for the positioning service UE.

8. The AMF transparently transmits the Positioning Information Request message to the gNodeB.
9. The gNodeB sends the AMF a Positioning Information Response message that contains the UE's SRS resource information.
10. The AMF transparently transmits the Positioning Information Response message to the LMF.
11. After obtaining the UE's SRS resource information, the LMF sends the AMF a Measurement Request message that contains information such as the SRS configuration, measurement reporting mode, and positioning measurement quantities. In on-demand mode, the *Report Characteristics* IE is set to **OnDemand**.
12. The AMF transparently transmits the Measurement Request message to the gNodeB. The transmission reception points (TRPs) involved in measurements may be under one or more gNodeBs, and the AMF routes the message to all the corresponding gNodeBs.
13. After receiving the Measurement Request message, the gNodeB measures the SRS from the UE in the indicated time-frequency position based on the **UL-RTOA** value contained in the *TRP Measurement Type* IE in the measurement quantities to obtain the TOA and RSRP measurement values, and then sends the results to the AMF through a Measurement Response message.
14. The AMF transparently transmits the Measurement Response message to the LMF.
15. The LMF parses the Measurement Response message to obtain the measured TOA and RSRP values. Based on the measurements, the LMF calculates the location and sends it to the AMF.
16. The AMF transparently transmits the calculated location to the GMLC.
17. The GMLC then sends it to the LCS client that initiates the positioning request.

Periodic Mode

In periodic mode, after the LCS client initiates a positioning request and the positioning signaling procedure is established, the gNodeB periodically reports the measurement results, enabling the LCS client to periodically obtain the UE's location. **Figure 6-6** illustrates this procedure.

Figure 6-6 Procedure for periodic positioning



The procedure for periodic positioning is described as follows:

- Steps 1–12 and 17–19 in **Figure 6-6** are similar to those in on-demand mode. The major difference lies in steps 11 and 12 where the value of the *Report Characteristics* IE in the Measurement Request message is **Periodic** and the message contains the *Measurement Periodicity* IE.
- Unlike that in on-demand mode, the Measurement Response message in steps 13 and 14 of **Figure 6-6** does not contain measurement results. Instead, it serves only as the response indicating successful processing of the Measurement Request message.

- Steps 15 and 16 of **Figure 6-6** are unique to the periodic mode. After measuring the TOA and RSRP, the gNodeB periodically sends the AMF a Measurement Report message containing the measurement results. The AMF then transparently transmits the message to the LMF. The LMF parses the Measurement Report message to obtain the TOA and RSRP measurement results, and then proceeds with the subsequent calculation.

6.1.3.2 LPHAP Procedure

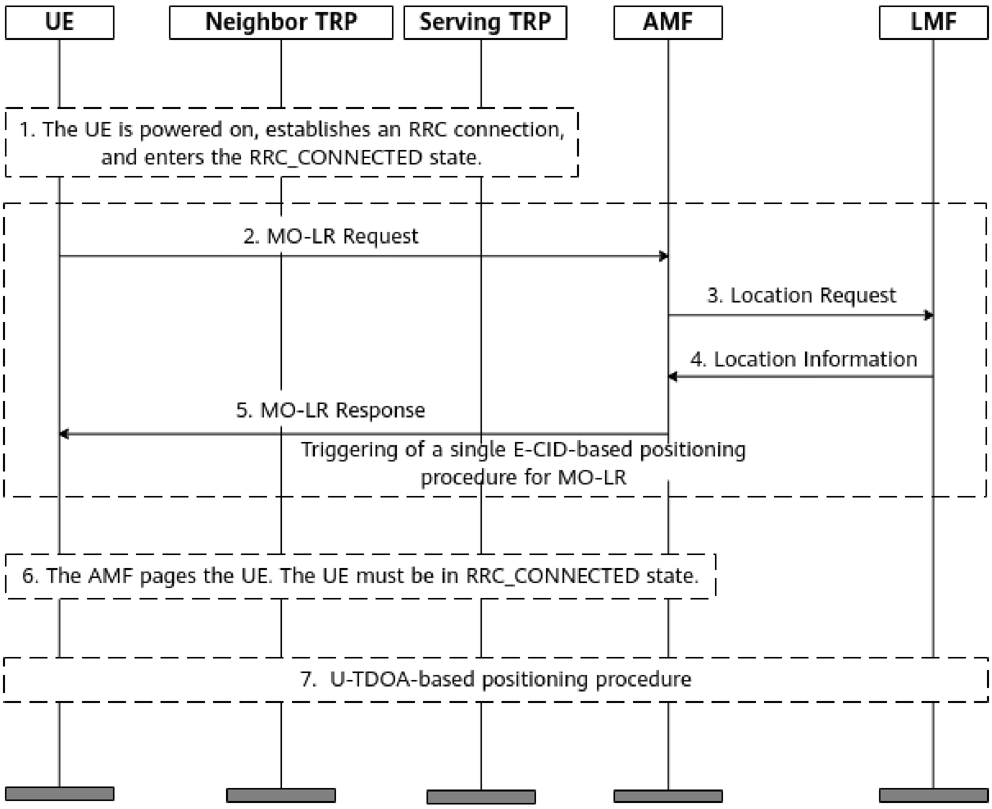
The overall LPHAP procedure is as follows:

1. A UE accesses the network after being powered on, establishes an RRC connection, and enters the RRC_CONNECTED state.
2. The UE sends a NAS message to trigger the positioning procedure as shown in **Figure 6-7**.
3. If the positioning server (LCS client) initiates a UE positioning subscription request, the U-TDOA-based positioning procedure in **Figure 6-8** is started. In addition, if data of sensors (such as battery level, barometric pressure, and motion sensors) is also subscribed to, the periodic procedure of positioning and sensing data reporting in **Figure 6-9** is started.

Positioning Triggering Procedure

- **Figure 6-7** shows the positioning triggering procedure.

Figure 6-7 Positioning triggering procedure



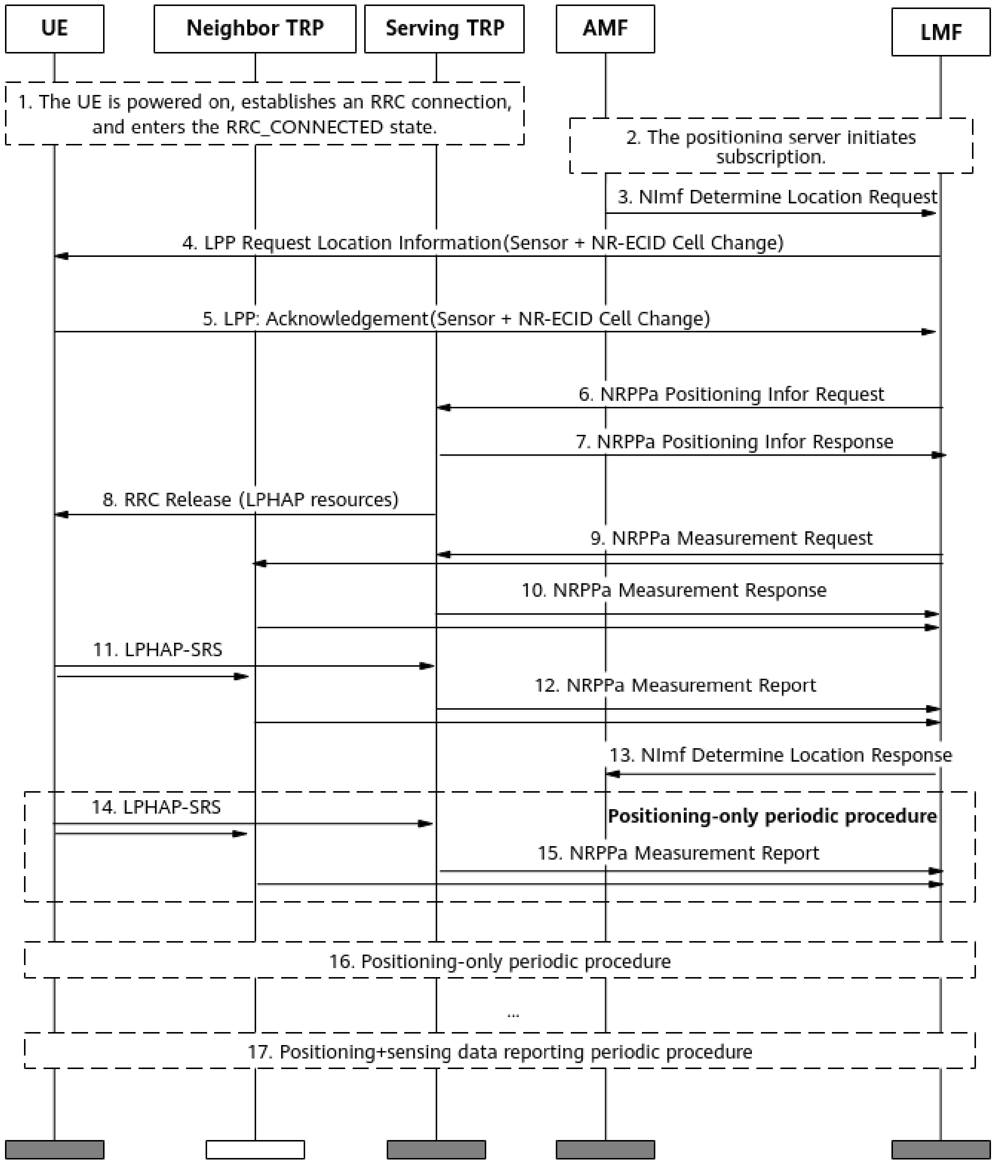
The procedure is described as follows:

a. The UE accesses the network after being powered on, establishes an RRC connection, and enters the RRC_CONNECTED state.
b. The UE sends a NAS message to trigger a single E-CID-based positioning procedure for MO-LR. The message type is **deferred** (LPHAP). An LPP message is embedded in the message to carry key parameters such as the UE capability, positioning period, and number of positioning times.
c. The AMF initiates a positioning request to the LMF. The message contains the GMLC address and is embedded with the LPP message in the MO-LR. The serving cell ID is carried. The LMF does not need to send a measurement request.
d. The LMF calculates the UE's location and reports the location to the GMLC and LCS client through the AMF.
e. The AMF sends an MO-LR response to the UE.
f. The AMF pages the UE. In this case, the UE must be in RRC_CONNECTED mode. The UE determines whether to enter the airplane mode or trigger positioning again based on whether it receives a positioning subscription request message from the positioning server (LCS client). If the UE has not received a subscription request message, it enters the airplane mode. Otherwise, the next step is performed.
g. If the positioning server initiates a UE positioning subscription request, the periodic U-TDOA-based positioning procedure is started.

Periodic U-TDOA-based Positioning Procedure

- **Figure 6-8** shows the periodic U-TDOA-based positioning procedure.

Figure 6-8 Periodic U-TDOA-based positioning procedure



The procedure is described as follows:

a. A UE accesses the network after being powered on, establishes an RRC connection, and enters the RRC_CONNECTED state.

b. The positioning server initiates a UE positioning subscription request to the LMF.

c. The AMF sends a periodic positioning request to the LMF. The message contains the GMLC address.

d. The LMF subscribes to periodic sensing data reporting from the UE based on the sensor capability supported by the UE. (If the UE does not have the sensor capability, the subscription is not performed.) The LMF subscribes to cell handover event reporting in NR E-CID-based positioning from the UE.

e. After receiving the periodic sensing data reporting subscription message, or receiving the cell handover event reporting subscription message of NR

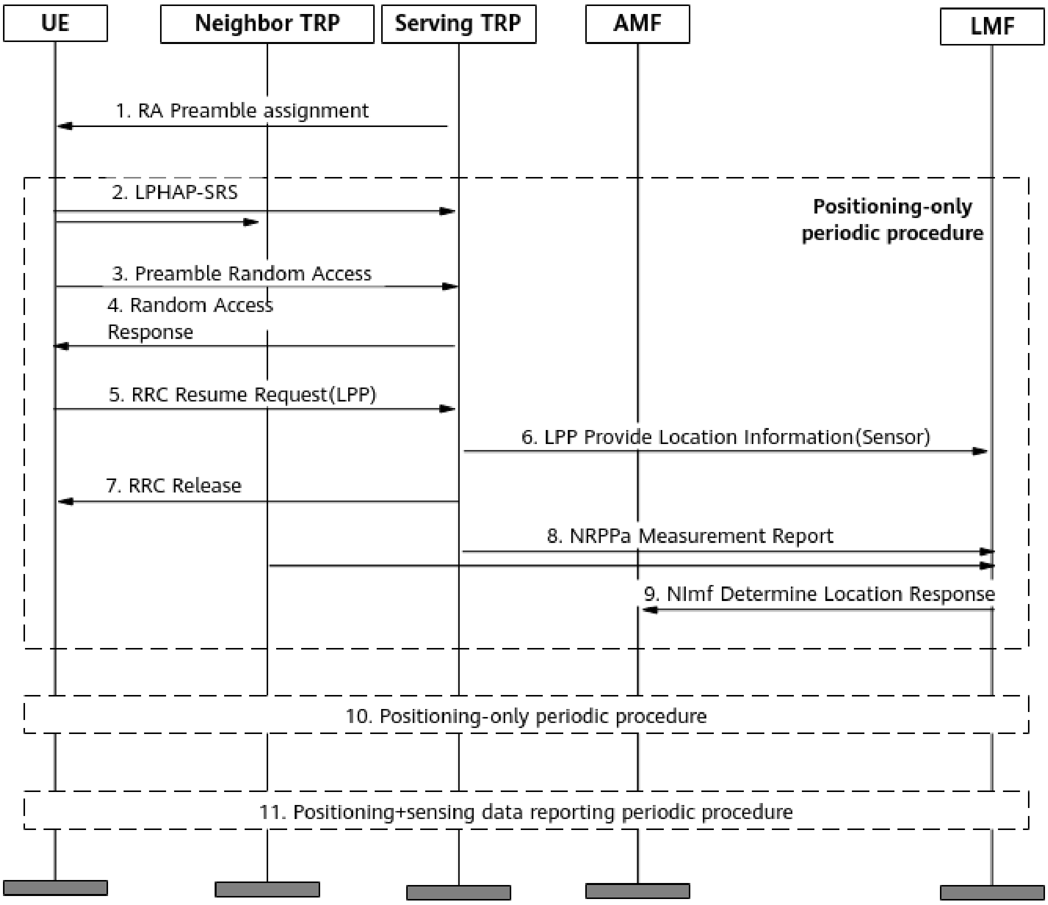
E-CID-based positioning, the UE returns an ACK message to the LMF. (The LPP message for periodic sensing data reporting subscription response is encapsulated in the LCS message. The LPP message is used for cell handover event reporting subscription response in NR E-CID-based positioning.)

f. The LMF applies to the serving cell of gNodeB for SRS resources for LPHAP and PUSCH resources for sensing data reporting. (If the UE does not have the sensor capability, PUSCH resources are not allocated.)
g. After resource allocation, the base station sends an NRPPa Positioning Infor Response message to the LMF to report the allocated resources.
h. After resource allocation, the base station sends an RRC Release message to the UE. The UE is released and enters the RRC_INACTIVE state.
i. The LMF instructs the serving cell and neighboring cells of the gNodeB to monitor the SRS at the corresponding location and perform TOA measurements.
j. After receiving the measurement request from the LMF, the serving cell of the gNodeB and neighboring cells respond to the request.
k. When the LPHAP-SRS transmission period arrives, the UE in the RRC_INACTIVE state sends an LPHAP-SRS to the serving cell of the gNodeB and neighboring cells. The LPP message needs to be encapsulated in the LCS Event Report message for transmission.
l. After receiving the LPHAP-SRS from the UE, the serving cell of the gNodeB and neighboring cells perform TOA measurement and report the TOA measurement result to the LMF.
m. After calculating the UE's location, the LMF reports the location to the GMLC and LCS client through the AMF.
n. When the positioning-only period arrives, the UE sends an LPHAP-SRS to start the positioning procedure.
o. After receiving the LPHAP-SRS from the UE, the serving cell of the gNodeB and neighboring cells perform TOA measurement and report the TOA measurement result to the LMF.
p. When the positioning period arrives, steps 14 and 15 are repeated.
q. The periodic procedure of positioning+sensing data reporting is started.

Periodic Procedure of Positioning+Sensing Data Reporting

Figure 6-9 shows the periodic procedure of positioning+sensing data reporting (based on the random access-based small data transmission (RA-SDT) procedure).

Figure 6-9 Periodic procedure of positioning+sensing data reporting



The procedure is described as follows:

1. The gNodeB allocates dedicated preamble resources to a low-power-consumption UE, and a saving operation is performed on the UE.
2. The UE sends an LPHAP-SRS in the RRC_INACTIVE state.
3. The UE performs non-contention-based random access through dedicated preamble resources.
4. The gNodeB responds to the random access request of the UE and delivers TA information and PUSCH resources to the UE.
5. The UE sends an RRC Resume Request message to the gNodeB.
6. The message encapsulates the LPP message's sensing data and the gNodeB sends it to the LMF.
7. After determining that the UE has no data to transmit, the gNodeB sends an RRC Release message to the UE. Then, the UE is released and enters the RRC_INACTIVE state.
8. After receiving the LPHAP-SRS from the UE, the serving cell of the gNodeB and neighboring cells perform TOA measurement and report the TOA measurement result to the LMF.
9. After calculating the UE's location, the LMF reports the location to the GMLC and LCS client through the AMF.
10. When the LPHAP-SRS transmission period arrives, the positioning-only periodic procedure is repeated.

11. When the LPHAP-SRS transmission period and the sensing data reporting period arrive, steps 1 to 9 are repeated.

6.2 Network Analysis

6.2.1 Benefits

This feature is recommended in 5GtoB campus scenarios if UEs' locations need to be determined by measuring radio signals through 5G networks or operators need to provide services based on UEs' locations.

In complex environments with various propagation paths or severe blockage, U-TDOA-based positioning and LPHAP may fail to take effect due to significant fluctuations in TOA measurement results. Therefore, it is recommended that SRS field strength-based positioning be enabled when U-TDOA-based positioning or LPHAP is enabled. For details about SRS field strength-based positioning, see **5 SRS Field Strength-based Positioning**.

Note that this feature involves the geographic locations of UEs during service provisioning or maintenance. As such, you must fully protect the personal data of users by complying with any and all applicable laws and user privacy policies in the concerned country.

6.2.2 Impacts

The impacts between this function and other functions are described in **6.3.2.3 Function Impacts**.

In high-load scenarios, the specifications of number of supported UEs may decrease. Activating this feature may affect the values of **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**.

After this feature and positioning-dedicated SRS transmission in uplink slots are enabled, SRSs for positioning UEs occupy symbol resources in uplink slots, which has the following impacts:

- The value of the **N.UL.Last.Symbol13.Pwr** counter may increase because the detection symbol is changed from symbol 13 to symbol 12.
- The UE access delay and handover delay may increase because the demodulation performance of the PUCCH and PRACH deteriorates.
- The PUCCH SINR distribution (**N.UL.SINR.PUCCH.Index0** to **N.UL.SINR.PUCCH.Index7**) may change.
- The number of UEs scheduled in the uplink per TTI and the total number of UEs scheduled in the uplink in a cell may decrease.
- The value of **Cell Uplink Average Throughput (DU)** may decrease.
- The values of counters related to CCE usage may change.
- The values of **N.RLC.ThpTime.UL.Cell** and **N.ThpTime.UL** may increase.
- The values of **N.PRB.PUSCH.Used.Avg**, **N.ChMeas.MIMO.UL.Pair.PRB**, and **N.ChMeas.MIMO.UL.Pair.Layer** may change.
- Reconfiguring the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter will cause the cell to restart.

6.3 Requirements

6.3.1 Licenses

The function related to U-TDOA-based positioning is under license control. To use this function, the operator must have purchased and activated the corresponding license.

Feature ID	Feature Name	Model	Sales Unit	Is Cell Activation Affected by License Insufficiency
FOFD-050207	Precise Positioning (LampSite NR) (trial)	NR0SMATPLS00	per Cell	No

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

Prerequisite functions of U-TDOA-based positioning/TOA fingerprint-based positioning

Function Name	Function Switch	Reference	Description
Distributed massive MIMO	**DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter	*Distributed Antenna Solutions (Low-Frequency TDD)*	When U-TDOA-based positioning/TOA fingerprint-based positioning needs to be enabled, the cell must be configured as a distributed massive MIMO cell.
Static SRS period configuration	**NRDUCellSrs**.***SrsPeriod***	*Channel Management*	When U-TDOA-based positioning/TOA fingerprint-based positioning needs to be enabled, the SRS period must be set to 40 ms or longer.
Cell networking mode	**NRDUCell**.***NrDuCellNetworkingMode***	None	When U-TDOA-based positioning/TOA fingerprint-based positioning needs to be enabled, the **NRDUCell**.***NrDuCellNetworkingMode*** parameter must be set to **NORMAL_CELL**.
Positioning-dedicated SRS transmission in uplink slots	**SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter	*Positioning*	When the **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter is selected, U-TDOA-based positioning/TOA fingerprint-based positioning depends on positioning-dedicated SRS transmission in uplink slots.

Prerequisite functions of LPHAP

Function Name	Function Switch	Reference	Description
U-TDOA-based high-precision positioning	**UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	When the **NRDUCell**.***LampSiteCellFlag*** parameter is set to **YES**, LPHAP depends on U-TDOA-based high-precision positioning.
RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	LPHAP requires RedCap Introduction to be enabled.
Positioning-dedicated SRS transmission in uplink slots	**SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter	*Positioning*	LPHAP requires positioning-dedicated SRS transmission in uplink slots to be enabled.

Prerequisite functions of the inter-base-station-transmission-based beacon-free function (specified by the **NRDUCellPosAlgo**.***InterTrpTransmissionSwitch*** parameter)

Function Name	Function Switch	Reference	Description
U-TDOA-based high-precision positioning	**UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	The positioning-dedicated beacon-free function requires U-TDOA-based high-precision positioning to be enabled.
Channel calibration	**INTER_RRU_CHN_CALIB_SW** option of the **NRDUCellAlgoSwitch**.***ChnCalibSwitch*** parameter	*MIMO (TDD)*	The positioning-dedicated beacon-free function requires channel calibration to be enabled.

6.3.2.2 Mutually Exclusive Functions

Mutually exclusive functions of U-TDOA-based positioning/TOA fingerprint-based positioning (specified by the **UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter)

Function Name	Function Switch	Reference	Description
User Service Type	**gNBRfspConfig**.***UserServiceType***	None	● When the **gNBRfspConfig**.***RfspIndex*** parameter is set to **0**, the corresponding **gNBRfspConfig**.***UserServiceType*** parameter must be set to **DS**. ● When positioning-dedicated SRS transmission in uplink slots is enabled for a cell under a base station, the **gNBRfspConfig**.***UserServiceType*** parameter cannot be set to **DS_LCS** for the base station.
LCS Scenario	**NRDUCellPosAlgo**.***LcsScenario***	None	When U-TDOA-based positioning/TOA fingerprint-based positioning is enabled, the **NRDUCellPosAlgo**.***LcsScenario*** parameter can only be set to **SCENARIO_3**, **SCENARIO_4**, **SCENARIO_5**, **SCENARIO_6**, **SCENARIO_146**, or **SCENARIO_189**.
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Hyper cells do not support U-TDOA-based positioning/TOA fingerprint-based positioning.
Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	*Cell Combination*	Cell Combination does not support U-TDOA-based positioning/TOA fingerprint-based positioning.
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	High-speed cells do not support U-TDOA-based positioning/TOA fingerprint-based positioning.

Function Name	Function Switch	Reference	Description
LTE and NR Uplink Spectrum Sharing	**LTE_NR_UL_SPECTRUM_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	None	LTE and NR Uplink Spectrum Sharing cannot work with U-TDOA-based positioning/TOA fingerprint-based positioning.
SRS period adaptation	**SRS_PERIOD_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	*Channel Management*	SRS period adaptation cannot work with U-TDOA-based positioning/TOA fingerprint-based positioning.
Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	Intra-base-station UL CoMP cannot work with U-TDOA-based positioning/TOA fingerprint-based positioning.
Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	Intra-base-station DL CoMP cannot work with U-TDOA-based positioning/TOA fingerprint-based positioning.
Intra-base-station overlapping area measurement	**INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	Intra-base-station overlapping area measurement cannot work with U-TDOA-based positioning/TOA fingerprint-based positioning.
Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	Coordinated interference management cannot work with U-TDOA-based positioning/TOA fingerprint-based positioning.
Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	Co-MM cannot work with U-TDOA-based positioning/TOA fingerprint-based positioning.

Function Name	Function Switch	Reference	Description
MU pairing enhancement	**NRDUCellAlgoSwitch**.***MuMimoOptSwith***	*Massive MIMO AHR (Low-Frequency TDD)*	MU pairing enhancement cannot work with U-TDOA-based positioning/TOA fingerprint-based positioning.
U-TDOA- and AoA-based high-precision positioning	**AOA_UTDOA_HYBRID_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	U-TDOA- and AoA-based high-precision positioning cannot be deployed on the same base station as U-TDOA-based positioning/TOA fingerprint-based positioning.
Uplink assistance	**UL_ASSISTANCE_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	None	U-TDOA-based positioning/TOA fingerprint-based positioning cannot work with uplink assistance.
PUSCH scheduling in self-contained slots	**PUSCH_SCH_IN_S_SLOT_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	None	U-TDOA-based positioning/TOA fingerprint-based positioning cannot work with PUSCH scheduling in self-contained slots.
Intra-frequency beam statistics	**INTRA_FREQ_BEAM_STAT_SW** option of the **NRDUCellCollabServ**.***NeighborNrCellMgmtSw*** parameter	None	Intra-frequency beam statistics cannot work with U-TDOA-based positioning/TOA fingerprint-based positioning.
GP symbol extension	**GP_SYMBOL_EXTENSION_SW** option of the **NRDUCellAlgoSwitch**.***AlgoCompatibility-SwitchExt*** parameter	None	GP symbol extension cannot work with SRS field strength-based positioning.

For details about the mutually exclusive functions of positioning-dedicated SRS transmission in uplink slots (specified by the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter), see **5.3.2.2 Mutually Exclusive Functions**.

Mutually exclusive function of LPHAP (specified by the **LPHAP_REDCAP_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter)

Function Name	Function Switch	Reference	Description
Inactive RA-SDT	**UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter	*UE Power Saving*	If the **LPHAP_REDCAP_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter is selected, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter cannot be selected.

Mutually exclusive functions of the inter-base-station-transmission-based beacon-free function (specified by the **NRDUCellPosAlgo**.***InterTrpTransmissionSwitch*** parameter)

Function Name	Function Switch	Reference	Description
Short-period inter-RRU calibration	**SHORT_PERIOD_INTERRRU_CALIB_SW** option of the **NRDUCellAlgoSwitch**.***ChnCalibSwitch*** parameter	None	If the **NRDUCellPosAlgo**.***InterTrpTransmissionSwitch*** parameter is set to **ON**, the **SHORT_PERIOD_INTERRRU_CALIB_SW** option of the **NRDUCellAlgoSwitch**.***ChnCalibSwitch*** parameter must be deselected.
Adaptive adjustment of calibration power	**CALIB_PWR_ADAPT_ADJ_SW** option of the **gNodeBAlgo**.***ChnCalibPolSw*** parameter	None	If the **NRDUCellPosAlgo**.***InterTrpTransmissionSwitch*** parameter is set to **ON**, the **CALIB_PWR_ADAPT_ADJ_SW** option of the **gNodeBAlgo**.***ChnCalibPolSw*** parameter must be deselected.

6.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	Power saving BWP does not take effect for UEs with U-TDOA-based positioning/TOA fingerprint-based positioning in effect.
DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	DRX does not take effect for UEs with U-TDOA-based positioning/TOA fingerprint-based positioning in effect.
Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	When the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is selected and the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected, the **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be selected. In this case, proactive avoidance of remote interference can take effect.
Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, downlink intra-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.

Function Name	Function Switch	Reference	Description
Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, downlink intra-FR inter-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.
Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch.** ***CaAlgoSwitch*** parameter	*CA Experience Improvement*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, inter-gNodeB CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.
Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, uplink intra-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.

Function Name	Function Switch	Reference	Description
Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, uplink intra-FR inter-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.
CA SRS carrier switching	**SRS_CARRIER_SWITCHING_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*CA Experience Improvement*	When both positioning-dedicated SRS transmission in uplink slots and CA SRS carrier switching are enabled: ● The setting of the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be the same between the PCC and SCCs. Otherwise, CA SRS carrier switching does not take effect. ● If PUCCH format 3 resources are insufficient, the **User Downlink Average Throughput (DU)** slightly decreases.

Function Name	Function Switch	Reference	Description
Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When both positioning-dedicated SRS transmission in uplink slots and intra-base-station UL CoMP are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, UL CoMP cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability and "positioning-dedicated SRS transmission in uplink slots" is enabled for the coordinating neighboring cell, UL CoMP can take effect for the UE and the performance gains are not affected. ● If a UE supports the positioning-dedicated SRS transmission capability and "positioning-dedicated SRS transmission in uplink slots" is not enabled for the coordinating neighboring cell, UL CoMP can take effect for the UE, but the performance gains decrease.

Function Name	Function Switch	Reference	Description
Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When both positioning-dedicated SRS transmission in uplink slots and intra-base-station DL CoMP are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, DL CoMP cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability, DL CoMP can take effect for the UE.
Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	When both positioning-dedicated SRS transmission in uplink slots and coordinated interference management are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, coordinated interference management cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability, coordinated interference management can take effect for the UE.

Function Name	Function Switch	Reference	Description
Super uplink	**SUPER_UL_TDM_SCH_SW** and **FLEX_SUPER_UPLINK_SW** options of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	When both positioning-dedicated SRS transmission in uplink slots and super uplink are enabled, the SUL cells must meet the SUL cell specifications of the corresponding baseband processing unit. Otherwise, positioning-dedicated SRS transmission in uplink slots cannot take effect. For details about the SUL cell specifications of different baseband processing units, see the table content where the **Frequency Band Type of a Cell** column includes **SUL** in **Product Specifications** > **BBU Technical Specifications** > **Technical Specifications of Baseband Processing Units** > **NR Technical Specifications of Baseband Processing Units** > **NR TDD Technical Specifications of Baseband Processing Units** in *3900 & 5900 Series Base Station Product Documentation*.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR TDD-capable main control boards and UBBPg3 support this function. Other BBPs do not support this function. For details, see *LampSite BBU Hardware Description* in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable 4T4R pRRUs support this function. For details, see the related pRRU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

The pRRU565*xAB* and pRRU575*xAB* RF modules must meet the following requirements:

- If a pRRU serves both an NR cell and an LTE cell working in the same frequency band, positioning-dedicated SRS transmission in uplink slots cannot be enabled.
- Up to four pRRUs can be deployed under a TRP.
- In intra-band NR multi-carrier scenarios, carriers under a TRP share the same sector configurations, and positioning-dedicated SRS transmission in uplink slots needs to be enabled.

NOTE

In LampSite scenarios, one TRP corresponds to one sector equipment group which may contain multiple pRRUs.

6.3.4 Cells

For U-TDOA-based positioning/TOA fingerprint-based positioning, the cell configuration must meet the following conditions:

- **NRDUCell**.***UlBandwidth*** must be set to 80 MHz or 100 MHz.
- **NRDUCell**.***SlotAssignment*** must be set to **8_2_DDDDDDDSUU** or **8_2_DDDSUUDDDD**.
- **NRDUCell**.***FrequencyBand*** must be set to **N41**, **N77**, **N78**, or **N79**.
- LampSite cells are configured (with **NRDUCell**.***LampSiteCellFlag*** set to **YES**).
- The cells are not in SUL mode (with **NRDUCell**.***DuplexMode*** set to a value other than **CELL_SUL**).

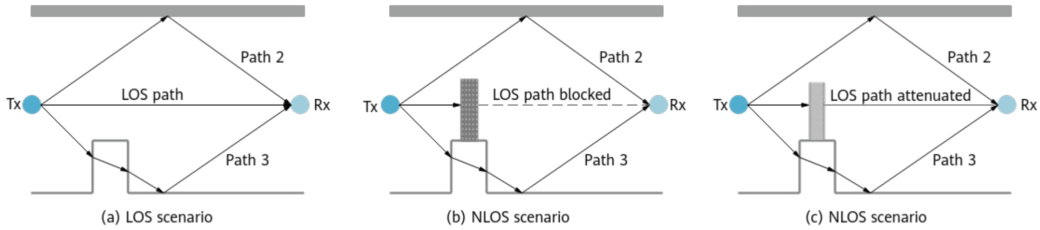
For the positioning-dedicated beacon-free function, the cell bandwidth can only be set to 100 MHz.

6.3.5 Networking

The accuracy of precise positioning is related to the proportion of LOS transmissions.

- In the LOS scenario shown in (a) of **Figure 6-10**, there is an unobstructed direct path between the TX and RX ends.
- In the NLOS scenarios shown in (b) and (c) of **Figure 6-10**, the direct path between the TX and RX ends is completely blocked, or signals on the direct path can penetrate the obstacle but arrive at the RX end with diminished strength. As such, the positioning accuracy decreases in NLOS scenarios.

Figure 6-10 LOS and NLOS scenarios



At least three pRRUs are needed to calculate the UE's location. LOS transmissions from a UE to at least three pRRUs are required to ensure the positioning accuracy. In U-TDOA-based positioning, the RSRP of SRS sent from the UE to each pRRU must be greater than –105 dBm to ensure the accuracy of TOA measurement results.

In addition, the pRRU installation mode, deployment envelope, spatial distribution, and hollow-out area must be considered in the planning and deployment of a network. After pRRU deployment is complete, accurate pRRU deployment positions need to be measured, and then configured on the LMF.

To ensure the accuracy of U-TDOA-based positioning, beacon UEs are deployed or U-TDOA-based beacon-free positioning is used to realize time synchronization between pRRUs. For details about the positioning network planning, simulation, and design, beacon UE deployment planning, as well as pRRU and beacon UE deployment position measurements, contact Huawei service engineers.

To ensure positioning accuracy, a maximum of four pRRUs can be deployed under a TRP.

UE positioning may fail in multi-cell-ID scenarios. Therefore, precise positioning cannot apply to multi-cell-ID scenarios.

LPHAP applies only to distributed massive MIMO networking scenarios.

The UE-based beacon-free function requires an inter-site distance (ISD) of 12 m to 18 m.

The inter-base-station-transmission-based beacon-free function requires pRRUs with built-in omnidirectional antennas and LOS transmission between pRRUs. The recommended ISD is 12 m to 18 m.

NOTE

A beacon UE synchronizes time as follows: A beacon UE periodically sends SRSs, adjacent pRRUs measure the TOA values of the received SRSs, and the gNodeB sends the measurement results to the LMF. Based on the TOA measurement result of each pRRU and the known coordinates of the pRRUs and beacon UE, the LMF calculates and saves the TDOA between every two pRRUs. The saved latency difference information is used for compensation during the location calculation.

The coordinates of a pRRU and beacon UE are configured by running the **ADD PRRUNODE** and **ADD BEACONUE** commands on the LMF, respectively.

6.3.6 Others

U-TDOA-based positioning/TOA fingerprint-based positioning:

- UEs must support the NR SA networking.
- The AMF and LMF must comply with 3GPP TS 38.455 V16.1.0 or later.
- If UEs capable of sending SRSs for positioning need to be positioned, the core network needs to support parsing of the *Positioning SRS* IE.

LPHAP:

- UEs must support the NR SA networking.
- The AMF and LMF must comply with 3GPP TS 38.455 V16.1.0 or later.
- If UEs capable of sending SRSs for positioning need to be positioned, the core network needs to support parsing of the *Positioning SRS* IE.
- The UE capability information elements of the UE must contain **ra-SDT-r17** and **srb-SDT-r17**.
- The UE capability information elements of the UE must contain **SRS-AllPosResourcesRRC-Inactive-r17**.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Table 6-1, Table 6-2, Table 6-3, and Table 6-4 describe the parameters used for function activation. No parameters are involved in function optimization. In addition, UE-based beacon-free function is controlled by the same parameter as U-TDOA-based high-precision positioning on the gNodeB side. Therefore, no additional parameter configuration is required for this function.

Table 6-1 Parameter used for activation (positioning TRP ID function)

Parameter Name	Parameter ID	Option	Setting Notes
Positioning TRP ID Switch	**gNodeBAlgo**.***PosTrpIdSwitch***	None	It is recommended that the positioning TRP ID function be enabled.

Table 6-2 Parameters used for activation (U-TDOA-based positioning/TOA fingerprint-based positioning)

Parameter Name	Parameter ID	Option	Setting Notes
User Service Type	**gNBRfspConfig**.***UserServiceType***	None	Set this parameter based on the planned user service type.

Parameter Name	Parameter ID	Option	Setting Notes
LCS Switch	**NRCellAlgoSwitch**.***LcsSwitch***	UTDOA_SW	Select this option when U-TDOA-based positioning/TOA fingerprint-based positioning is required. **gNBRfspConfig**.***UserServiceType*** must be set to **LCS** for U-TDOA-based positioning accuracy tests or high-accuracy positioning services.
SRS for Positioning in UL Slot Switch	**NRDUCellPosAlgo**.***LcsMeasureAlgoSw***	SRS_FOR_POSITION_ON_ULSLOT_SW	Select this option if UEs capable of sending SRSs for positioning need to be positioned.
LCS Measure Algorithm Switch	**NRDUCellPosAlgo**.***LcsMeasureAlgoSw***	COMMON_USER_LCS_SW	Select this option for a common UE (non-low-power-consumption positioning service UE) not configured with an RFSP index during registration.

Table 6-3 Parameters used for activation (LPHAP)

Parameter Name	Parameter ID	Option	Setting Notes
User Service Type	**gNBRfspConfig**.***UserServiceType***	None	Set this parameter based on the planned user service type.
LCS Switch	**NRCellAlgoSwitch**.***LcsSwitch***	LPHAP_REDCAP_SW	Select this option if LPHAP is required.

Table 6-4 Parameters used for activation (inter-base-station-transmission-based beacon-free function)

Parameter Name	Parameter ID	Option	Setting Notes
Inter-TRP Transmission Max Distance	**NRDUCellPosAlgo**.***InterTrpTransMaxDistance***	None	It is recommended that this parameter be set to two times the ISD.
Channel Calibration Switch	**NRDUCellAlgoSwitch**.***ChnCalibSwitch***	INTER_RRU_CHN_CALIB_SW	Select this option before enabling the positioning-dedicated beacon-free function.
Inter-TRP Transmission Switch	**NRDUCellPosAlgo**.***InterTrpTransmissionSwitch***	None	It is recommended that this parameter be set to **ON** when inter-TRP transmission is required for improving the positioning accuracy and eliminating the need of beacon deployment.
Channel RSSI Threshold	**NRDUCellChnPwr**.***ChnRssiThld***	None	The inter-base-station-transmission-based beacon-free function depends on the connectivity check between pRRUs. It is recommended that the **NRDUCellChnPwr**.***ChnRssiThld*** parameter be set to **-65** in scenarios with inter-base-station-transmission-based beacon-free function enabled.

NOTE

For details about positioning parameter configurations related to the cloud core network, see the UEG LCS product documentation of the cloud core network.

6.4.1.2 Using MML Commands

Before using MML commands, refer to **6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- If RFSP indexes have been configured during UE registration, the **gNBRfspConfig**.***UserServiceType*** parameter can be configured based on RFSP indexes. The following gives an example of configuration:
  a. Planning by customers based on actual service requirements: Assume that UEs in campus A have three types of services. Three UE types need to be planned: Type-I UEs perform only data services. Type-II UEs support non-LPHAP services and a small number of data services. Type-III UEs support LPHAP services.
  b. SIM card registration with the operator: According to the preceding requirements, customers in campus A register SIM cards with the operator, and the operator assigns three **gNBRfspConfig**.***RfspIndex*** values (for example, **100**, **101**, and **102**) which are configured in the users' registration information, to distinguish the three UE types.
  c. SIM card-service mapping by customers: After obtaining registered SIM cards from the operator, the customers in campus A determine and maintain the mapping between the three types of SIM cards and three types of services. For example, **100** is allocated to the UEs performing only data services, **101** to the UEs performing common positioning services and a small number of data services, and **102** to the UEs performing LPHAP services.
  d. Adding MML-based configurations on the gNodeB: After obtaining the mapping between SIM cards and services, network maintenance engineers run MML commands and configure the **gNBRfspConfig**.***UserServiceType*** parameter based on the service types planned for the three types of SIM cards (with different **gNBRfspConfig**.***RfspIndex*** values). (The default value of **gNBRfspConfig**.***UserServiceType*** is **DS**. If precise positioning is enabled, the value of **gNBRfspConfig**.***UserServiceType*** must be **LCS** or **LPHAP_LCS**.) The MML-based configuration examples are as follows:
```
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=100, UserServiceType=DS;
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=101, UserServiceType=LCS;
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=102, UserServiceType=LPHAP_LCS;
```
- If a common UE (non-low-power-consumption positioning service UE) is not configured with an RFSP index during registration, turn on the common UE positioning switch:
```
//Enabling positioning-dedicated SRS transmission in uplink slots
MOD NRDUCELLPOSALGO: NrDuCellId=1, LcsMeasureAlgoSw=SRS_FOR_POSITION_ON_USLOT_SW-1;
//Turning on the common UE positioning switch
MOD NRDUCELLPOSALGO: NrDuCellId=1, LcsMeasureAlgoSw=COMMON_USER_LCS_SW-1;
```
- Positioning TRP ID function:
  a. Find the RRU or port associated with an engineering parameter ID based on the engineering parameter configuration of the positioning service.

b. Find the **SectorEqmId** corresponding to the RRU or port based on **MO SECTOREQM**.

c. Find the **NrDuCellTrpId** and **NrDuCellCoverageId** corresponding to **SectorEqmId** based on **MO NRDUCELLCOVERAGE**.

d. Configure a positioning TRP ID (controlled by **NRDUCellLcsProperty**.***PosTrpId***) for every combination of **NrDuCellTrpId**, **NrDuCellCoverageId**, and engineering parameter ID. The positioning TRP ID ranges from 1 to 65535 and is unique within a gNodeB. This parameter takes effect only after the positioning TRP ID function is enabled. The following is an MML command example:

```
//Enabling the positioning TRP ID function
MOD GNODEBALGO: PosTrpIdSwitch=ON;
```

- U-TDOA-based positioning/TOA fingerprint-based positioning:

```
//Adding gNodeB RFSP configurations with UserServiceType set to LCS for UEs performing only positioning services or UEs performing both positioning and MBB services and requiring positioning accuracy guarantee. This ensures that full-bandwidth SRS resources are allocated to the UEs performing positioning services.
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=10, UserServiceType=LCS;
//Enabling U-TDOA-based positioning
MOD NRCELLALGOSWITCH: NrCellId=1, LcsSwitch=UTDOA_SW-1;
```

- LPHAP:

```
//Setting the user service type to LPHAP if the service type planned for the SIM card is LPHAP
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=102, UserServiceType=LPHAP_LCS;
//Enabling LPHAP
MOD NRCELLALGOSWITCH: NrCellId=1, LcsSwitch=LPHAP_REDCAP_SW-1;
```

- Inter-base-station-transmission-based beacon-free function:

```
//Turning on the channel calibration switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, ChnCalibSwitch=INTER_RRU_CHN_CALIB_SW-1;
//Configuring the maximum inter-TRP transmission distance
MOD NRDUCELLPOSALGO: NrDuCellId=1, InterTrpTransMaxDistance=24;
//Enabling the positioning-dedicated beacon-free function
MOD NRDUCELLPOSALGO: NrDuCellId=1, InterTrpTransmissionSwitch=ON;


//(Optional) Configuring Channel RSSI Threshold
MOD NRDUCELLCHNPWR: NrDuCellId=0, ChnRssiThld=-65;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Positioning TRP ID function:

```
//Disabling the positioning TRP ID function
MOD GNODEBALGO: PosTrpIdSwitch=OFF;
```

- U-TDOA-based positioning/TOA fingerprint-based positioning:

```
//Disabling U-TDOA-based positioning
MOD NRCELLALGOSWITCH: NrCellId=1, LcsSwitch=UTDOA_SW-0;
```

- LPHAP:

```
//Disabling LPHAP
MOD NRCELLALGOSWITCH: NrCellId=1, LcsSwitch=LPHAP_REDCAP_SW-0;
```

- Positioning-dedicated beacon-free function:

```
//Disabling the positioning-dedicated beacon-free function
MOD NRDUCELLPOSALGO: NrDuCellId=1, InterTrpTransmissionSwitch=OFF;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Verify whether U-TDOA-based positioning has been activated based on signaling tracing or performance counters.

Based on signaling tracing:

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left, choose **NG Interface Trace**.

Step 3 Enable the LCS client to initiate a positioning request. After processing by the AMF, the LMF triggers the U-TDOA-based positioning procedure.

Step 4 On the **NG Interface Trace** page, check for the following messages:

- A POSITIONING INFORMATION REQUEST or MEASUREMENT REQUEST message sent from the AMF to the gNodeB
- A POSITIONING INFORMATION RESPONSE, MEASUREMENT RESPONSE, or MEASUREMENT REPORT message sent from the gNodeB to the AMF

  If the MEASUREMENT RESPONSE/MEASUREMENT REPORT message contains **UL RTOA**, U-TDOA-based positioning has been activated.

----End

Based on related performance counters:

Step 1 On the MAE-Access, initiate gNodeB and cell performance measurement-related tasks to obtain the values of the **N.Positioning.Info.Req.Att**, **N.Positioning.Info.Req.Succ**, **N.Positioning.Meas.Req.Att**, **N.Positioning.Meas.Req.Succ**, and **N.Positioning.Meas.Rpt** counters.

Step 2 Enable the LCS client to initiate a positioning request during the measurement period. After processing by the AMF, the LMF triggers the U-TDOA-based positioning procedure.

Step 3 Check the counter values.

1. In on-demand positioning mode, U-TDOA-based positioning has been activated if none of the values of the following counters is 0: **N.Positioning.Info.Req.Att**, **N.Positioning.Info.Req.Succ**, **N.Positioning.Meas.Req.Att**, and **N.Positioning.Meas.Req.Succ**.
2. In periodic positioning mode, U-TDOA-based positioning has been activated if none of the values of the following counters is 0: **N.Positioning.Info.Req.Att**, **N.Positioning.Info.Req.Succ**, **N.Positioning.Meas.Req.Att**, **N.Positioning.Meas.Req.Succ**, and **N.Positioning.Meas.Rpt**.

----End

6.4.3 Network Monitoring

After Precise Positioning (LampSite NR) is activated, the performance can be monitored by using the following counters.

Counter ID	Counter Name	Counter Description
1911829380	**N.Positioning.Info.Req.Att**	Number of positioning information requests
1911829379	**N.Positioning.Info.Req.Succ**	Number of positioning information responses
1911829377	**N.Positioning.Meas.Req.Att**	Number of positioning measurement requests
1911829376	**N.Positioning.Meas.Req.Succ**	Number of successful positioning measurements
1911829375	**N.Positioning.Meas.Rpt**	Number of positioning measurement reporting times
1911840425	**N.User.Positioning.LPHAP.Avg**	Average number of UEs performing LPHAP services
1911842612	**N.User.Positioning.Avg**	Average number of UEs performing positioning services

When a large number of positioning requests are initiated simultaneously, flow control may be triggered, causing NRPPa messages to be discarded. The performance can be monitored by using the following counters.

Counter ID	Counter Name	Counter Description
1911829378	**N.NRPPa.Ng.Msg.Disc.FlowCtrl**	Number of times NRPPa messages over the NG interface are discarded due to flow control

7 Precise Positioning (NR) (Trial)

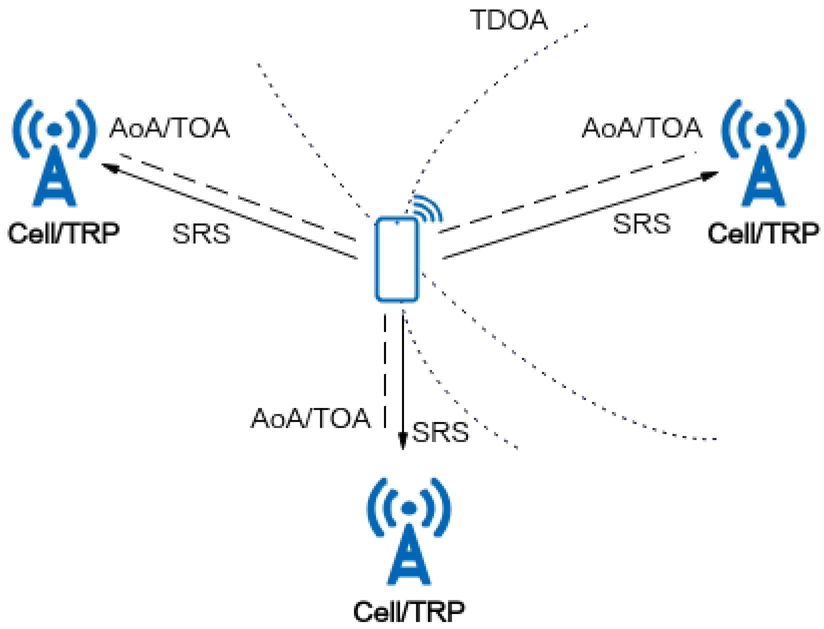
7.1 U-TDOA- and AoA-based High-Precision Positioning

7.1.1 Principles

7.1.1.1 U-TDOA- and AoA-based High-Precision Positioning

In U-TDOA- and AoA-based high-precision positioning, the gNodeB measures the time of arrival (TOA) and AoA of the SRSs from a UE to multiple TRPs and sends the measurement results to the LMF. Then, the LMF calculates the UE's location based on the AoA, time difference of arrival (TDOA) between every two TRPs, and known TRP locations. See **Figure 7-1**.

Figure 7-1 U-TDOA- and AoA-based high-precision positioning principles



This function is controlled by the **NRDUCellPosAlgo**.***ApplicationScenario*** parameter and the **AOA_UTDOA_HYBRID_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter. U-TDOA- and AoA-based high-precision positioning has been enabled if both of the following conditions are met:

- The **AOA_UTDOA_HYBRID_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter is selected.
- The **NRDUCellPosAlgo**.***ApplicationScenario*** parameter is set to **NORMAL_SCENARIO**.

For details about the positioning procedure of this function, see **7.1.1.3 U-TDOA- and AoA-based High-Precision Positioning Procedure**.

U-TDOA- and AoA-based high-precision positioning depends on high-precision TOA and AoA measurements. The measurement accuracy is closely related to the bandwidth for SRS transmission. A gNodeB can select bandwidth resources for positioning services based on user service types (specified by the **gNBRfspConfig**.***UserServiceType*** parameter).

- For UEs mainly performing positioning services:
  - If they are common UEs (non-low-power-consumption positioning service UEs), the user service type can be set to **LCS** or the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter can be selected. In this way, full-bandwidth SRS resources are configured for the UEs to ensure high positioning accuracy for them.
  - If they are low-power-consumption positioning service UEs, the user service type needs to be set to **LPHAP_LCS**. In this way, full-bandwidth SRS resources are configured for the low-power-consumption positioning service UEs in inactive state to ensure high positioning accuracy for them.
- For UEs mainly performing data services, the user service type can be set to **DS**, indicating that SRS resources are allocated based on the traditional SRS resource allocation algorithm. For UEs that are not configured with a service type, the user service type is **DS** by default.

When U-TDOA- and AoA-based high-precision positioning is enabled, the following operations are also required:

- Enable positioning-dedicated SRS transmission in uplink slots (by selecting the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter). After this function is enabled, the gNodeB allocates SRS resources for positioning in fixed symbols of uplink slots.
- Enable inter-base-station-transmission-based beacon-free function. For details, see **7.1.1.2 Inter-Base-Station-Transmission-based Beacon-free Function**.

U-TDOA- and AoA-based high-precision positioning is also applicable to long narrow corridors where TRPs are deployed linearly. After the function is enabled, two linearly deployed TRPs are used for U-TDOA-based positioning. The two TRPs of the gNodeB measure the TOA values by receiving SRSs and report the TOA values to the LMF. Then, the LMF calculates the UE location based on the coordinates of TRPs receiving the signals and the difference of distances between the UE and the two TRPs. The calculated UE coordinates indicate a relative location on the straight line between the two TRPs.

7.1.1.2 Inter-Base-Station-Transmission-based Beacon-free Function

In U-TDOA-based positioning, TRPs take different time to process received signals. As a result, an error is introduced during the calculation of TDOA over the air interface, affecting the positioning accuracy. To improve positioning accuracy, a fixed-location beacon UE deployment solution can be used. Specifically, a beacon UE is deployed in the positioning area and its location is known. However, in this solution, the beacon UE needs to be placed at a fixed location in the center of the positioning area, and sustained power supply is required. Therefore, it is difficult to promote the related positioning service on a large scale. As such, the inter-base-station-transmission-based beacon-free function is introduced.

After this function is enabled, signals are exchanged between TRPs for calculating the inter-TRP differences in the time of processing received signals based on their known coordinates. During the TOA measurement of the to-be-positioned UE, the differences in the time of processing received signals are eliminated. With this function, the positioning accuracy requirement can be met without deploying beacon UEs. The inter-base-station-transmission-based beacon-free function is controlled by the **NRDUCellPosAlgo**.***InterTrpTransmissionSwitch*** parameter.

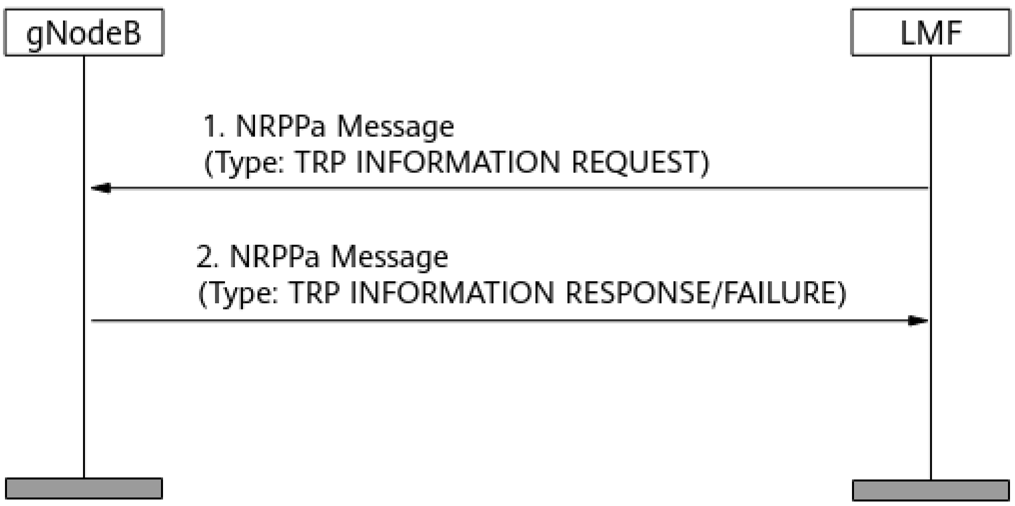
7.1.1.3 U-TDOA- and AoA-based High-Precision Positioning Procedure

The U-TDOA- and AoA-based high-precision positioning measurement procedure includes **On-Demand Mode** and **Periodic Mode**. The mode is specified in the request from the core network.

Positioning TRP Coordinate Configuration

The LMF sends a TRP INFORMATION REQUEST message to trigger the gNodeB to report TRP information through a TRP INFORMATION RESPONSE message. The reported information is used by the LMF for UE location calculation, as shown in **Figure 7-2**.

Figure 7-2 TRP information exchange



The positioning function requires TRP coordinates to be configured on the base station and sent to the LMF through the TRP INFORMATION RESPONSE message. On the base station, either of the following configuration methods can be used:

- Configure TRP physical coordinates by setting LCS engineering parameters for TRPs in the positioning area (**ADD LCSENGINEERINGPARA**) and LCS engineering parameter binding relationships (**ADD LCSENGPARABIND**).
- Configure TRP physical coordinates by setting the actual longitudes and latitudes of base stations in the positioning area (**ADD LOCATION**), LCS engineering parameters for TRPs in the positioning area (**ADD LCSENGINEERINGPARA**), and LCS engineering parameter binding relationships (**ADD LCSENGPARABIND**).

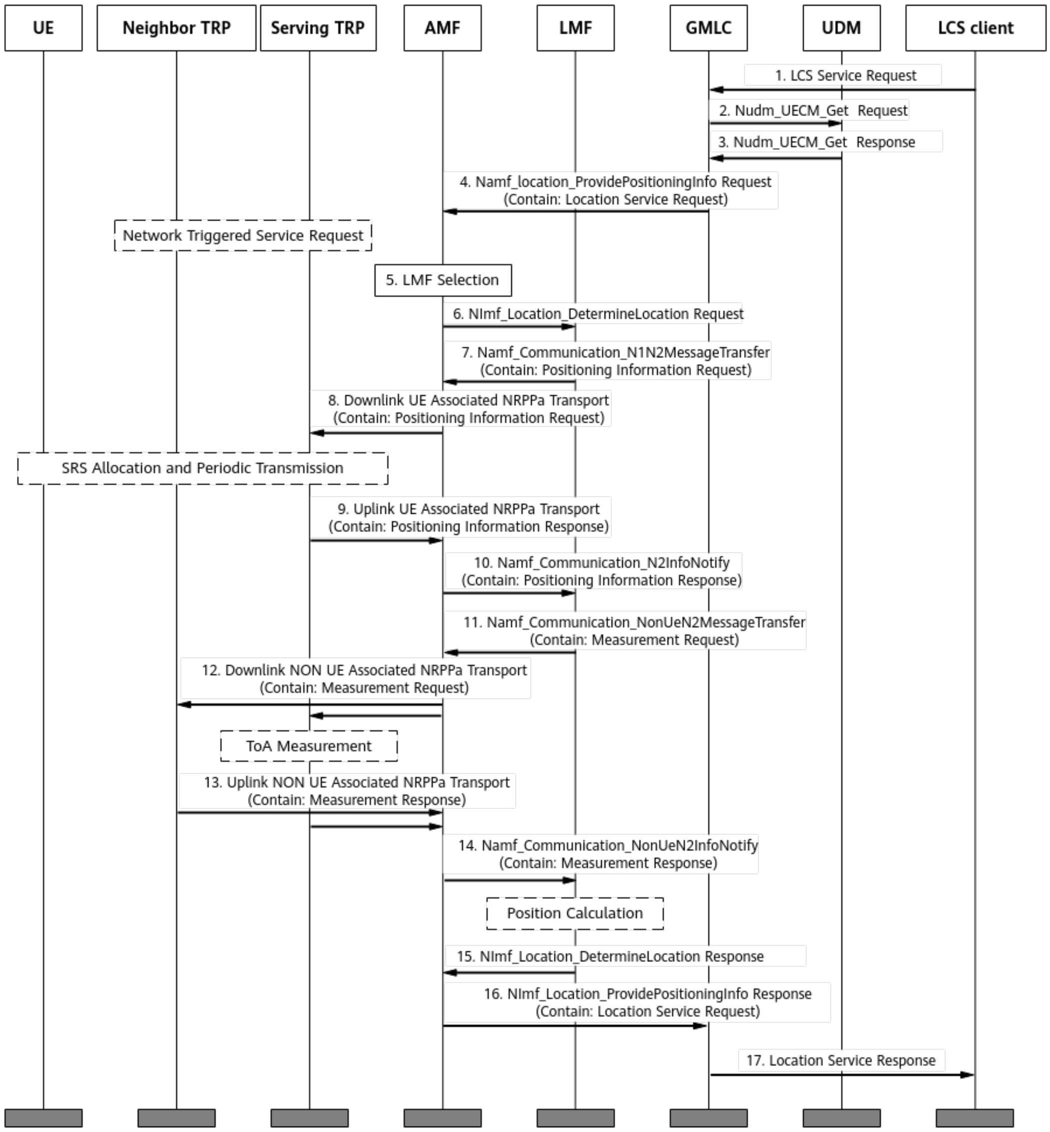
NOTE

- As defined in section 9.2.24 "TRP ID" of 3GPP TS 38.455 V17.5.0, the value range of **TRP ID** carried in the TRP INFORMATION REQUEST and TRP INFORMATION RESPONSE messages is (1, 65535). If the positioning TRP ID function (specified by **gNodeBAlgo**.***PosTrpIdSwitch***) is disabled, change the TRP ID to a non-zero value before enabling the Precise Positioning feature.
- If the positioning TRP ID is required, run the corresponding MML command to configure the positioning TRP ID and set **gNodeBAlgo**.***PosTrpIdSwitch*** to **ON**. For details, see **7.1.4.1.2 Using MML Commands**.
- Either **LCSENGINEERINGPARA**.***LOCATIONNAME*** or **LCSENGINEERINGPARA**.***MAPID*** must be set to a valid value.

On-Demand Mode

In on-demand mode, each time the LCS client sends a positioning request, it initiates the procedure in **Figure 7-3** and obtains only one location.

Figure 7-3 On-demand positioning



The procedure for on-demand positioning is described as follows:

1. The LCS client sends the GMLC a positioning request where the target UE is identified by a generic public subscription identifier (GPSI) or subscription permanent identifier (SUPI).
2. The GMLC invokes a Nudm_UECM_Get service operation that carries the GPSI or SUPI of the target UE to initiate a positioning request toward the home UDM of this UE.
3. The UDM returns the network addresses of the AMF that serves the UE.
4. The GMLC sends a positioning request to the AMF. If the UE is in idle state, the AMF initiates a network-triggered service request procedure to establish a signaling connection with the UE before proceeding to step 5. If the UE is in connected state, it skips directly to step 5.
5. The AMF selects an LMF to serve the target UE.
6. The AMF sends a positioning request to the LMF.

7. The LMF sends a Positioning Information Request message to the AMF, requesting SRS resource information for the UE performing positioning services.
8. The AMF transparently transmits the Positioning Information Request message to the gNodeB.
9. The gNodeB sends the AMF a Positioning Information Response message that contains information about the SRS resources for the UE.
10. The AMF transparently transmits the Positioning Information Response message to the LMF.
11. After obtaining information about the SRS resources for the UE, the LMF sends a Measurement Request message to the AMF. The message contains information such as the SRS configuration, measurement reporting mode, and positioning measurement quantity. In on-demand mode, the *Report Characteristics* IE is set to **OnDemand**.
12. The AMF transparently transmits the Measurement Request message to the gNodeB. The TRPs involved in measurements may be under one or more gNodeBs, and the AMF routes the message to all the corresponding gNodeBs.
13. After receiving the Measurement Request message, the gNodeB measures the SRS time-frequency position of the UE and obtains the TOA, AoA, and RSRP measurement results. The **TRP Measurement Type** IE carries the values **UL-RTOA** and **UL-AOA**. The gNodeB reports the measurement results to the AMF through a Measurement Response message. In 64T64R/32T32R scenarios, the gNodeB can also report **LOS/NLOS Information** to assist the LMF in location calculation.
14. The AMF transparently transmits the Measurement Response message to the LMF.
15. The LMF parses the Measurement Response message to obtain the measured TOA, AoA, and RSRP values. The LMF calculates the location and sends it to the AMF.
16. The AMF transparently transmits the calculated location to the GMLC.
17. The GMLC sends it to the positioning request-initiating LCS client.

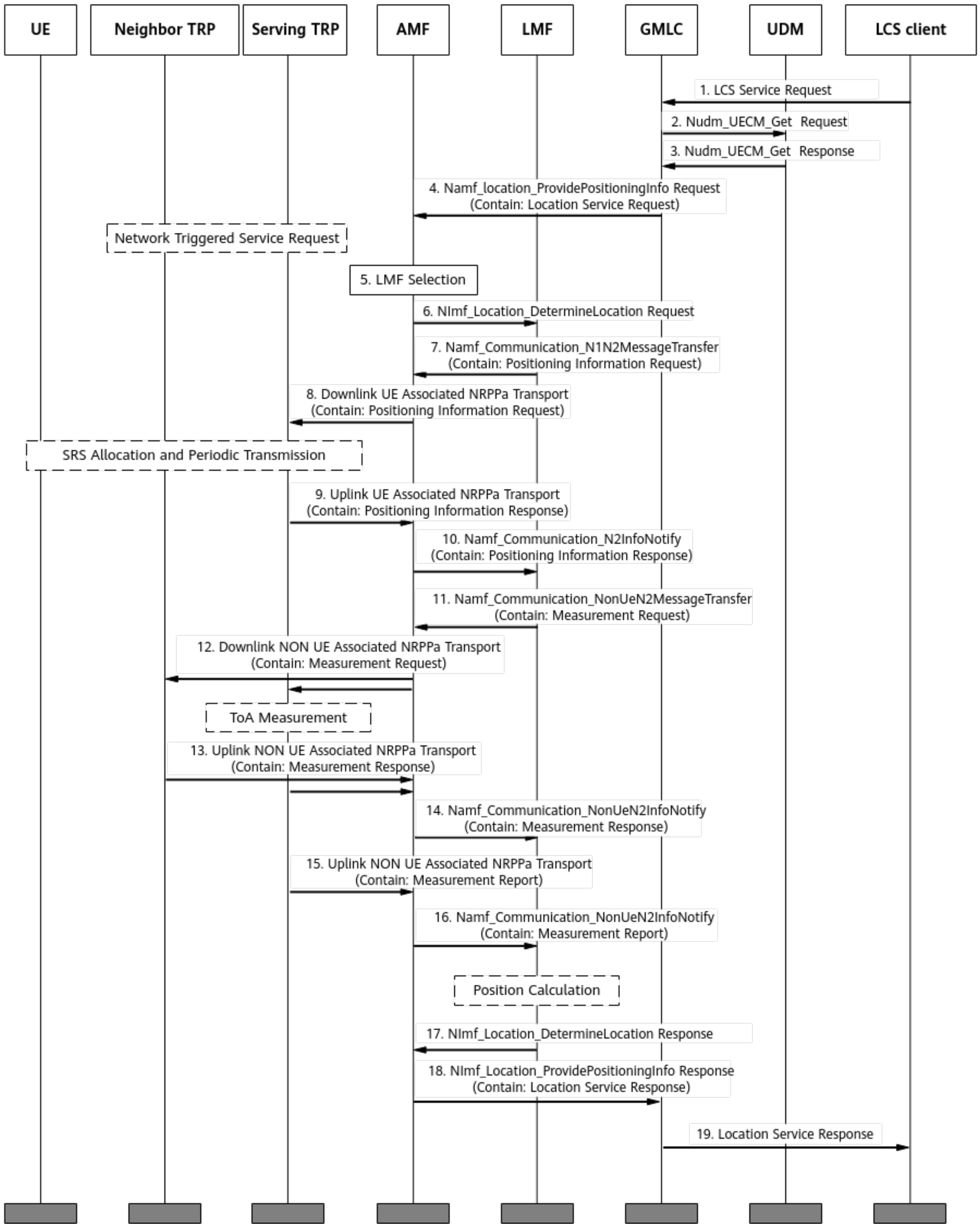
NOTE

When detecting that the SRS of the UE changes, the gNodeB sends a Positioning Information Update message containing the *SRS Configuration* IE to the LMF. When detecting that the SRS transmission of the UE stops, the gNodeB sends a Positioning Information Update message containing the *SRS Transmission Status* IE to the LMF.

Periodic Mode

In periodic mode, after the LCS client initiates a positioning request and the positioning signaling procedure is established, the gNodeB periodically reports the measurement results, enabling the LCS client to periodically obtain the UE's location. **Figure 7-4** shows this procedure.

Figure 7-4 Periodic positioning



The procedure for periodic positioning is described as follows:

- Steps 1–12 and 17–19 in **Figure 7-4** are similar to those in the on-demand positioning procedure. The major difference lies in steps 11 and 12 where the value of the *Report Characteristics* IE in the Measurement Request message is **Periodic** and the message contains the *Measurement Periodicity* IE.
- The Measurement Response message of steps 13 and 14 in **Figure 7-4**, unlike that in on-demand positioning, does not contain measurement results. Instead, the message serves only as the response indicating successful processing of the Measurement Request message.

- Steps 15 and 16 in **Figure 7-4** are unique to periodic positioning. After measuring the TOA, AoA, and RSRP, the gNodeB periodically sends the AMF a Measurement Report message containing the measurement results. The AMF transparently transmits the message to the LMF. The LMF parses the message to obtain the TOA, AoA, and RSRP measurement results, and then proceeds with the subsequent calculation operations.

NOTE

The current version supports only the measurement periods of 1280 ms, 2560 ms, 5120 ms, and 10240 ms.

7.1.2 Network Analysis

7.1.2.1 Benefits

This feature is recommended in 5GtoB campus scenarios if UEs' locations need to be determined by measuring radio signals through 5G networks or operators need to provide UE-location-based services.

The horizontal positioning accuracy is used as the accuracy of U-TDOA- and AoA-based high-precision positioning:

- In LOS scenarios, the positioning accuracy is 3 m @ 90% and closely related to the proportion of LOS scenarios. LOS transmission is closely related to the physical environment and inter-TRP distance. In this situation, the number and installation positions of TRPs need to be planned. A higher proportion of LOS scenarios leads to higher positioning accuracy.
- In NLOS scenarios, the positioning accuracy is 10 m @ 90% if NLOS positioning enhancement is enabled.

Note that this feature involves the geographical locations of UEs during service provisioning or maintenance. As such, you must fully protect the personal data of users by complying with any and all applicable laws and user privacy policies in the concerned country or territory.

7.1.2.2 Impacts

The impacts between this function and other functions are described in **7.1.3.2.3 Function Impacts**.

In high-load scenarios, enabling this feature decreases the specifications of number of UEs supported by the network. After this feature is activated, average uplink and downlink cell throughputs (specified by **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**) may decrease.

After this feature is enabled, SRSs for positioning UEs occupy symbol resources in uplink slots, which has the following impacts:

- The value of the **N.UL.Last.Symbol13.Pwr** counter may change because the detection symbol is changed from symbol 13 to symbol 12.
- The UE access delay and handover delay may increase because the demodulation performance of the PUCCH and PRACH deteriorates.
- The PUCCH SINR distribution (**N.UL.SINR.PUCCH.Index0** to **N.UL.SINR.PUCCH.Index7**) may change.

- The number of UEs scheduled in the uplink per TTI and the total number of UEs scheduled in the uplink in a cell may decrease.
- The average uplink cell throughput (specified by **Cell Uplink Average Throughput (DU)**) may decrease.
- The values of counters related to CCE usage may change.
- The total duration of uplink data transmission at the RLC layer in a cell (**N.RLC.ThpTime.UL.Cell**) and the duration of uplink data transmission at the RLC layer in a cell (**N.ThpTime.UL**) may increase.
- The values of **N.PRB.PUSCH.Used.Avg**, **N.ChMeas.MIMO.UL.Pair.PRB**, **N.ChMeas.MIMO.UL.Pair.Layer**, and counters related to uplink interference may change.
- Reconfiguring the switch for positioning-dedicated SRS transmission in uplink slots (specified by the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter) will cause the cell to restart.

After this feature is activated and the LCS client initiates a positioning request, neighboring cells will assist in measurements and create UE measurement instances. As a result, the value of **VS.NRBoard.User.Max** increases.

7.1.3 Requirements

7.1.3.1 Licenses

The Precise Positioning (NR) feature requires that the following license be purchased and activated.

Feature ID	Feature Name	Model	Sales Unit
FOFD-070210	Precise Positioning (NR) (trial)	NR0S00H PPU00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.1.3.2.1 Prerequisite Functions

Prerequisite functions of U-TDOA- and AoA-based high-precision positioning

Function Name	Function Switch	Reference	Description
Cell networking mode	**NRDUCell**.***NrDuCellNetworkingMode***	None	When the **NRDUCellPosAlgo**.***ApplicationScenario*** parameter is set to **NORMAL_SCENARIO**, the **NRDUCell**.***NrDuCellNetworkingMode*** parameter can only be set to **NORMAL_CELL**.
Positioning-dedicated SRS transmission in uplink slots	**SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter	*Positioning*	U-TDOA- and AoA-based high-precision positioning requires positioning-dedicated SRS transmission in uplink slots to be enabled.

Prerequisite function of positioning-dedicated SRS transmission in uplink slots

Function Name	Function Switch	Reference	Description
UE-characteristic-based SRS period adaptation	**USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	*Channel Management*	Positioning-dedicated SRS transmission in uplink slots requires UE-characteristic-based SRS period adaptation to be enabled.

Prerequisite functions of the inter-base-station-transmission-based beacon-free function

Function Name	Function Switch	Reference	Description
U-TDOA- and AoA-based high-precision positioning	**AOA_UTDOA_HYBRID_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	The inter-base-station-transmission-based beacon-free function requires U-TDOA- and AoA-based high-precision positioning to be enabled.

Function Name	Function Switch	Reference	Description
Inter-RRU channel calibration for distributed massive MIMO	**DM_MIMO_INTER_RRU_CHN_CALIB_SW** option of the **NRDUCellAlgoSwitch**.***DmMimoSwitch*** parameter	*Distributed Antenna Solutions (Low-Frequency TDD)*	The inter-base-station-transmission-based beacon-free function requires "inter-RRU channel calibration for distributed massive MIMO" to be enabled.
Transmit and Receive Mode	**NRDUCellTrp**.***TxRxMode*** set to **8T8R**	None	The inter-base-station-transmission-based beacon-free function supports only 8T8R.

7.1.3.2.2 Mutually Exclusive Functions

Mutually exclusive functions of U-TDOA- and AoA-based high-precision positioning

Function Name	Function Switch	Reference	Description
Location Name	**LCSENGINEERINGPARA**.***LOCATIONNAME***	None	Either **Location Name** or **Map ID** (controlled by **LCSENGINEERINGPARA**.***MAPID***) must be set to a valid value.
SRS field strength-based positioning	**SRS_RSRP_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	If the **SRS_RSRP_LOCATION_SW** or **UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter is selected for a cell, U-TDOA- and AoA-based high-precision positioning cannot be enabled for other cells served by the same base station.
U-TDOA-based positioning	**UTDOA_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	
Multi-cell SRS measurement	**MULTI_CELL_SRS_MEAS_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	None	Multi-cell SRS measurement cannot work with U-TDOA- and AoA-based high-precision positioning.

Function Name	Function Switch	Reference	Description
LCS Scenario	**NRDUCellPosAlgo**.***LcsScenario***	None	When U-TDOA- and AoA-based high-precision positioning is enabled, the **NRDUCellPosAlgo**.***LcsScenario*** parameter can only be set to **SCENARIO_1**, **SCENARIO_2**, **SCENARIO_8**, **SCENARIO_100**, or **SCENARIO_145**.
Intra-frequency beam statistics	**INTRA_FREQ_BEAM_STAT_SW** option of the **NRDUCellCollabServ**.***NeighborNrCellMgmtSw*** parameter	None	Intra-frequency beam statistics cannot work with U-TDOA- and AoA-based high-precision positioning.

Mutually exclusive functions of positioning-dedicated SRS transmission in uplink slots

Function Name	Function Switch	Reference	Description
User Service Type	**gNBRfspConfig**.***UserServiceType***	None	● When the **gNBRfspConfig**.***RfspIndex*** parameter is set to **0**, the corresponding **gNBRfspConfig**.***UserServiceType*** parameter must be set to **DS**. ● When positioning-dedicated SRS transmission in uplink slots is enabled for a cell served by a base station, the **gNBRfspConfig**.***UserServiceType*** parameter cannot be set to **DS_LCS** for the base station.
Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	*Hyper Cell*	Positioning-dedicated SRS transmission in uplink slots cannot work in hyper cells.

Function Name	Function Switch	Reference	Description
LTE and NR Uplink Spectrum Sharing	**LTE_NR_UL_SPECTRUM_SHARING_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	None	LTE and NR Uplink Spectrum Sharing cannot work with positioning-dedicated SRS transmission in uplink slots.
PUSCH scheduling in self-contained slots	**PUSCH_SCH_IN_S_SLOT_SW** option of the **NRDUCellPusch**.***UlPuschAlgoSwitch*** parameter	None	PUSCH scheduling in self-contained slots cannot work with positioning-dedicated SRS transmission in uplink slots.
SRS period shortening for downlink 1T4R UEs	**DL_1T4R_SRS_PERIOD_SHORTEN_SW** option of the **NRDUCellPdschPrecode**.***SuMimoSrsPeriodShortSw*** parameter	None	SRS period shortening for downlink 1T4R UEs cannot work with positioning-dedicated SRS transmission in uplink slots.
SRS period shortening for downlink 2T4R UEs	**DL_2T4R_SRS_PERIOD_SHORTEN_SW** option of the **NRDUCellPdschPrecode**.***SuMimoSrsPeriodShortSw*** parameter	None	SRS period shortening for downlink 2T4R UEs cannot work with positioning-dedicated SRS transmission in uplink slots.
SRS period shortening for downlink 1T2R UEs	**DL_1T2R_SRS_PRD_SHORTEN_SW** option of the **NRDUCellPdschPrecode**.***SuMimoSrsPeriodShortSw*** parameter	None	SRS period shortening for downlink 1T2R UEs cannot work with positioning-dedicated SRS transmission in uplink slots.
Speed turbo UE guarantee	**ST_UE_GUARANTEE_SW** option of the **NRCellAlgoSwitch**.***SpeedTurboSw*** parameter	None	Speed turbo UE guarantee cannot work with positioning-dedicated SRS transmission in uplink slots.
SRS interference coordination based on self-contained and uplink slots	**SRS_INTRF_COORD_S_U_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Channel Management*	SRS interference coordination based on self-contained and uplink slots cannot work with positioning-dedicated SRS transmission in uplink slots.

Function Name	Function Switch	Reference	Description
IAB Deployment	**NRDUCELL**.***DeployPosition*** set to **DEPLOY_IAB**	None	If the local cell is deployed in IAB mode, positioning-dedicated SRS transmission in uplink slots cannot be enabled.
Uplink narrowband interference avoidance	**UL_NARROWBAND_INTRF_AVOID** option of the **NRDUCellIntrfIdent**.***IntrfOptimizationMethod*** parameter	None	Uplink narrowband interference avoidance cannot work with positioning-dedicated SRS transmission in uplink slots.
SRS capability optimization	**SRS_CAPB_OPT_SW** option of the **NRDUCellSrs**.***SrsAlgoSwitch*** parameter	None	SRS capability optimization cannot work with positioning-dedicated SRS transmission in uplink slots.
Light-load time sequence optimization	**LIGHT_LOAD_TIME_SEQ_OPT_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter	None	Light-load time sequence optimization cannot work with positioning-dedicated SRS transmission in uplink slots.
Differentiated eMBB QoS	**EMBB_QOS_DIFF_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter	None	Differentiated eMBB QoS cannot work with positioning-dedicated SRS transmission in uplink slots.
Bi-Carrier	**BI_CARRIER_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	None	When the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is selected, Bi-Carrier cannot be enabled.
SRS resource allocation for UEs performing downlink large-packet services	**DL_BIG_PKT_SRS_RES_ALLOC_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	None	When the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is selected, SRS resource allocation for UEs performing downlink large-packet services cannot be enabled.

Function Name	Function Switch	Reference	Description
Uplink symbol power saving	**UL_SYMBOL_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch**.***PowerSavingSwitch*** parameter	*Energy Conservation and Emission Reduction*	Uplink symbol power saving cannot work with positioning-dedicated SRS transmission in uplink slots.
Intra-frequency beam statistics	**INTRA_FREQ_BEAM_STAT_SW** option of the **NRDUCellCollabServ**.***NeighborNrCellMgmtSw*** parameter	None	Intra-frequency beam statistics cannot work with positioning-dedicated SRS transmission in uplink slots.
Intra-frequency assistant cell	**NRDUCell**.***SupplementaryCellInd*** set to **INTRA_FREQ_ASSISTANT_CELL**	None	Positioning-dedicated SRS transmission in uplink slots cannot work in intra-frequency assistant cells.
Flexible spectrum access	**FLEX_SPCT_ACCESS_SW** option of the **NRDUCellCarrMgmt**.***MultiBandFlexSchSwitch*** parameter	*Uplink Flexible Spectrum Scheduling*	Positioning-dedicated SRS transmission in uplink slots cannot work with flexible spectrum access.
Inter-FR CA coverage enhancement	**PUCCH_GROUP_ADAPT_SW** option of the **NRDUCellCarrMgmt**.***InterFrCaEnhSw*** parameter	*Carrier Aggregation*	Positioning-dedicated SRS transmission in uplink slots cannot work with inter-FR CA coverage enhancement.
TDD-only uplink intra-FR1 inter-band CA	**FR1_INTER_BAND_TT_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	None	Positioning-dedicated SRS transmission in uplink slots cannot work with TDD-only uplink intra-FR1 inter-band CA.
Wideband SRS measurement collection	**WIDEBAND_SRS_COLLECT_SW** option of the **NRCellFeatureSwitch**.***CellMeasCollectSwitch*** parameter	None	Positioning-dedicated SRS transmission in uplink slots cannot work with wideband SRS measurement collection.
Downlink avoidance enhancement mode 2	**DL_AVOID_ENH_MODE2_SW** option of the **NRDUCellTimeIntrf**.***TimeIntrfAvoidSw*** parameter	None	Positioning-dedicated SRS transmission in uplink slots cannot work with downlink avoidance enhancement mode 2.

Function Name	Function Switch	Reference	Description
PUSCH coordinated power control	**PUSCH_COORD_PWR_CTRL_SW** option of the **NRDUCellUlPcConfig**.***UlPwrCtrlAlgoExtSwitch*** parameter	*Massive MIMO Uplink Boosting (Low-Frequency TDD)*	PUSCH coordinated power control cannot work with positioning-dedicated SRS transmission in uplink slots.
SRS and PUSCH/PUCCH concurrency compatibility	**SRS_PUSCH_CONCURRENCY_COMPT_SW** option of the **NRDUCellCarrMgmt**.***CaCompatibilitySwitch*** parameter	*Carrier Aggregation*	When the SRS and PUSCH/PUCCH concurrency compatibility function is enabled, positioning-dedicated SRS transmission in uplink slots cannot be enabled together with proactive avoidance of remote interference.
Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	

7.1.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
MetaAAU beam sensing	**NRDUCellBeamAlgo**.***BeamPerceiveMode*** set to **DISTRIBUTED_MODE**	*MIMO (TDD)*	MetaAAU beam sensing does not take effect when U-TDOA- and AoA-based high-precision positioning is in effect.
Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving**.***BwpPwrSavingSw*** parameter	*UE Power Saving*	Power saving BWP does not take effect for UEs with U-TDOA- and AoA-based high-precision positioning in effect.
DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	DRX does not take effect for UEs with U-TDOA- and AoA-based high-precision positioning in effect.

Function Name	Function Switch	Reference	Description
Proactive avoidance of remote interference	**SRS_RIM_INTRF_AVOID_SW** and **SRS_INTRF_COORD_S_SLOT_SW** options of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter	*Remote Interference Management (Low-Frequency TDD)*	If the **SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is selected and the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter is selected, the **SRS_INTRF_COORD_S_SLOT_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be selected.
Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, downlink intra-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.
Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, downlink intra-FR inter-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.
Inter-gNodeB CA	**INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*CA Experience Improvement*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, inter-gNodeB CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.

Function Name	Function Switch	Reference	Description
Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, uplink intra-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.
Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	If the **COMMON_USER_LCS_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter is deselected, uplink intra-FR inter-band CA does not take effect for positioning service UEs for which the **gNBRfspConfig**.***UserServiceType*** parameter is set to LCS.
CA SRS carrier switching	**SRS_CARRIER_SWITCHING_SW** option of the **NRDUCellCarrMgmt**.***CaEnhancedAlgoSwitch*** parameter	*CA Experience Improvement*	When both positioning-dedicated SRS transmission in uplink slots and CA SRS carrier switching are enabled: ● The setting of the **SRS_RIM_INTRF_AVOID_SW** option of the **NRDUCellSrs**.***SrsAlgoExtSwitch*** parameter must be the same between the PCC and SCCs. Otherwise, CA SRS carrier switching does not take effect. ● If PUCCH format 3 resources are insufficient, the **User Downlink Average Throughput (DU)** slightly decreases.

Function Name	Function Switch	Reference	Description
Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When both positioning-dedicated SRS transmission in uplink slots and intra-base-station UL CoMP are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, UL CoMP cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability and "positioning-dedicated SRS transmission in uplink slots" is enabled for the coordinating neighboring cell, UL CoMP can take effect for the UE and the performance gains are not affected. ● If a UE supports the positioning-dedicated SRS transmission capability and "positioning-dedicated SRS transmission in uplink slots" is not enabled for the coordinating neighboring cell, UL CoMP can take effect for the UE, but the performance gains decrease.
Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	When both positioning-dedicated SRS transmission in uplink slots and intra-base-station DL CoMP are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, DL CoMP cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability, DL CoMP can take effect for the UE.

Function Name	Function Switch	Reference	Description
Inter-cell interference avoidance	**INTER_CELL_INTRF_AVOID_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Massive MIMO iBeam (Low-Frequency TDD)*	When both positioning-dedicated SRS transmission in uplink slots and inter-cell interference avoidance are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, inter-cell interference avoidance cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability, inter-cell interference avoidance can take effect for the UE.
Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	When both positioning-dedicated SRS transmission in uplink slots and Co-MM are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, Co-MM cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability, Co-MM can take effect for the UE.
Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	When both positioning-dedicated SRS transmission in uplink slots and coordinated interference management are enabled: ● If a UE does not support the positioning-dedicated SRS transmission capability, coordinated interference management cannot take effect for the UE. ● If a UE supports the positioning-dedicated SRS transmission capability, coordinated interference management can take effect for the UE.

Function Name	Function Switch	Reference	Description
Super uplink	**SUPER_UL_TDM_SCH_SW** and **FLEX_SUPER_UPLINK_SW** options of the **NRDUCellAlgoSwitch**.***SuperUplinkSwitch*** parameter	*Super Uplink*	When both positioning-dedicated SRS transmission in uplink slots and super uplink are enabled, the SUL cells must meet the SUL cell specifications of the corresponding baseband processing unit. Otherwise, positioning-dedicated SRS transmission in uplink slots cannot take effect. For details about the SUL cell specifications of different baseband processing units, see the table content where the **Frequency Band Type of a Cell** column includes **SUL** in **Product Specifications** > **BBU Technical Specifications** > **Technical Specifications of Baseband Processing Units** > **NR Technical Specifications of Baseband Processing Units** > **NR TDD Technical Specifications of Baseband Processing Units** in *3900 & 5900 Series Base Station Product Documentation*.

7.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

U-TDOA- and AoA-based high-precision positioning

- All NR TDD-capable main control boards support this function.
- NR TDD-capable baseband processing units: UBBPg3 and UBBPg3a

RF Modules

U-TDOA- and AoA-based high-precision positioning

- Massive MIMO modules: 32T32R or 64T64R low-frequency TDD modules
- Easy Macro modules: 8T8R low-frequency TDD modules (Easy Macro modules support only TOA measurement.)
- RRU modules: 2T2R, 2T4R, 4T4R, or 8T8R low-frequency TDD modules (RRU modules support only TOA measurement.)

7.1.3.4 Cells

For U-TDOA- and AoA-based high-precision positioning, the cell configuration must meet the following requirements:

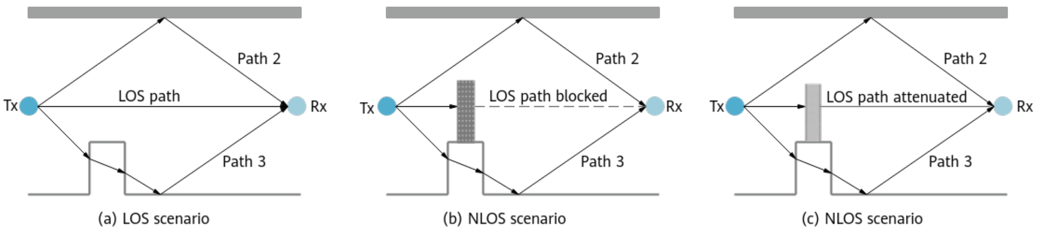
- **NRDUCell**.***UlBandwidth*** must be set to **CELL_BW_100M**.
- The same frequency band and slot configuration need to be configured for the cells covering the positioning area. In addition, the frequency band and slot configuration must meet the following requirements:
  - **NRDUCell**.***FrequencyBand*** must be set to **N40**, **N41**, **N77**, **N78**, or **N79**.
  - **NRDUCell**.***SlotAssignment*** must be set to **8_2_DDDDDDDSUU** or **8_2_DDDSUUDDDD**.
- The cells must be non-LampSite cells (with **NRDUCell**.***LampSiteCellFlag*** set to a value other than **YES**).
- The cells are not in SUL mode (with **NRDUCell**.***DuplexMode*** set to a value other than **CELL_SUL**).

7.1.3.5 Networking

The accuracy of precise positioning is related to the proportion of LOS transmissions.

- In the LOS scenario shown in (a) of **Figure 7-5**, there is an unobstructed direct path between the TX and RX ends.
- In the NLOS scenarios shown in (b) and (c) of **Figure 7-5**, the direct path between the TX and RX ends is completely blocked, or signals on the direct path can penetrate the obstacle but arrive at the RX end with diminished strength. As such, the positioning accuracy decreases in NLOS scenarios.

Figure 7-5 LOS and NLOS scenarios



To ensure the positioning accuracy:

- The receive time must be precisely synchronized among TRPs. Therefore, the TRPs used for positioning must be deployed on the same gNodeB.
- For U-TDOA- and AoA-based high-precision positioning, the propagation environment between a UE and three or more TRPs must meet the LOS conditions.

- To ensure the accuracy of TOA/AoA measurement results, the RSRP of SRS received by each TRP from the UE must be greater than –105 dBm, and the SRS SINR must be greater than 0 dB.

The inter-base-station-transmission-based beacon-free function is supported only in distributed massive MIMO scenarios and requires LOS transmission. It is recommended that sites be deployed at a height of 6 m to 15 m with an ISD of 50 m and a downtilt of 6° to 12°.

In addition, TRP networking distribution needs to be considered during network planning and deployment. After the site is deployed, accurate TRP deployment positions need to be measured, and then configured on the LMF. For details about the positioning network planning, simulation, and design, as well as the TRP installation position measurement, contact Huawei service engineers.

UE positioning may fail in multi-cell-ID scenarios. Therefore, precise positioning cannot apply to multi-cell-ID scenarios.

7.1.3.6 Others

Before U-TDOA- and AoA-based high-precision positioning is enabled:

- UEs must support the NR SA networking.
- The AMF and LMF versions must be V24.1.0 or later.
- If UEs capable of sending SRSs for positioning need to be positioned, the core network needs to support parsing of the *Positioning SRS* IE.

Before the inter-base-station-transmission-based beacon-free function is enabled, TRP coordinates need to be configured on the base station side.

7.1.4 Operation and Maintenance

7.1.4.1 Data Configuration

7.1.4.1.1 Data Preparation

Table 7-1, **Table 7-2**, and **Table 7-3** describe the parameters used for function activation. No parameters are involved in function optimization.

Table 7-1 Parameter used for activation (positioning TRP ID function)

Parameter Name	Parameter ID	Option	Setting Notes
Positioning TRP ID Switch	**gNodeBAlgo**.***PosTrpIdSwitch***	None	It is recommended that the positioning TRP ID function be enabled.

Table 7-2 Parameters used for activation (U-TDOA- and AoA-based high-precision positioning)

Parameter Name	Parameter ID	Option	Setting Notes
User Service Type	**gNBRfspConfig**.***UserServiceType***	None	● For UEs mainly performing positioning services: – If they are common UEs (non-low-power-consumption positioning service UEs), this parameter can be set to **LCS**. In this way, SRSs are transmitted over the full bandwidth to ensure high positioning accuracy. – If they are low-power-consumption positioning service UEs, this parameter needs to be set to **LPHAP_LCS** and full-bandwidth SRS resources for positioning are configured for UEs in inactive state to ensure LPHAP resources. ● For UEs mainly performing data services, this parameter can be set to **DS**, so that the traditional SRS resource allocation algorithm is used

Parameter Name	Parameter ID	Option	Setting Notes
			for SRS resource allocation.
SRS for Positioning in UL Slot Switch	**NRDUCellPosAlgo**.***LcsMeasureAlgoSw***	SRS_FOR_POSITION_ON_USLOT_SW	Select this option if U-TDOA- and AoA-based high-precision positioning is required.
LCS Measure Algorithm Switch	**NRDUCellPosAlgo**.***LcsMeasureAlgoSw***	COMMON_USER_LCS_SW	Select this option for a common UE (non-low-power-consumption positioning service UE) not configured with an RFSP index during registration.
LCS Switch	**NRCellAlgoSwitch**.***LcsSwitch***	AOA_UTDOA_HYBRID_LOCATION_SW	Select this option if U-TDOA- and AoA-based high-precision positioning is required.
Application Scenario	**NRDUCellPosAlgo**.***ApplicationScenario***	None	Set this parameter to **NORMAL_SCENARIO**.

Table 7-3 Parameters used for activation (inter-base-station-transmission-based beacon-free function)

Parameter Name	Parameter ID	Option	Setting Notes
Inter-TRP Transmission Max Distance	**NRDUCellPosAlgo**.***InterTrpTransMaxDistance***	None	It is recommended that this parameter be set to two times the ISD.
Distributed MIMO Switch	**NRDUCellAlgoSwitch**.***DmMimoSwitch***	DM_MIMO_INTER_RRU_CHN_CALIB_SW	Select this option before enabling the inter-base-station-transmission-based beacon-free function.

Parameter Name	Parameter ID	Option	Setting Notes
Inter-TRP Transmission Switch	**NRDUCellPosAlgo**.***InterTrpTransmissionSwitch***	None	It is recommended that this parameter be set to **ON** when inter-TRP transmission is required for improving the positioning accuracy and eliminating the need of beacon deployment.

7.1.4.1.2 Using MML Commands

Before using MML commands, refer to **7.1.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- If RFSP indexes have been configured during UE registration, the **gNBRfspConfig**.***UserServiceType*** parameter can be configured based on RFSP indexes. The following gives an example of configuration:
  a. Planning by customers based on actual service requirements: Assume that UEs in campus A have three types of services. Three UE types need to be planned: Type-I UEs perform only data services. Type-II UEs support non-LPHAP services and a small number of data services. Type-III UEs support LPHAP services.
  b. SIM card registration with the operator: According to the preceding requirements, customers in campus A register SIM cards with the operator, and the operator assigns three **gNBRfspConfig**.***RfspIndex*** values (for example, **100**, **101**, and **102**) which are configured in the users' registration information, to distinguish the three UE types.
  c. SIM card-service mapping by customers: After obtaining registered SIM cards from the operator, the customers in campus A determine and maintain the mapping between the three types of SIM cards and three types of services. For example, **100** is allocated to the UEs performing only data services, **101** to the UEs performing positioning services, and **102** to the UEs performing services of low power enhancement for positioning.

d. Adding MML-based configurations on the gNodeB: After obtaining the mapping between SIM cards and services, network maintenance engineers run MML commands and configure the **gNBRfspConfig**.***UserServiceType*** parameter based on the service types planned for the three types of SIM cards (with different **gNBRfspConfig**.***RfspIndex*** values). (The default value of **gNBRfspConfig**.***UserServiceType*** is **DS**. If U-TDOA- and AoA-based high-precision positioning is enabled, the value of **gNBRfspConfig**.***UserServiceType*** must be **LCS** or **LPHAP_LCS**.) The MML-based configuration examples are as follows:

```
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=100, UserServiceType=DS;
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=101, UserServiceType=LCS;
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=102, UserServiceType=LPHAP_LCS;
```

- If a common UE (non-low-power-consumption positioning service UE) is not configured with an RFSP index during registration, turn on the common UE positioning switch:

```
//Enabling positioning-dedicated SRS transmission in uplink slots
MOD NRDUCELLPOSALGO: NrDuCellId=1, LcsMeasureAlgoSw=SRS_FOR_POSITION_ON_USLOT_SW-1;
//Turning on the common UE positioning switch
MOD NRDUCELLPOSALGO: NrDuCellId=1, LcsMeasureAlgoSw=COMMON_USER_LCS_SW-1;
```

- Positioning TRP ID function

a. Find the RRU or port associated with an engineering parameter ID based on the engineering parameter configuration of the positioning service.

b. Find the **SectorEqmId** corresponding to the RRU or port based on **MO SECTOREQM**.

c. Find the **NrDuCellTrpId** and **NrDuCellCoverageId** corresponding to **SectorEqmId** based on **MO NRDUCELLCOVERAGE**.

d. Configure a positioning TRP ID (controlled by **NRDUCellLcsProperty**.***PosTrpId***) for every combination of **NrDuCellTrpId**, **NrDuCellCoverageId**, and engineering parameter ID. The positioning TRP ID ranges from 1 to 65535 and is unique within a gNodeB. This parameter takes effect only after the positioning TRP ID function is enabled. The following is an MML command example:

```
//Enabling the positioning TRP ID function
MOD GNODEBALGO: PosTrpIdSwitch=ON;
```

- U-TDOA- and AoA-based high-precision positioning function

```
//Adding gNodeB RFSP configurations for UEs requiring positioning services with UserServiceType set to LCS to ensure SRS resource allocation for positioning services
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=10, UserServiceType=LCS;
//Enabling positioning-dedicated SRS transmission in uplink slots
MOD NRDUCELLPOSALGO: NrDuCellId=1, LcsMeasureAlgoSw=SRS_FOR_POSITION_ON_USLOT_SW-1;
//Enabling U-TDOA- and AoA-based high-precision positioning
MOD NRCELLALGOSWITCH: NrCellId=1, LcsSwitch=AOA_UTDOA_HYBRID_LOCATION_SW-1;
//Configuring an application scenario
MOD NRDUCELLPOSALGO: NrDuCellId=1, ApplicationScenario=NORMAL_SCENARIO;
```

- Inter-base-station-transmission-based beacon-free function

```
//Turning on the switch for inter-RRU channel calibration for distributed massive MIMO
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
DmMimoSwitch=DM_MIMO_INTER_RRU_CHN_CALIB_SW-1;
//Configuring the maximum inter-TRP transmission distance
MOD NRDUCELLPOSALGO: NrDuCellId=1, InterTrpTransMaxDistance=100;
//Turning on the inter-TRP transmission switch
MOD NRDUCELLPOSALGO: NrDuCellId=1, InterTrpTransmissionSwitch=ON;
```


Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Positioning TRP ID function
```
//Disabling the positioning TRP ID function
MOD GNODEBALGO: PosTrpIdSwitch=OFF;
```
- Inter-base-station-transmission-based beacon-free function
```
//Turning off the inter-TRP transmission switch
MOD NRDUCELLPOSALGO: NrDuCellId=1, InterTrpTransmissionSwitch=OFF;
```
- U-TDOA- and AoA-based high-precision positioning function
```
//Disabling U-TDOA- and AoA-based high-precision positioning
MOD NRCELLALGOSWITCH: NrCellId=1, LcsSwitch=AOA_UTDOA_HYBRID_LOCATION_SW-0;
```

7.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.1.4.2 Activation Verification

Activation of Precise Positioning (NR) can be verified by tracing signaling messages or measuring performance counters.

Based on signaling tracing:

Step 1 On the MAE-Access, choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**.

Step 2 In the navigation tree on the left, choose **NG Interface Trace**.

Step 3 Enable the LCS client to initiate a positioning request. After the AMF processes the request, the LMF triggers the U-TDOA- and AoA-based high-precision positioning procedure.

Step 4 On the **NG Interface Trace** page, check for the following messages:

- A message that contains POSITIONING INFORMATION REQUEST or MEASUREMENT REQUEST sent from the AMF to the gNodeB
- A message that contains POSITIONING INFORMATION RESPONSE, MEASUREMENT RESPONSE, or MEASUREMENT REPORT sent from the gNodeB to the AMF

  If these messages can be traced, Precise Positioning (NR) has been successfully activated.

----End

Based on performance counters:

Step 1 On the MAE-Access, start tasks for measuring counters related to **Measurement of gNodeB Performance** and **Measurement of Cell Performance**, including **N.Positioning.Info.Req.Att**, **N.Positioning.Info.Req.Succ**,

N.Positioning.Meas.Req.Att, **N.Positioning.Meas.Req.Succ**, and **N.Positioning.Meas.Rpt**.

Step 2 Enable the LCS client to initiate a positioning request during the measurement period. After the AMF processes the request, the LMF triggers the U-TDOA- and AoA-based high-precision positioning procedure.

Step 3 Check the counter values.

1. In on-demand positioning mode, if the values of the **N.Positioning.Info.Req.Att**, **N.Positioning.Info.Req.Succ**, **N.Positioning.Meas.Req.Att**, and **N.Positioning.Meas.Req.Succ** counters are not 0, U-TDOA- and AoA-based high-precision positioning has been activated.
2. In periodic positioning mode, if the values of the **N.Positioning.Info.Req.Att**, **N.Positioning.Info.Req.Succ**, **N.Positioning.Meas.Req.Att**, **N.Positioning.Meas.Req.Succ**, and **N.Positioning.Meas.Rpt** counters are not 0, Precise Positioning (NR) has been activated.

----End

7.1.4.3 Network Monitoring

After Precise Positioning (NR) is activated, the performance can be monitored by using the following counters.

Counter ID	Counter Name	Counter Description
1911829380	**N.Positioning.Info.Req.Att**	Number of positioning information requests
1911829379	**N.Positioning.Info.Req.Succ**	Number of positioning information responses
1911829377	**N.Positioning.Meas.Req.Att**	Number of positioning measurement requests
1911829376	**N.Positioning.Meas.Req.Succ**	Number of successful positioning measurements
1911829375	**N.Positioning.Meas.Rpt**	Number of positioning measurement reporting times
1911842612	**N.User.Positioning.Avg**	Average number of UEs performing positioning services
1911840425	**N.User.Positioning.LPHAP.Avg**	Average number of UEs performing LPHAP services

When a large number of positioning requests are initiated simultaneously, flow control may be triggered, causing NRPPa messages to be discarded. The performance can be monitored by using the following counter.

Counter ID	Counter Name	Counter Description
1911829378	**N.NRPPa.Ng.Msg.Disc.FlowCtrl**	Number of times NRPPa messages over the NG interface are discarded due to flow control

7.2 LPHAP

7.2.1 Principles

7.2.1.1 LPHAP

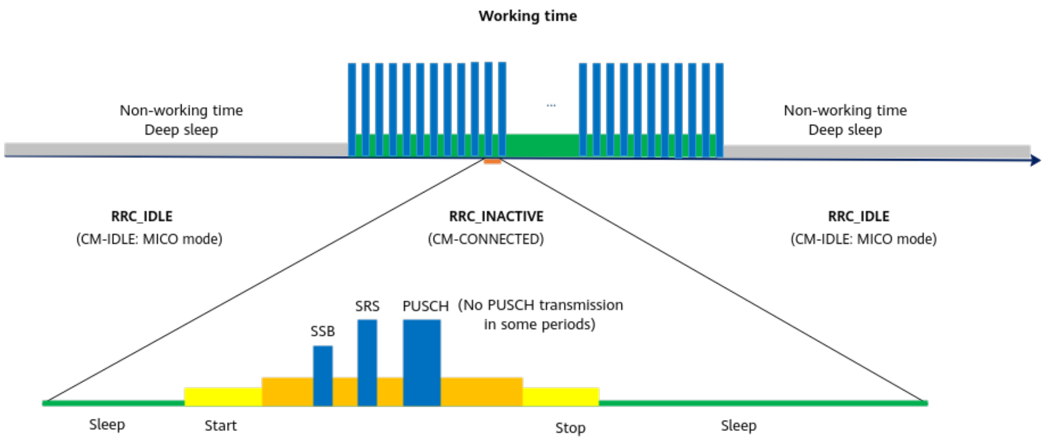
Positioning services in campuses/factories—such as valuable assets positioning—require a certain level of positioning accuracy and necessitate positioning tags powered by batteries that meet battery life requirements. Therefore, LPHAP is introduced.

LPHAP is controlled by the **LPHAP_REDCAP_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter. After this function takes effect, the following operations are performed for a RedCap UE:

- The LMF triggers periodic U-TDOA-based positioning on the UE.
- The UE needs to set up an RRC connection only once. After obtaining the PUSCH and SRS resources allocated by the gNodeB for LPHAP, the UE sends SRSs for positioning and performs PUSCH small data transmission (SDT) in the RRC_INACTIVE state, and then enters the sleep state.
- The gNodeB measures SRSs for positioning and reports the measurement results for location calculation by the LMF.
- After the next positioning period starts, the UE reads SSB information, completes time synchronization, sends positioning information, and then enters the sleep state again. This positioning procedure repeats periodically.

This function saves power by keeping the UE in the sleep state most of the time as shown in **Figure 7-6**.

Figure 7-6 LPHAP



During LPHAP, SRS bandwidth resources are selected for positioning services based on the user service type (**gNBRfspConfig**.***UserServiceType***). For UEs performing LPHAP services, this parameter needs to be set to **LPHAP_LCS** so that SRS resources for positioning are used to ensure positioning accuracy.

This function is recommended in 5GtoB scenarios where the geographical locations of low-power-consumption UEs need to be determined by measuring radio signals on the 5G network or where operators need to support the services that provide locations of low-power-consumption UEs and have certain low-power consumption requirements. This function saves UE power.

7.2.1.2 LPHAP Procedure

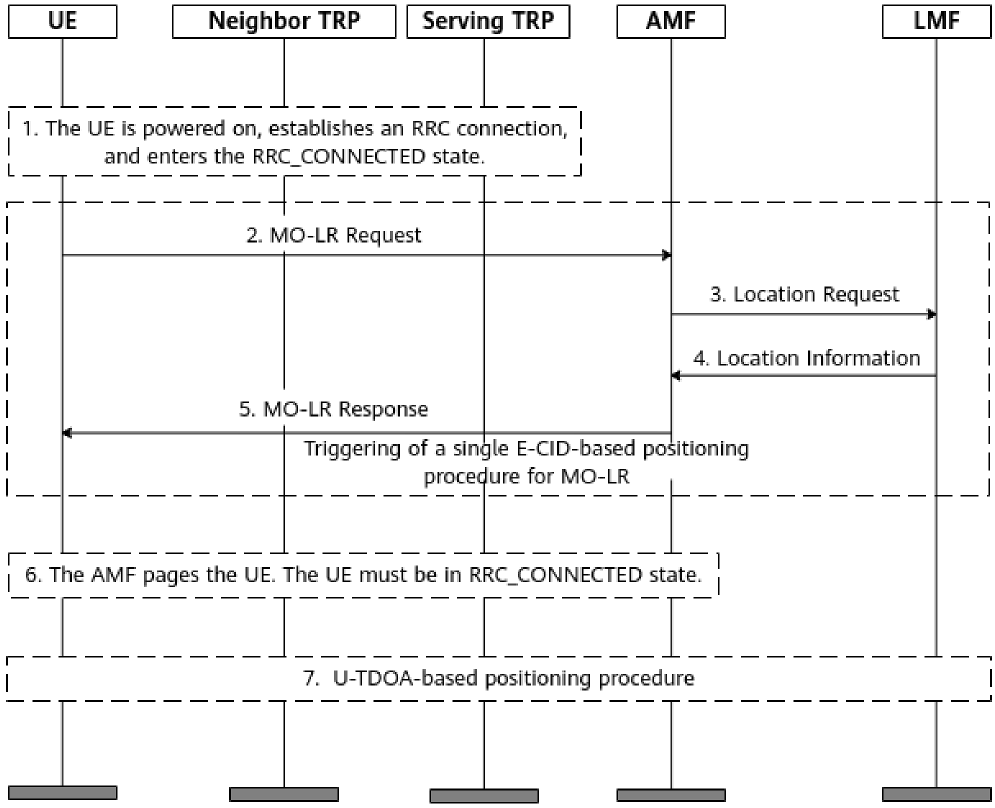
The overall LPHAP procedure is as follows:

1. A UE accesses the network after being powered on, establishes an RRC connection, and enters the RRC_CONNECTED state.
2. The UE sends a NAS message to trigger the positioning procedure as shown in **Figure 7-7**.
3. If the positioning server (LCS client) initiates a UE positioning subscription request, the U-TDOA-based positioning procedure in **Figure 7-8** is started. In addition, if data of sensors (such as battery level, barometric pressure, and motion sensors) is also subscribed to, the periodic procedure of positioning and sensing data reporting in **Figure 7-9** is started.

Positioning Triggering Procedure

- **Figure 7-7** shows how the positioning procedure is triggered.

Figure 7-7 Positioning triggering procedure



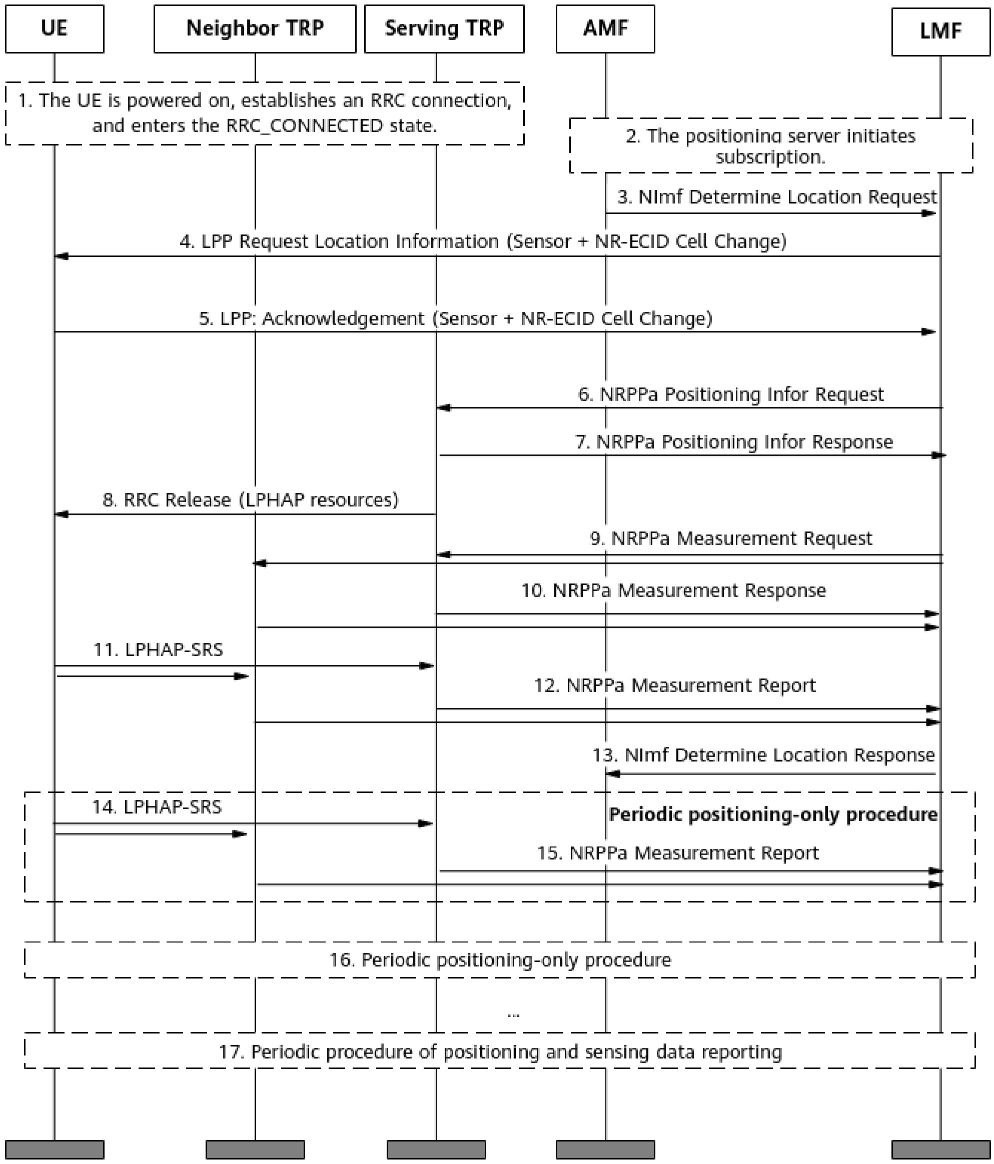
The procedure is described as follows:

a. A UE accesses the network after being powered on, establishes an RRC connection, and enters the RRC_CONNECTED state.
b. The UE sends a NAS message to trigger a single E-CID-based positioning procedure for MO-LR. The message type is **deferred** (LPHAP). An LPP message is embedded in the message to carry key parameters such as the UE capability, positioning period, and number of positioning times.
c. The AMF initiates a positioning request to the LMF. The message contains the GMLC address and is embedded with the LPP message in the MO-LR. The serving cell ID is carried. The LMF does not need to send a measurement request.
d. The LMF calculates the UE's location and reports the location to the GMLC and LCS client through the AMF.
e. The AMF sends an MO-LR response to the UE.
f. The AMF pages the UE. In this case, the UE must be in RRC_CONNECTED mode. The UE determines whether to enter the airplane mode or trigger positioning again based on whether it receives a positioning subscription request message from the LCS client. If the UE has not received a subscription request message, it enters the airplane mode. Otherwise, the next step is performed.
g. If the positioning server initiates a UE positioning subscription request, the periodic U-TDOA-based positioning procedure is started.

Periodic U-TDOA-based Positioning Procedure

- Figure 7-8 shows the periodic U-TDOA-based positioning procedure.

Figure 7-8 Periodic U-TDOA-based positioning procedure



The procedure is described as follows:

a. A UE accesses the network after being powered on, establishes an RRC connection, and enters the RRC_CONNECTED state.

b. The positioning server initiates a UE positioning subscription request to the LMF.

c. The AMF sends a periodic positioning request to the LMF. The message contains the GMLC address.

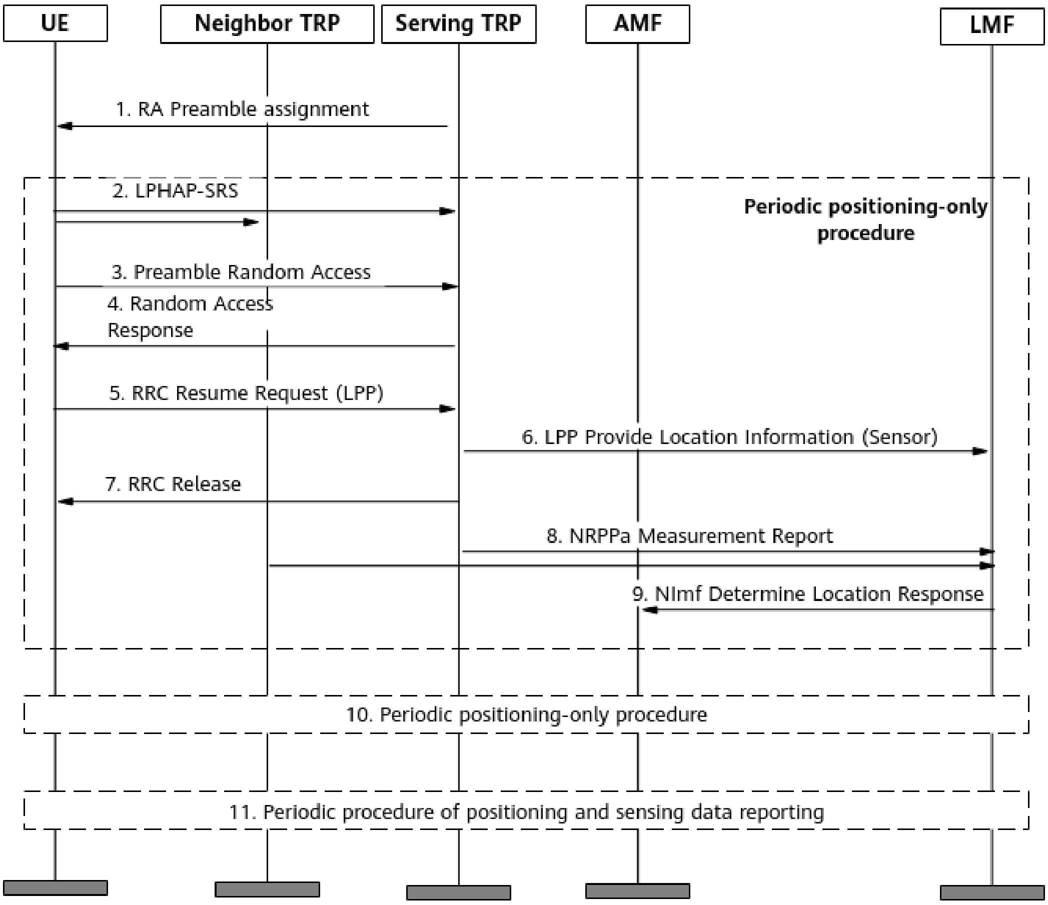
d. The LMF subscribes to periodic sensing data reporting from the UE based on the sensor capability supported by the UE. (If the UE does not have the sensor capability, the subscription is not performed.) The LMF subscribes to cell handover event reporting in NR E-CID-based positioning from the UE.

e. After receiving the periodic sensing data reporting subscription message, or receiving the cell handover event reporting subscription message of NR E-CID-based positioning, the UE returns an ACK message to the LMF. (The LPP message for periodic sensing data reporting subscription response is encapsulated in the LCS message. The LPP message is used for cell handover event reporting subscription response in NR E-CID-based positioning.)
f. The LMF applies to the serving cell of gNodeB for SRS resources for LPHAP and PUSCH resources for sensing data reporting. (If the UE does not have the sensor capability, PUSCH resources are not allocated.)
g. After resource allocation, the base station sends an NRPPa Positioning Infor Response message to the LMF to report the allocated resources.
h. After resource allocation, the base station sends an RRC Release message to the UE. The UE is released and enters the RRC_INACTIVE state.
i. The LMF instructs the serving cell and neighboring cell of the gNodeB to monitor the SRS at the corresponding location and perform TOA measurements.
j. After receiving the measurement request from the LMF, the serving cell and neighboring cell of the gNodeB respond to the request.
k. When the LPHAP-SRS transmission period starts, the UE in the RRC_INACTIVE state sends an LPHAP-SRS to the serving cell and neighboring cell of the gNodeB. The LPP message needs to be encapsulated in the LCS Event Report message for transmission.
l. After receiving the LPHAP-SRS from the UE, the serving cell and neighboring cell of the gNodeB perform TOA measurement and report the TOA measurement result to the LMF.
m. After calculating the UE's location, the LMF reports the location to the GMLC and LCS client through the AMF.
n. When the period of periodic positioning-only procedure starts, the UE sends an LPHAP-SRS to perform positioning.
o. After receiving the LPHAP-SRS from the UE, the serving cell and neighboring cell of the gNodeB perform TOA measurement and report the TOA measurement result to the LMF.
p. When the positioning period starts, steps 14 and 15 are repeated.
q. The periodic procedure of positioning and sensing data reporting is started.

Periodic Procedure of Positioning and Sensing Data Reporting

Figure 7-9 shows the periodic procedure of positioning and sensing data reporting (based on the random access-based small data transmission (RA-SDT) procedure).

Figure 7-9 Periodic procedure of positioning and sensing data reporting



The procedure is described as follows:

1. The gNodeB allocates dedicated preamble resources to a low-power-consumption UE, and a saving operation is performed on the UE.
2. The UE sends an LPHAP-SRS in the RRC_INACTIVE state.
3. The UE performs non-contention-based random access through dedicated preamble resources.
4. The gNodeB responds to the random access request of the UE and delivers TA information and PUSCH resources to the UE.
5. The UE sends an RRC Resume Request message to the gNodeB.
6. The message encapsulates the LPP message's sensing data and the gNodeB sends it to the LMF.
7. After determining that the UE has no data to transmit, the gNodeB sends an RRC Release message to the UE. Then, the UE is released and enters the RRC_INACTIVE state.
8. After receiving the LPHAP-SRS from the UE, the serving cell and neighboring cell of the gNodeB perform TOA measurement and report the TOA measurement result to the LMF.
9. After calculating the UE's location, the LMF reports the location to the GMLC and LCS client through the AMF.
10. When the LPHAP-SRS transmission period starts, the periodic positioning-only procedure is repeated.

11. When the LPHAP-SRS transmission period and the sensing data reporting period start, steps 1 to 9 are repeated.

7.2.2 Network Analysis

7.2.2.1 Benefits

This feature is recommended in 5GtoB campus scenarios if UEs' locations need to be determined by measuring radio signals through 5G networks or operators need to provide UE-location-based services.

Note that this feature involves the geographical locations of UEs during service provisioning or maintenance. As such, you must fully protect the personal data of users by complying with any and all applicable laws and user privacy policies in the concerned country or territory.

After U-TDOA-based positioning is enabled, activating the positioning-dedicated beacon-free function improves the positioning accuracy and eliminates the need to deploy beacon UEs.

7.2.2.2 Impacts

The impacts between this function and other functions are described in **7.2.3.2.3 Function Impacts**.

For details about network impacts, see **7.1.2.2 Impacts**.

7.2.3 Requirements

7.2.3.1 Licenses

The Precise Positioning (NR) feature requires that the following license be purchased and activated.

Feature ID	Feature Name	Model	Sales Unit
FOFD-070210	Precise Positioning (NR) (trial)	NR0S00H PPU00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.2.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
U-TDOA- and AoA-based high-precision positioning	**AOA_UTDOA_HYBRID_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	*Positioning*	When **NRDUCell**.***LampSiteCellFlag*** is set to **NO**, LPHAP depends on U-TDOA- and AoA-based high-precision positioning.
RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	LPHAP depends on RedCap Introduction.
Positioning-dedicated SRS transmission in uplink slots	**SRS_FOR_POSITION_ON_USLOT_SW** option of the **NRDUCellPosAlgo**.***LcsMeasureAlgoSw*** parameter	*Positioning*	LPHAP depends on positioning-dedicated SRS transmission in uplink slots.
RedCap identification strategy	**NRDUCellRedCap**.***RedCapIdentifyStrategy***	None	When LPHAP is enabled, **NRDUCellRedCap**.***RedCapIdentifyStrategy*** cannot be set to **DEDICATED_RO**.

7.2.3.2.2 Mutually Exclusive Functions

Mutually exclusive functions of LPHAP

Function Name	Function Switch	Reference	Description
Extended Cell Range (TDD)	**NRDUCell**.***CellRadius*** set to a value greater than 14500 m and less than or equal to 60000 m	*Extended Cell Range*	None
Extended Cell Range of 90 km (TDD)	**NRDUCell**.***CellRadius*** set to a value greater than 60000 m and less than or equal to 90000 m	*Extended Cell Range*	None

Function Name	Function Switch	Reference	Description
Inactive RA-SDT	**UE_PWR_SAVING_ENHANCE_SW** option of the **NRDUCellUePwrSaving**.***UePowerSavingSwitch*** parameter and **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter	*UE Power Saving*	If the **LPHAP_REDCAP_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter is selected, the **RRC_INACTIVE_RA_SDT_SW** option of the **NRCellAlgoSwitch**.***InactiveStrategySwitch*** parameter cannot be selected.
High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	For an NR TDD cell, when the **NRDUCell**.***CellRadius*** parameter is set to a value greater than or equal to **4610** and the **NRDUCell**.***HighSpeedFlag*** parameter is set to **HIGH_SPEED**, LPHAP cannot be enabled.
Master TRP of a Light DM MIMO Cell	**NRDUCellTrp**.***TrpType*** set to **MASTER_DM_MIMO_LIGHT**	None	When the **NRDUCellTrp**.***TrpType*** parameter is set to **MASTER_DM_MIMO_LIGHT**, LPHAP cannot be enabled.

For details about the mutually exclusive functions of positioning-dedicated SRS transmission in uplink slots, see **7.1.3.2.2 Mutually Exclusive Functions**.

7.2.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
MetaAAU beam sensing	**NRDUCellBeamAlgo**.***BeamPerceiveMode*** set to **DISTRIBUTED_MODE**	*MIMO (TDD)*	MetaAAU beam sensing does not take effect when LPHAP is in effect.

7.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations)

Boards

- All NR TDD-capable main control boards support this function.
- NR TDD-capable baseband processing units: UBBPg3 and UBBPg3a

RF Modules

- Massive MIMO modules: 32T32R or 64T64R low-frequency TDD modules
- Easy Macro modules: 8T8R low-frequency TDD modules (Easy Macro modules support only TOA measurement.)
- RRU modules: 2T2R, 2T4R, 4T4R, or 8T8R low-frequency TDD modules (RRU modules support only TOA measurement.)

7.2.3.4 Cells

For LPHAP, the cell configuration must meet the following requirements:

- **NRDUCell**.***UlBandwidth*** must be set to 100 MHz.
- The same frequency band and slot configuration need to be configured for the cells covering the positioning area. In addition, the frequency band and slot configuration must meet the following requirements:
  - **NRDUCell**.***FrequencyBand*** must be set to **N40**, **N41**, **N77**, **N78**, or **N79**.
  - **NRDUCell**.***SlotAssignment*** must be set to **8_2_DDDDDDDSUU** or **8_2_DDDSUUDDDD**.
- The cells must be non-LampSite cells (with **NRDUCell**.***LampSiteCellFlag*** set to a value other than **YES**).
- The cells are not in SUL mode (with **NRDUCell**.***DuplexMode*** set to a value other than **CELL_SUL**).
- **NRDUCellPrach**.***PrachConfigurationIndex*** must be set to a value other than **0**, **1**, **2**, and **200**; in addition, when **NRDUCell**.***CellRadius*** is set to a value greater than or equal to 4610, **NRDUCellPrach**.***PrachConfigurationIndex*** must be set to a value other than **195**, **200**, **202**, **206**, and **210**.

7.2.3.5 Networking

LPHAP is supported only in distributed massive MIMO networking. In this situation, the networking requirements are the same as those for U-TDOA- and AoA-based high-precision positioning. For details, see **7.1.3.5 Networking**.

7.2.3.6 Others

- UEs must support the NR SA networking.
- The AMF and LMF versions must be V24.1.0 or later.
- If UEs capable of sending SRSs for positioning need to be positioned, the core network needs to support parsing of the *Positioning SRS* IE.
- The UE capability information elements of the UE must contain **ra-SDT-r17** and **srb-SDT-r17**.

- The UE capability information elements of the UE must contain **SRS-AllPosResourcesRRC-Inactive-r17**.

7.2.4 Operation and Maintenance

7.2.4.1 Data Configuration

7.2.4.1.1 Data Preparation

This function depends on U-TDOA- and AoA-based high-precision positioning. For details about the data preparation for U-TDOA- and AoA-based high-precision positioning, see **7.1.4.1.1 Data Preparation**. **Table 7-4** describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-4 Parameters used for activation (LPHAP)

Parameter Name	Parameter ID	Option	Setting Notes
User Service Type	**gNBRfspConfig**.***UserServiceType***	None	For UEs performing LPHAP services, this parameter needs to be set to **LPHAP_LCS** so that SRS resources for positioning are used to ensure positioning accuracy.
LCS Switch	**NRCellAlgoSwitch**.***LcsSwitch***	LPHAP_REDCAP_SW	Select this option if LPHAP is required.

7.2.4.1.2 Using MML Commands

Before using MML commands, refer to **7.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- This function depends on U-TDOA- and AoA-based high-precision positioning. For details about the activation command examples for U-TDOA- and AoA-based high-precision positioning, see **7.1.4.1.2 Using MML Commands**.
- User service type configuration (specified by **gNBRfspConfig**.***UserServiceType***)


```
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=102, UserServiceType=LPHAP_LCS;
```

- LPHAP function

```
//Setting the user service type to LPHAP if the service type planned for the SIM card is LPHAP
ADD GNBRFSPCONFIG: OperatorId=0, RfspIndex=102, UserServiceType=LPHAP_LCS;
//Enabling the LPHAP function
MOD NRCELLALGOSWITCH: NrCellId=1, LcsSwitch=LPHAP_REDCAP_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

LPHAP function

```
//Disabling the LPHAP function
MOD NRCELLALGOSWITCH: NrCellId=1, LcsSwitch=LPHAP_REDCAP_SW-0;
```

7.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.2.4.2 Activation Verification

For details about the methods of activation verification for Precise Positioning (NR), see **7.1.4.2 Activation Verification**.

7.2.4.3 Network Monitoring

After Precise Positioning (NR) is activated, the performance can be monitored by using the following counters.

Counter ID	Counter Name	Counter Description
1911829380	**N.Positioning.Info.Req.Att**	Number of positioning information requests
1911829379	**N.Positioning.Info.Req.Succ**	Number of positioning information responses
1911829377	**N.Positioning.Meas.Req.Att**	Number of positioning measurement requests
1911829376	**N.Positioning.Meas.Req.Succ**	Number of successful positioning measurements
1911829375	**N.Positioning.Meas.Rpt**	Number of positioning measurement reporting times

Counter ID	Counter Name	Counter Description
1911840425	**N.User.Positioning.LPHAP.Avg**	Average number of UEs performing LPHAP services
1911842612	**N.User.Positioning.Avg**	Average number of UEs performing positioning services

When a large number of positioning requests are initiated simultaneously, flow control may be triggered, causing NRPPa messages to be discarded. The performance can be monitored by using the following counter.

Counter ID	Counter Name	Counter Description
1911829378	**N.NRPPa.Ng.Msg.Disc.FlowCtrl**	Number of times NRPPa messages over the NG interface are discarded due to flow control

8 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

9 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

10 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

11 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.413: "NR; NG Application Protocol (NGAP)"
- 3GPP TS 38.455: "NR; NR Positioning Protocol A (NRPPa)"
- *High Speed Mobility*
- *Hyper Cell*
- *Cell Combination*
- *Channel Management*
- *UE Power Saving*
- *DRX*
- *CoMP*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *CA Experience Improvement*
- *Co-MM (Low-Frequency TDD)*
- *Super Uplink*
- *Massive MIMO AHR (Low-Frequency TDD)*
- *Carrier Aggregation*
- *Remote Interference Management (Low-Frequency TDD)*
- *MIMO (TDD)*
- *Energy Conservation and Emission Reduction*
- *RedCap*
- *Extended Cell Range*
- *Uplink Flexible Spectrum Scheduling*
- *License Control Item Lists*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*